

AZ-104

Microsoft Azure Administrator
Enterprise-Grade Study Guide, Architecture Reference & Exam Cheat Sheet

[Identity & Governance](#)

[Storage](#)

[Compute](#)

[Networking](#)

[Monitor & Backup](#)

[Scenario Bank](#)

[CLI + PowerShell Cheat Sheet](#)

Ownership & visibility maps at every scope · architecture diagrams · hub-spoke & identity flows · 20+ practical scenarios · full command reference

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1. Exam Overview, Strategy & Domain Weights

The AZ-104: Microsoft Azure Administrator exam validates the skills needed to implement, manage, and monitor an organization's Azure environment: identity, governance, storage, compute, virtual networking, and resource monitoring/backup. It targets administrators who perform day-to-day operational tasks rather than architects who design solutions from scratch (that's AZ-305) — expect questions phrased as "how do you configure/troubleshoot/implement X" rather than "which architecture should you design."

Attribute	Detail
Question count	Typically 40-60 questions
Duration	120-150 minutes (varies by region/language)
Passing score	700 / 1000 (scaled score, not a raw percentage)
Question formats	Multiple choice/multiple answer, drag-and-drop ordering, hot-area/mark review, case studies with tabs (requirements/existing environment/proposed solution), occasional labs
Prerequisite knowledge	6+ months hands-on Azure administration experience recommended; comfort with PowerShell and/or CLI scripting
Retake policy	24-hour wait after first attempt, 14 days after second, 5 attempts per 12 months

Domain weights and what each really tests

Domain	Weight	Core Services	What it really tests
Manage Azure identities and governance	20-25%	Entra ID, RBAC, Subscriptions, Management Groups, Policy, Locks, Tags, Cost Management, PIM	Whether you understand scope and can secure access without over- or under-granting permission
Implement and manage storage	15-20%	Storage Accounts, Blob/File/Queue/Table, Redundancy, Lifecycle, Import/Export, Backup	Whether you can choose the right redundancy/tier/access method for a stated durability and cost requirement
Deploy and manage Azure compute resources	20-25%	VMs, Availability Sets/Zones, VMSS, ARM/Bicep, Containers, App Service	Whether you can build resilient, right-sized, automatable compute
Configure and manage virtual networking	15-20%	VNet, Subnet, NSG, Peering, VPN Gateway, Load Balancer, Private Link, DNS	Whether you understand traffic flow, isolation, and hybrid connectivity
Monitor and back up Azure resources	10-15%	Azure Monitor, Log Analytics, Alerts, Azure Backup, Site Recovery	Whether you can detect problems and recover from them

Study allocation tip: Because Identity/Governance and Compute together make up 40-50% of the exam, and because governance concepts (scope, inheritance, RBAC) resurface inside every other domain's scenario questions, master Section 2 of this guide before moving on — it is the highest-leverage material in the entire syllabus.

Exam-day strategy

- **Read the "existing environment" tab first** in case studies — the correct answer is almost always constrained by something already deployed (an existing Availability Set, an existing Policy, an existing on-prem AD).
- **Eliminate answers that violate a hard technical fact** before comparing the "best" remaining options — e.g., "Management Groups can contain resources directly" is simply false and eliminates that option instantly.
- **When two answers both "work," pick the one with the narrower blast radius / least privilege** — Azure exams consistently reward the least-permissive, most-scoped solution that still satisfies every stated requirement.
- **Watch for absolute words** ("always," "never," "automatically") in answer options — Azure behavior is full of exceptions, and absolutist answers are frequently the distractor.
- **Budget your time:** if a case-study question is taking more than 3 minutes, mark it for review and move on — case studies often have repeated tabs that count against your clock even when you're not making progress.

Common traps that recur across every domain

Trap	Reality
"Owner on a resource group grants blob data access"	Management-plane roles (Owner/Contributor) do not grant data-plane access to Storage, Key Vault secrets, or Cosmos DB data by default — separate data-plane RBAC roles or keys/SAS are required
"VNet Peering is transitive"	It is not. A-B and B-C peered does not let A reach C without a route via a hub/NVA
"Stopping a VM stops billing"	Only Deallocating a VM stops compute billing; a guest-OS "Stop" from within Windows/Linux still leaves it billed
"A Management Group is a"	Only Subscriptions are billing boundaries; Management Groups are pure governance containers

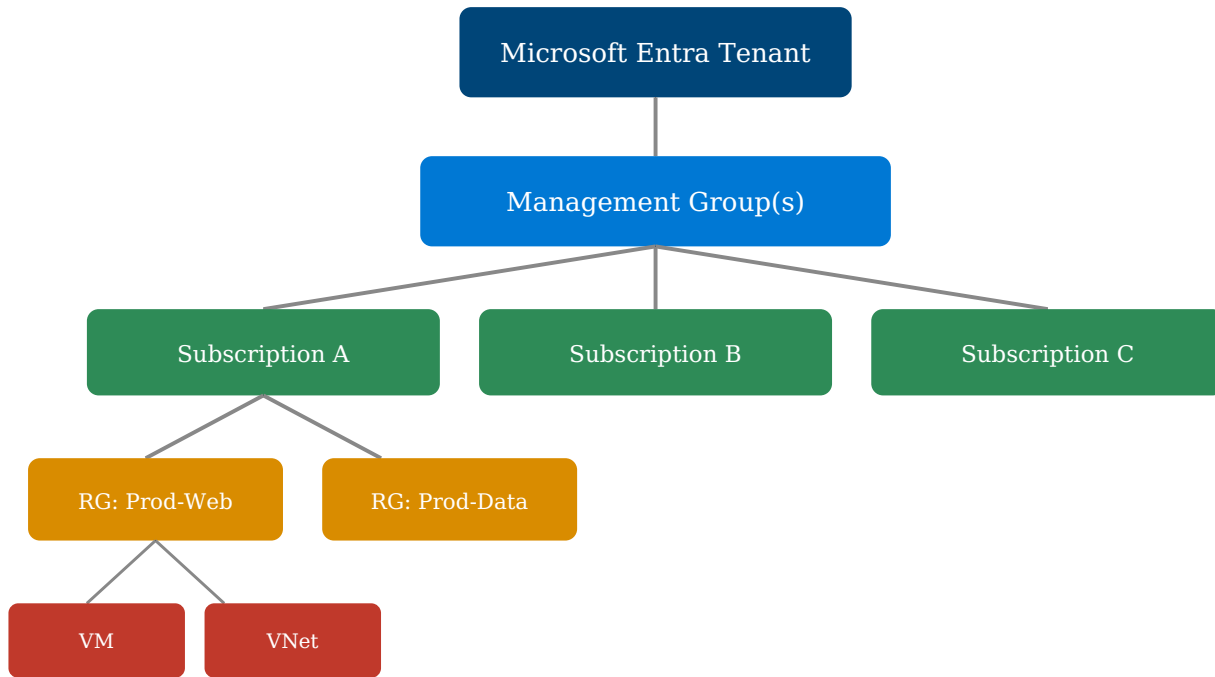
billing boundary"	
"Locks can be bypassed by an Owner"	Resource Locks apply regardless of RBAC role — even a subscription Owner must remove the lock first
"GRS is a disaster recovery / backup solution"	GRS is an <i>availability</i> feature for the storage service itself; it does not protect against accidental deletion or corruption the way Azure Backup does

"Every AZ-104 scenario question is really asking one of three things: where does this apply (scope), who can see or touch it (access), and what happens if it fails (resilience). Answer those three questions first and the correct option usually becomes obvious."

2. Identity & Governance

2.1 The Scope Hierarchy & Inheritance

Every governance decision in Azure — RBAC, Policy, Locks, Cost, Blueprints/Template Specs — is applied at a **scope**, and scopes nest in a strict tree. This is the single most important mental model for the entire exam: nearly every domain (storage security, network rules, backup policy) is ultimately "a thing applied at some scope, inherited by everything beneath it."



Inheritance flows **DOWN** only: Tenant → Mgmt Group → Subscription → Resource Group → Resource
 RBAC/Policy/Locks assigned higher up automatically apply to everything nested beneath

Figure 2.1 — Azure scope hierarchy and inheritance direction

Level-by-level deep dive

TENANT Microsoft Entra Tenant

The tenant is the top-level container for identity: every user, group, device, application registration, and external (B2B/B2C) identity lives here. It is **not** an Azure resource itself — it is the Entra ID directory that one or more Azure subscriptions trust for authentication. A tenant is identified by a unique tenant ID (GUID) and one or more verified domains (e.g., `contoso.com`).

- A person can be a **member** of one tenant (their home organization) but a **guest** in many others via B2B collaboration.
- A subscription trusts **exactly one** tenant for authentication at any given point in time — this can be changed (subscription-to-tenant transfer) but never split across two tenants simultaneously.
- The tenant also holds directory-wide settings: Conditional Access policies, MFA registration, external collaboration settings, and the root Management Group.

MGMT GROUP Management Groups

Management Groups exist purely to let you apply RBAC and Policy to **many subscriptions at once** without repeating the assignment on each one. They form a tree beneath a single, always-present **Root Management Group** (named after the tenant ID by default).

- Up to **6 levels** of management groups beneath the root (root + 6 = 7 total addressable levels including subscriptions as leaves).
- Up to **10,000** management groups per directory.
- A subscription can belong to exactly **one** management group at a time, but can be moved (with sufficient permission on source and destination).
- The root management group cannot be deleted or moved, and by default every new subscription lands directly under it until explicitly organized.

SUBSCRIPTION Subscriptions

The subscription is the fundamental **billing and default policy/RBAC boundary**. Resource quotas (e.g., number of vCPUs per region, number of storage accounts) are enforced per subscription. Common enterprise pattern: separate subscriptions per environment (Prod/Dev/Test) or per business unit, so that billing, quotas, and blast radius are cleanly separated.

- Types: Free Trial, Pay-As-You-Go, Enterprise Agreement (EA), Microsoft Customer Agreement (MCA), CSP (Cloud Solution Provider) — each has different billing hierarchies (e.g., EA: Enrollment → Department → Account → Subscription).
- A subscription's **Access control (IAM)** blade is where you typically start assigning RBAC in day-to-day admin work.

RESOURCE GROUP Resource Groups

A logical container and the standard **unit of lifecycle management** — most deployments target a resource group, and deleting a resource group deletes everything inside it. Resource groups themselves have a "location" property, but that only stores the RG's own metadata; the resources inside can be deployed to different regions than the RG's metadata location.

- Resource groups cannot be nested inside other resource groups.
- A resource belongs to exactly one resource group, but can reference/interact with resources in other resource groups, subscriptions, or even regions (e.g., a VM in RG-A can attach a disk from... no, actually disks must be same-region but can be a different RG).

RESOURCE Resources

The actual deployed object — VM, VNet, storage account, etc. This is the most granular scope for RBAC/Policy, giving the narrowest possible blast radius when you need to grant access to exactly one thing.

2.2 Ownership, Visibility & Accessibility per Level

The exam repeatedly tests a subtle distinction: **who owns** a level (can change its configuration), **who can see** it (visibility), and **what is accessible** from within it (accessibility of child/sibling resources). These three are not the same thing, and conflating them is a very common source of wrong answers.

Level	Ownership (who configures it)	Visibility (who can see it exists)	Accessibility (what it can reach)
Tenant	Global Administrator (Entra role) — distinct from any Azure RBAC role	Any authenticated user can see the tenant's public info (name, domains); directory objects are visible per Entra RBAC/roles (e.g., a Guest has restricted directory read by default)	Trusts 1+ subscriptions; users/groups/apps defined here are referenced by RBAC at every lower scope
Management Group	Owner/Contributor via RBAC scoped to that MG; the very first owner is granted by a Global Admin elevating themselves	Visible to anyone with Reader (or higher) at that MG or an ancestor MG	Any Policy/RBAC assigned here cascades to every subscription and resource beneath it
Subscription	Account Administrator (classic) / Owner (RBAC)	Visible to anyone with a role assignment at the subscription or an ancestor MG	Default scope for most day-to-day RBAC/Policy assignments; quotas and billing are enforced here
Resource Group	Owner/Contributor scoped to the RG	Visible to anyone with Reader+ at the RG, subscription, or ancestor MG	Deployment/lifecycle boundary; a "Move resources" operation changes this scope
Resource	Owner/Contributor scoped to the resource itself, or inherited from above	Visible to anyone with Reader+ at the resource or any ancestor scope	Data-plane accessibility (e.g., blob contents) is a <i>separate</i> concern from management-plane visibility — see §3.3

Exam gotcha: "Visibility" of a resource in the portal/CLI (can you see it in a list) requires only **Reader** at some scope covering it. "Accessibility" of its *data* (e.g., reading blob contents, connecting to a VM's disk) is governed by entirely separate mechanisms (data-plane RBAC, network rules, OS-level auth) that management-plane Reader/Owner do NOT automatically grant.

How to reason about a "can user X do Y to resource Z" question

- Find every scope** above resource Z: the resource itself, its RG, its subscription, and every ancestor management group up to root.
- Collect every role assignment** for user X (directly, or via any Entra group X belongs to) at each of those scopes.
- Union all the permitted actions** (RBAC is additive) — the highest-privilege applicable action at the narrowest scope typically decides borderline cases, but remember it's cumulative, not "closest wins."
- Check for Deny Assignments** (rare, usually from Blueprints/managed apps) and **Azure Policy deny effects** — these can block an action regardless of RBAC.
- Check for Resource Locks** at any level in the chain — these block delete/modify regardless of RBAC.
- For data-plane operations** (blob read, VM RDP/SSH, SQL query), separately verify a data-plane role, key/SAS, or network path exists — management-plane RBAC alone is not sufficient.

Cross-subscription and cross-tenant visibility

Scenario	What's visible	Mechanism
User with Reader on Subscription A only	Everything in Sub A; nothing in Sub B, even in the same tenant	RBAC scope is per-subscription unless assigned at a covering Management Group
Guest user from Tenant B invited into Tenant A	Only what's explicitly assigned via RBAC in Tenant A; default directory read is limited for guests	Entra B2B + RBAC assignment
MSP managing a customer's subscription	Only the delegated scope/roles defined in the Lighthouse offer, visible under "My customers" in the MSP's own tenant — without ever becoming a guest in the customer tenant	Azure Lighthouse (ARM-based cross-tenant delegation)
Two peered VNETs in different subscriptions	Network connectivity flows (if peering + NSGs allow), but management-plane visibility of one subscription's resources is <u>not</u> granted to the other's Owner unless RBAC is separately assigned	VNet Peering (network only) vs. RBAC (management) are independent

Tip: Network connectivity (peering, VPN) and management-plane access (RBAC) are always independent axes. Two resources can talk to each other over the network while their administrators have zero visibility into each other's subscriptions, and vice versa (an Owner might have full management rights over a VNet they can't route traffic through because of NSGs/UDRs).

2.3 RBAC — Roles, Scopes, Assignments, Effective Permissions

RBAC anatomy: the three-part assignment



Figure 2.2 — A role assignment = WHO + WHAT they can do + WHERE it applies

Built-in role tiers (memorize these four first)

Role	Manage access (RBAC)?	Manage resources?	Read?	Typical use
Owner	✓ Yes	✓ Full	✓	Subscription/RG admins; can delegate access to others
Contributor	✗ No	✓ Full (create/modify/delete)	✓	Engineers who deploy/manage resources but shouldn't grant access
Reader	✗ No	✗ No	✓	Auditors, dashboards, read-only visibility
User Access Administrator	✓ Yes (RBAC only)	✗ No	✓	IAM specialists who only manage who-can-access-what

Other frequently tested built-ins

Role	Scope of power
Virtual Machine Contributor	Manage VMs (start/stop/create) but cannot manage the VNet they attach to or grant access
Virtual Machine Administrator Login / User Login	Data-plane RBAC for Entra-based VM sign-in (Admin = local admin rights, User = standard user rights) — separate from Owner/Contributor
Network Contributor	Manage all network resources (VNet, NSG, Public IP, Load Balancer) but not VMs
Storage Account Contributor	Manage storage accounts (create/config) but does NOT grant data-plane access to blobs by default
Storage Blob Data Reader/Contributor/Owner	Data-plane access to blob contents — separate from the management-plane Storage Account Contributor role
Backup Contributor / Backup Operator	Manage Recovery Services vault backup policies/jobs (Operator cannot delete backup data)
Security Reader / Security Admin	View or manage Microsoft Defender for Cloud posture & policies
Key Vault Administrator / Secrets User / Certificates Officer	RBAC-model data-plane roles for Key Vault (modern replacement for Key Vault access policies)
Managed Identity Contributor / Operator	Create user-assigned identities vs. only assign them to resources
Log Analytics Reader / Contributor	Query vs. configure a Log Analytics workspace
Classic Administrator (Co-Admin/Service Admin)	Legacy Azure Service Management (ASM) model, largely deprecated but may appear in exam legacy questions

Classic exam gotcha — Management vs. Data plane: Being **Owner** or **Contributor** on a Storage Account does not automatically let you read blob data via Entra ID auth (data plane). You need an explicit data role like **Storage Blob Data Contributor**, OR use account keys/SAS which bypass RBAC entirely.

Effective permissions: Azure RBAC is additive

Azure RBAC is **additive by default**: your effective permission at a resource is the union of every role assigned to you (directly or via group) at that scope and every scope above it. There is no explicit "Deny" role in classic RBAC — the only way to truly deny is an **Azure RBAC Deny Assignment** (used internally by Blueprints/managed apps, not user-creatable directly) or an **Azure Policy** "deny" effect (which blocks actions, not identity).

Inherited access example

Alice = Reader at Subscription, Contributor at Resource Group "Web".

Effective at RG Web: **Contributor** (higher of the two, additive)
Effective at RG "Data" (sibling RG): **Reader only** (inherited from subscription)

Group-based assignment (best practice)

Assign roles to **Entra ID Groups**, not individual users.
Add/remove users from the group; RBAC updates automatically.
Reduces role-assignment sprawl (default limit: 4,000 role assignments per subscription).

Ownership model: who can assign what

- Only **Owner** or **User Access Administrator** at a scope can create new role assignments at that scope.
- You cannot grant a role you don't yourself have effective visibility into at that scope (no privilege escalation beyond your own assignable scope, except via a Custom Role's `AssignableScopes` misconfiguration — a known audit risk).
- **Global Administrator** (Entra ID role) is separate from Azure RBAC Owner, but a Global Admin can elevate themselves to **User Access Administrator** at root scope ("/") via Entra ID → Properties → "Access management for Azure resources" toggle — this is how a locked-out root management group gets its first owner.

Assigning roles: CLI and PowerShell side by side

```
# ---- Azure CLI ----
# Assign a role at resource group scope
az role assignment create --assignee user@contoso.com \
  --role "Contributor" \
  --resource-group Prod-Web

# List a user's effective roles across all scopes
az role assignment list --assignee user@contoso.com --all -o table

# Remove a role assignment
az role assignment delete --assignee user@contoso.com --role "Contributor" --resource-group Prod-Web

# ---- PowerShell (Az module) ----
# Assign a role at resource group scope
New-AzRoleAssignment -SignInName user@contoso.com `
  -RoleDefinitionName "Contributor" `
  -ResourceGroupName "Prod-Web"

# List a user's effective roles
Get-AzRoleAssignment -SignInName user@contoso.com

# Remove a role assignment
Remove-AzRoleAssignment -SignInName user@contoso.com `
  -RoleDefinitionName "Contributor" -ResourceGroupName "Prod-Web"
```

Custom roles

When built-ins don't fit, create a Custom Role from a JSON definition combining `Actions`, `NotActions`, `DataActions`, `NotDataActions`, and `AssignableScopes`.

```
{
  "Name": "VM Operator - Start/Stop Only",
  "IsCustom": true,
  "Description": "Can start, restart, and deallocate VMs but not create/delete",
  "Actions": [
    "Microsoft.Compute/virtualMachines/start/action",
    "Microsoft.Compute/virtualMachines/restart/action",
    "Microsoft.Compute/virtualMachines/deallocate/action",
    "Microsoft.Compute/virtualMachines/read"
  ],
  "NotActions": [],
  "DataActions": [],
  "NotDataActions": [],
  "AssignableScopes": ["/subscriptions/xxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx"]
}

# CLI: create the custom role from the JSON file
az role definition create --role-definition vm-operator-role.json

# PowerShell equivalent
New-AzRoleDefinition -InputFile 'vm-operator-role.json'
```

Tip: Up to 5,000 custom roles per tenant. Wildcards (`*`) are allowed in Actions but should be scoped narrowly for least privilege. Use `DataActions` / `NotDataActions` when the custom role needs to control data-plane operations (e.g., blob read but not write).

Checking "what can I do here" — a diagnostic flow



Figure 2.3 — The complete diagnostic sequence for "can user X do Y here"

2.4 Azure Policy, Initiatives & Governance at Scale

RBAC vs. Policy vs. Locks — know the difference cold

Mechanism	Controls	Question it answers	Applies to
RBAC	Who can do what	"Can this identity perform this action?"	Identities (users/groups/SPs/managed identities)
Azure Policy	What configurations are allowed, audited, or auto-remediated	"Is this resource's configuration compliant?"	Resources, regardless of who deploys them
Resource Locks	Prevent delete or delete+modify, irrespective of RBAC	"Can this resource be changed/removed at all?"	Resources/RGs/Subscriptions
Tags	Metadata for cost, ownership, automation	"Who owns this / what cost center?"	Most resource types

Policy effects (memorize order of "strength")

Effect	Behavior
Deny	Blocks the non-compliant request at deployment time (strongest, preventive)
DeployIfNotExists (DINE)	Triggers a remediation deployment (e.g., auto-install monitoring agent) if a related resource is missing
Modify	Adds/updates/removes properties or tags on existing resources (needs a Managed Identity with write rights)
Append	Adds fields/tags to the request before creation
Audit	Allows the request but flags it non-compliant in reports (detective, not preventive)
AuditIfNotExists	Flags non-compliance if a related resource is missing, without remediating
Disabled	Policy rule is turned off without deleting the definition (useful for staged rollout/testing)

Anatomy of a policy definition

```
{
  "properties": {
    "displayName": "Require HTTPS-only on storage accounts",
    "policyType": "Custom",
    "mode": "Indexed",
    "policyRule": {
      "if": {
        "allOf": [
          { "field": "type", "equals": "Microsoft.Storage/storageAccounts" },
          { "field": "Microsoft.Storage/storageAccounts/supportsHttpsTrafficOnly", "notEquals": "true" }
        ]
      },
      "then": { "effect": "deny" }
    }
  }
}
```

```
# CLI: create and assign
az policy definition create --name require-https-storage --rules policy.json --mode Indexed
az policy assignment create --name enforce-https --policy require-https-storage --scope "/subscriptions/xxxx"
```

```
# PowerShell equivalent
New-AzPolicyDefinition -Name "require-https-storage" -Policy 'policy.json'
New-AzPolicyAssignment -Name "enforce-https" -PolicyDefinition $def -Scope "/subscriptions/xxxx"
```

Scenario: "All storage accounts must disallow public blob access, and any non-compliant existing account should be fixed automatically." → Use a Policy with effect **Modify** or **DeployIfNotExists**, assigned a **system-assigned managed identity** with **Storage Account Contributor** on the remediation scope, then trigger a **Remediation task** for existing non-compliant resources (new resources are enforced going forward automatically).

Initiatives (Policy Sets) & Blueprints/Template Specs

- Initiative** = a group of related policy definitions assigned together (e.g., "CIS Azure Benchmark," "ISO 27001"). Simplifies assigning dozens of policies as one unit and reporting compliance as one aggregate score.
- Azure Blueprints** (legacy, being retired in favor of **Template Specs + Policy + deployment stacks**) packaged RG structure + RBAC + Policy + ARM templates as one versioned artifact for repeatable landing zones. Blueprint assignments "lock" their artifacts from casual deletion.
- Management Group policy assignments** are the standard way to enforce guardrails tenant-wide (e.g., "deny resource creation

outside approved regions" at the root management group).

- **Exemptions:** a specific resource or scope can be exempted from a policy/initiative assignment (with an expiry date and justification)
— useful for legitimate exceptions without weakening the overall policy.

Compliance evaluation triggers

Trigger	When it fires
Resource creation/update (ARM request)	Immediate, synchronous — this is how <code>Deny</code> blocks a deployment
Policy assignment/definition change	Triggers an evaluation cycle for all resources in scope within ~30 minutes
Standard compliance cycle	Every 24 hours automatically
On-demand	<code>az policy state trigger-scan</code> / <code>Start-AzPolicyComplianceScan</code>

"In Azure governance, RBAC decides who, Policy decides what shape, and Locks decide whether at all. A well-governed landing zone uses all three together, applied as high in the hierarchy as the business boundary allows."

2.5 Locks, Tags & Cost Management

Resource Locks

Lock type	Effect
CanNotDelete (Delete lock)	Authorized users can still read/modify, but cannot delete
ReadOnly	Authorized users can read but cannot modify or delete — can break dependent operations (e.g., can't scale a VM behind a ReadOnly-locked RG)

Gotcha: Locks inherit down and apply regardless of RBAC role — even a subscription Owner cannot delete a resource with a `CanNotDelete` lock until the lock is removed. A `ReadOnly` lock on a Resource Group can silently break automatic tasks like Key Vault soft-delete purge, VM disk resize, or Autoscale (which needs write access to scale instance count) because those need write access even though they look like "normal operation."

```
# CLI: create a delete lock on a resource group
az lock create --name DontDeleteProdRG --resource-group Prod-Web --lock-type CanNotDelete

# PowerShell equivalent
New-AzResourceLock -LockName "DontDeleteProdRG" -LockLevel CanNotDelete -ResourceGroupName "Prod-Web"
```

Tags

- Key-value pairs (max 50 per resource, 512 chars for key, 256 for value) used for cost allocation, automation, and governance.
- Enforce mandatory tags via Policy `Modify` (auto-append) or `Deny` (reject if missing) effect.
- Tags can be inherited to resources from their resource group/subscription using the **Inherit a tag from the subscription or resource group** built-in policy (does not retroactively apply — only on new/updated resources unless remediated).
- Not all resource types support tags (e.g., some classic/legacy resource types).

Cost Management + Billing

Feature	Purpose
Cost Analysis	Interactive breakdown of spend by resource, tag, service, or resource group over a time range
Budgets	Set a spend threshold with alert notifications at defined percentages (e.g., 80%, 100%, 120% forecasted)
Azure Advisor (Cost)	Rightsizing recommendations for underutilized VMs, unattached disks, idle resources
Reservations	1 or 3-year commitment for VM/DB/other compute in exchange for a significant discount (up to ~70%)
Savings Plans	Commit to a \$/hour spend across eligible compute (more flexible than Reservations, slightly lower discount)
Cost allocation / Cross-charge	Group and re-bill costs to internal departments using tags or management groups

Tip: Enterprise Agreement (EA) hierarchy (*Enrollment* → *Department* → *Account* → *Subscription*) is a billing hierarchy, distinct from the Entra/Management Group governance hierarchy — organizations often map them 1:1 for simpler reporting, but the exam may test that they are technically independent systems.

2.6 Privileged Identity Management & Conditional Access

Privileged Identity Management (PIM)

PIM converts **standing** (always-on) privileged access into **just-in-time** (JIT), time-bound, approval-gated access — a core Zero Trust practice and an increasingly common exam topic.

PIM concept	What it does
Eligible assignment	User can <i>activate</i> a role for a limited window rather than holding it permanently
Active assignment	Traditional standing assignment (still supported, used sparingly for break-glass or automation accounts)
Activation requirements	Can require MFA re-authentication, justification text, approval from a designated approver, and a maximum activation duration
Access reviews	Periodic recertification — are these eligible/active assignments still needed?
Alerts	PIM flags anomalies like roles activated too frequently or assigned outside PIM entirely

PIM applies to both **Entra ID directory roles** (Global Admin, User Admin, etc.) and **Azure resource roles** (Owner, Contributor at any scope) — the two are configured in separate blades but follow the same eligible/active/activation model.

Scenario: "Production subscriptions must require a change ticket reviewed by a second person before anyone gets Contributor access." → Configure the Contributor role assignment on the production subscription as **PIM-eligible** rather than active/standing, require approval from a designated approver group, require MFA and a business justification on activation, and set a maximum activation duration (e.g., 8 hours) so access automatically expires.

Conditional Access

Conditional Access (CA) is an Entra ID feature that enforces *if-then* access rules based on signals: user/group, application, device state, location/network, and sign-in risk. It answers "under what conditions is a sign-in allowed" — complementary to, but distinct from, RBAC (which answers "what can an already-signed-in identity do").

Signal (condition)	Example
User/Group	Applies only to the "Azure Admins" group
Cloud app	Applies only when accessing the Azure portal / Azure Management API
Location	Block or require MFA when sign-in originates outside named trusted IP ranges
Device state	Require the device be marked compliant (Intune) or hybrid Entra-joined
Sign-in risk (Entra ID Protection)	Require MFA or block when risk is Medium/High (impossible travel, leaked credentials, etc.)

Grant controls (then...)	Effect
Block access	Sign-in denied outright
Require MFA	Additional factor required even with valid password
Require compliant/hybrid-joined device	Blocks unmanaged/BYOD devices
Require approved client app / app protection policy	Enforces managed-app data controls (Intune)

Tip: Conditional Access requires Entra ID P1 (included in most Microsoft 365/EA bundles); PIM requires Entra ID P2. Both are frequently paired with RBAC in enterprise scenario questions — CA controls *whether you can sign in at all*, PIM controls *whether your privileged role is currently active*, RBAC controls *what that role lets you do*.

2.7 Scenario: Acquiring a New Organization

This is one of the most realistic "senior admin" scenarios and maps directly to exam case studies: your company (Contoso) acquires another company (Fabrikam), which already runs its own Entra tenant and Azure subscriptions. You must bring Fabrikam under governance while keeping it operational.

Decision tree: how much do you merge?

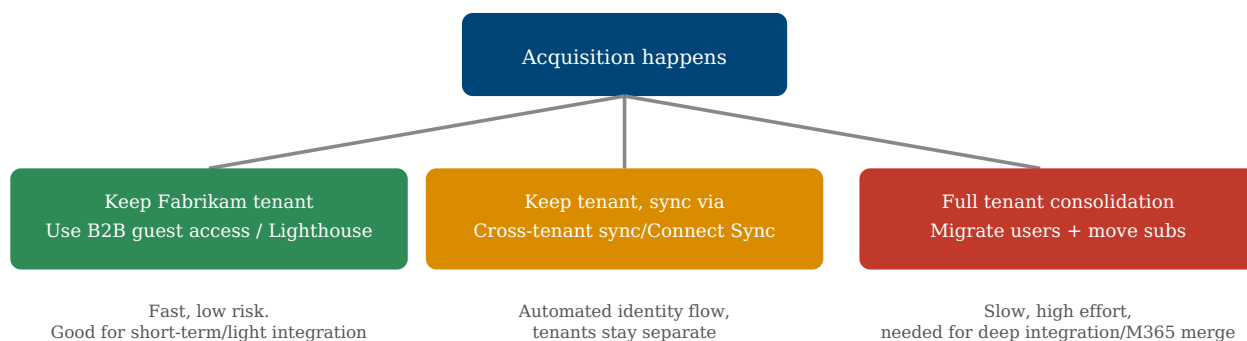


Figure 2.4 — Three common acquisition integration strategies, lightest to heaviest

Option A — Azure Lighthouse / Guest (B2B) access (no tenant change)

1. Invite key Fabrikam admins as **Entra B2B guest users** into the Contoso tenant, or use **Azure Lighthouse** to let Contoso admins manage Fabrikam's subscriptions *without moving them* — Lighthouse projects RBAC across tenants via a delegated resource management template.
2. Contoso engineers get role assignments (e.g., Contributor) onto Fabrikam's subscription scope, visible in the Contoso "My customers" blade — the subscription's identity/billing stays with Fabrikam's tenant.
3. Best for: MSP-style oversight, short transition periods, or when Fabrikam must remain a legally separate entity.

```
# Azure Lighthouse delegation is deployed as an ARM template from the managing tenant
az deployment sub create --location eastus \
  --template-file lighthouse-delegation.json \
  --parameters mspOfferName="Contoso Managed Services" \
    mspTenantId="<contoso-tenant-id>" \
    authorizations='[{"principalId":"<contoso-group-id>","roleDefinitionId":"b24988ac-6180-42a0-ab88-20f7382dd24c"}]'

# PowerShell equivalent
New-AzSubscriptionDeployment -Location eastus -TemplateFile lighthouse-delegation.json `
  -mspOfferName "Contoso Managed Services" -mspTenantId "<contoso-tenant-id>"
```

Option B — Cross-tenant synchronization (tenants stay separate, identities flow)

- **Cross-tenant synchronization** (Entra ID feature) automatically provisions/de-provisions Fabrikam users as B2B guests in Contoso's tenant on a schedule — avoids manual invites at scale.
- Combine with **Entra ID cross-tenant access settings** to control inbound/outbound trust: which Fabrikam users can access Contoso resources, whether MFA/device compliance from Fabrikam's tenant is trusted (trust settings), and B2B collaboration defaults.
- Resources stay governed by Fabrikam's own RBAC/Policy at their subscription, while cross-tenant access settings gate what flows between tenants.

Option C — Full consolidation: moving subscriptions into the Contoso tenant

1. **Transfer billing ownership** of the Fabrikam subscription to a Contoso account (Cost Management + Billing → Transfer subscription), or for EA/MCA scenarios use the **subscription-to-tenant transfer** so the subscription's Entra directory association changes to Contoso.
2. All RBAC role assignments in the subscription that reference Fabrikam-tenant identities become invalid post-move (principals no longer resolve) — **re-create role assignments** using Contoso-tenant identities (or synced Fabrikam users) before or immediately after cutover.
3. Re-parent the subscription under the correct **Management Group** in Contoso's hierarchy so Contoso's Policy/RBAC guardrails apply automatically.
4. Migrate/merge users: bulk-provision Fabrikam users into Contoso Entra ID (or keep Entra Connect/Cloud Sync hybrid sync from Fabrikam's on-prem AD if it must continue operating), then decommission the old tenant once cutover is verified.

5. Update DNS/custom domains, re-issue app registrations/service principals that were tenant-specific, and rotate any secrets/certs tied to the old tenant.

Exam gotcha: Moving a subscription to a new tenant does not move the resources' RBAC assignments with it in a usable way — assignments referencing old-tenant object IDs silently stop working because those object IDs don't exist in the new tenant. Always audit and re-assign RBAC after a tenant change.

Governance after acquisition — landing zone pattern

Step	Action
1	Create/confirm a management group for the acquired org, e.g. MG-Fabrikam , nested under Contoso's root or a "Subsidiaries" node
2	Move Fabrikam's subscription(s) into that management group
3	Assign Contoso's baseline Policy initiative (tagging, allowed regions, security baseline) at that management group
4	Assign RBAC at the management group for the Contoso platform team (e.g., Reader for visibility, Contributor for a transition team)
5	Review/consolidate Log Analytics workspaces, Recovery Services vaults, and budgets for the acquired subscriptions

Tip: Prefer **Azure Lighthouse or cross-tenant sync over a hard tenant merge** whenever the acquired company must retain operational independence (common immediately post-acquisition). Full consolidation is a longer project usually done 6-18 months later.

2.8 Subscription Transfers & RBAC Reassignment

Types of subscription transfer

Transfer type	What changes	What stays
Billing ownership transfer (same tenant)	Who pays; the Account Admin	Tenant association, RBAC assignments (identities still resolve)
Subscription-to-tenant transfer (cross-tenant)	Which Entra tenant authenticates the subscription; all existing RBAC assignments become orphaned (identities from the old tenant no longer resolve)	Resource configuration/data itself is untouched — only identity/auth context changes
Management group re-parenting	Which Policy/RBAC guardrails cascade down from above	Subscription's own direct role assignments and resources

Step-by-step: cross-tenant subscription transfer checklist

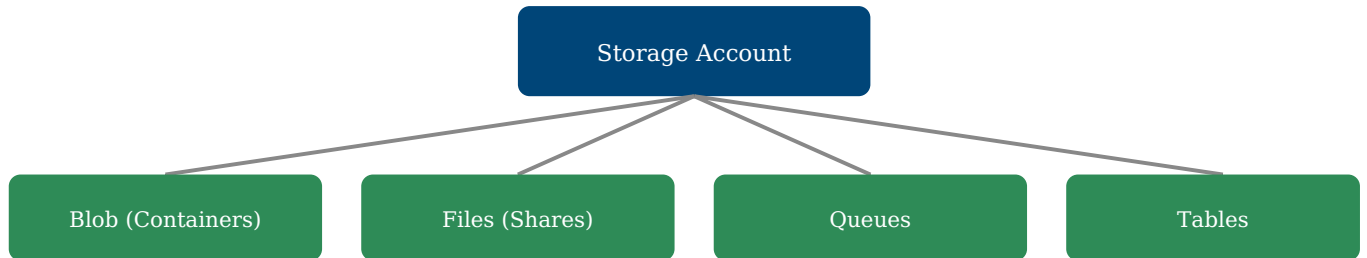
1. **Pre-transfer audit:** export all role assignments (`az role assignment list --all`) and all Key Vault access policies/managed identity references — anything tied to the old tenant's object IDs will break.
2. **Initiate transfer** via Cost Management + Billing → Subscriptions → Transfer billing ownership, or via the Azure Account Center for MCA/PAYG; accept in the destination tenant.
3. **Immediately re-create RBAC assignments** using destination-tenant identities (users, groups, or newly-created service principals) — do this as close to cutover as possible to minimize the access gap.
4. **Re-associate Managed Identities:** user-assigned identities and their role assignments may need re-creation since role assignments reference the identity's object ID under the old tenant.
5. **Update service principals / app registrations** used by CI/CD pipelines — these live in the tenant, not the subscription, and must be re-created or re-referenced in the new tenant.
6. **Re-parent under the correct Management Group** in the destination tenant so inherited Policy/RBAC applies.
7. **Validate:** confirm Log Analytics, Backup vaults, and any cross-resource references (e.g., Private DNS Zone links, VNet peerings to other subscriptions) still function.

Gotcha: VNet peerings to VNets in *other* subscriptions still work after a tenant transfer (peering is a subscription-to-subscription network construct, not tenant-bound) — but any RBAC needed to *manage* that peering from the new tenant's identities must be re-granted.

3. Storage

3.1 Storage Account Model & Service Relationships

Storage account hierarchy



One storage account = one namespace; each service type is a sibling child, not nested inside another

Service-by-service relationships & typical pairings

Service	Data model	Typical consumer	Association with other services
Blob	Flat namespace of containers holding unstructured objects (blobs); hierarchical namespace optional (ADLS Gen2)	VMs, web apps, Data Factory, Databricks, backup targets	Container = folder-like grouping; a blob's URL is <code>https://<account>.blob.core.windows.net/<container>/<blob></code> ; often paired with Event Grid for change notifications (e.g., trigger a Function on blob upload)
Files	Fully managed SMB/NFS file shares	VMs mounting shares, lift-and-shift of on-prem file servers, Azure File Sync targets	Can be the cloud endpoint of an Azure File Sync topology; NTFS-like permissions via identity-based auth (Entra Kerberos, Entra DS, on-prem AD DS)
Queues	Simple FIFO-ish messaging (at-least-once delivery, visibility timeout model)	Decoupling app tiers, work-item distribution (e.g., web tier enqueues, worker tier dequeues)	Often paired with VMSS/Functions for autoscaling workers based on queue length
Tables	NoSQL key-attribute store (PartitionKey + RowKey)	Lightweight structured data, device/telemetry logs, session state	Largely superseded by Cosmos DB Table API for new designs needing global distribution, but classic Table service remains supported and cheaper

Storage account kinds & performance tiers

Account kind	Supports	Notes
StorageV2 (general purpose v2)	Blob, Files, Queues, Tables	Default recommended kind; supports all tiers (Hot/Cool/Cold/Archive) and lifecycle management
BlockBlobStorage	Blob only, Premium performance	High-transaction-rate workloads, low latency
FileStorage	Files only, Premium performance	Required for NFS 4.1 Azure Files shares
Storage (classic / legacy v1)	Legacy, deprecated for new deployments	Should be migrated to StorageV2 when encountered

Performance tier	Media	Use case
Standard	HDD-backed	General purpose, cost-sensitive
Premium	SSD-backed	Low-latency, high-IOPS workloads (VM disks, databases, high-transaction blob)

Naming, namespace & hierarchical namespace (ADLS Gen2)

- Storage account name: globally unique across all of Azure, 3-24 lowercase letters/numbers only.
- Enabling **Hierarchical Namespace (HNS)** on a StorageV2 account turns it into **Azure Data Lake Storage Gen2** — adds real directory semantics (atomic rename/delete of folders) and POSIX-like ACLs on top of Blob storage, used heavily by analytics engines (Synapse, Databricks, HDInsight).
- HNS must be chosen at account creation and generally cannot be toggled off afterward.

3.2 Redundancy Options & Access Tiers

Redundancy options (SLA/durability ladder)

SKU	Copies	Scope	Durability	Use when
LRS (Locally redundant)	3 copies	Single datacenter	11 nines	Cheapest; dev/test, reproducible data
ZRS (Zone redundant)	3 copies	3 Availability Zones, same region	12 nines	Need zone-level resilience, data must stay in-region
GRS (Geo redundant)	6 copies (3+3)	Primary region + paired secondary region	16 nines	Regional disaster protection; secondary not readable unless RA-GRS
RA-GRS	6 copies	Primary + secondary, secondary is readable	16 nines	Read access needed even during a regional outage
GZRS / RA-GZRS	6 copies	ZRS in primary + geo copy to secondary	16 nines	Best combined zone + region resilience

Gotcha: Geo-failover (GRS → secondary becomes primary) is a manual customer-initiated action (or Microsoft-initiated in a true regional loss) and is not instantaneous — it's an availability feature, not a real-time DR mechanism. It is not a substitute for Azure Backup. Replication itself is asynchronous, so a small amount of the most recent writes (Recovery Point Objective, typically minutes) can be lost on failover.

Access tiers (Blob only) — cost vs. latency tradeoff

Tier	Storage cost	Access cost	Retrieval latency	Min. storage duration
Hot	Highest	Lowest	Immediate	None
Cool	Lower	Higher than Hot	Immediate	30 days
Cold	Lower still	Higher than Cool	Immediate	90 days
Archive	Lowest	Highest	Hours (rehydrate required)	180 days

Automate tier transitions with a **Lifecycle Management policy** (rule-based JSON, e.g., "move to Cool after 30 days of no access, Archive after 90, delete after 365"). Rules can filter by blob prefix/container and by last-modified or last-accessed time (if access tracking is enabled).

```
# CLI: set a blob's tier
az storage blob set-tier --account-name stgprod01 --container-name uploads --name file.zip --tier Cool

# PowerShell equivalent
Set-AzStorageBlobTier -Container "uploads" -Blob "file.zip" -Tier "Cool" -Context $ctx

# Rehydrate a blob from Archive (priority: Standard = up to 15h, High = under 1h)
az storage blob set-tier --account-name stgprod01 --container-name archive --name old.zip \
  --tier Hot --rehydrate-priority High
```

Choosing redundancy + tier together — worked examples

Workload	Recommended SKU	Recommended tier	Reasoning
Compliance archive, 7-year retention, rarely read	GRS	Archive	Cheapest storage, cross-region protection, retrieval time acceptable since access is rare
Active web app content, must survive a zone outage	ZRS	Hot	Zone resilience without cross-region latency/cost, frequent access
Financial transaction logs, must be readable during regional outage	RA-GRS	Hot or Cool depending on access pattern	Readable secondary is a hard requirement
Dev/test scratch data	LRS	Hot	Cheapest option, resilience not required

3.3 Ownership: Management Plane vs. Data Plane

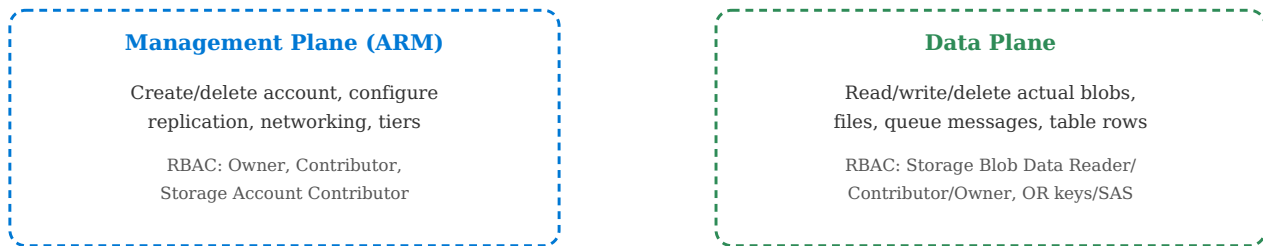


Figure 3.1 — Two independent authorization systems for the same storage account

This distinction is one of the most frequently misunderstood concepts on the exam. A storage account has **two separate control surfaces**:

- **Management plane** (Azure Resource Manager): operations on the storage account resource itself — create, delete, configure redundancy/networking/tiers, rotate keys, view metrics. Governed entirely by standard Azure RBAC roles like Owner, Contributor, or the narrower Storage Account Contributor.
- **Data plane**: operations on the actual content — reading a blob, writing a queue message, listing table rows. Governed by *either*: (a) Entra ID-based data RBAC roles (Storage Blob Data Reader/Contributor/Owner , Storage Queue Data ... , Storage Table Data ...), or (b) shared-secret methods that bypass Entra ID entirely: account keys and SAS tokens derived from them.

Exam gotcha: A user with Owner on a storage account can rotate its keys and therefore *indirectly* gain data access that way — but they cannot read blob data *directly* via *their Entra ID identity* without an explicit data-plane role. This "indirect path via key rotation" is itself worth knowing: it's why disabling account-key access entirely (`allowSharedKeyAccess = false`) is a meaningful security hardening step even for accounts already using RBAC.

Who "owns" data-plane access, summarized

Access method	Effectively controlled by	Revocation
Storage Blob Data Contributor (RBAC)	Whoever manages RBAC at that scope (Owner/User Access Admin)	Remove the role assignment — immediate
Account key	Whoever has Owner/Contributor and can read keys, or anyone who already has a key	Rotate/regenerate the key — invalidates ALL SAS tokens derived from it too
SAS token (account-key based)	Whoever generated it; valid until expiry or until the underlying key is rotated	Rotate the parent key, or use a stored access policy you can revoke
User delegation SAS (Entra-ID based)	Entra ID; tied to the delegating user's own RBAC permissions and token lifetime	Revoke the delegating identity's role assignment, or let the (short) token expire

3.4 VM-to-Storage Access Paths

Every path a VM (or app) can use to reach storage

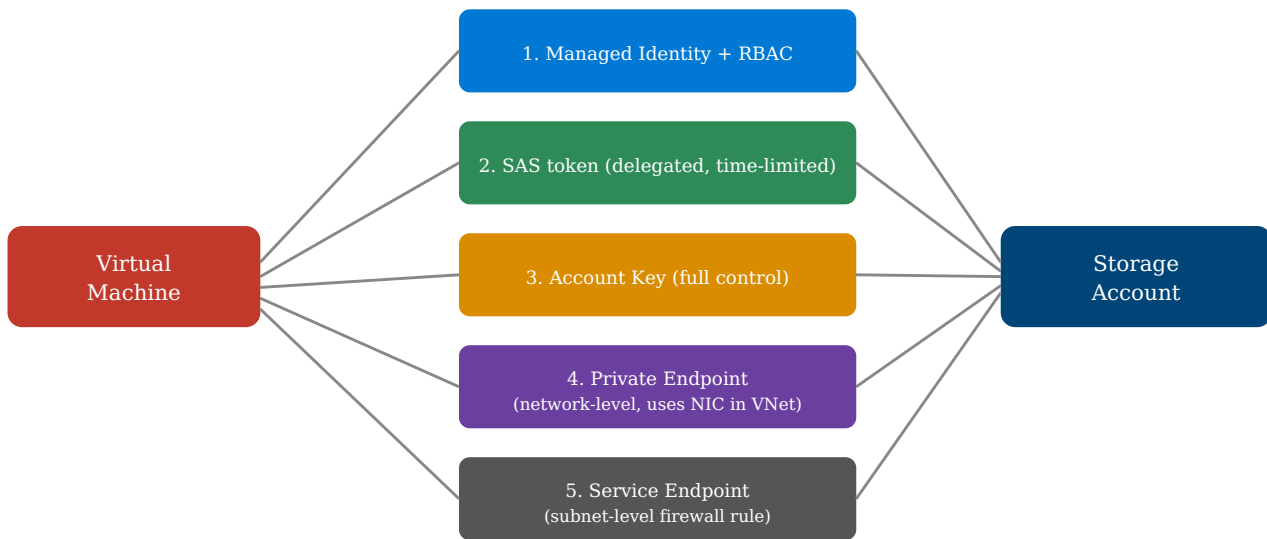


Figure 3.2 — Five ways a VM/app reaches a storage account, identity-based to network-based

Method	Auth model	Best practice rating	Notes
System/User-assigned Managed Identity + RBAC data role	Entra ID token, no secrets	★★★★★ Best	VM's identity is granted e.g. Storage Blob Data Contributor; SDK/CLI fetches token automatically from IMDS (169.254.169.254)
SAS token (Shared Access Signature)	Signed URL, scoped permissions + expiry	★★★★	Good for delegating temporary/external access without sharing keys; prefer user delegation SAS (backed by Entra ID) over account-key SAS
Storage account key	Shared secret, full account control	★★ Avoid	Bypasses RBAC entirely; rotate regularly; disable via "Allow storage account key access = No" to force Entra ID auth tenant-wide
Private Endpoint	Network isolation (NIC gets a private IP in your VNet)	★★★★★	Traffic never traverses the public internet; pair with Private DNS zone <code>privatelink.blob.core.windows.net</code>
Service Endpoint	Subnet-level route optimization + firewall allow-list	★★★	Traffic stays on Microsoft backbone but storage account still has a public endpoint; simpler than Private Link but less isolated

```

# ---- CLI ----
# Enable a system-assigned managed identity on a VM
az vm identity assign -g Prod-Web -n WebVM01

# Grant that identity Blob data access on the storage account
az role assignment create \
  --assignee-object-id $(az vm show -g Prod-Web -n WebVM01 --query identity.principalId -o tsv) \
  --assignee-principal-type ServicePrincipal \
  --role "Storage Blob Data Contributor" \
  --scope $(az storage account show -g Prod-Data -n stgprod01 --query id -o tsv)

# Generate a short-lived user-delegation SAS (Entra-ID backed, no account key needed)
az storage container generate-sas --account-name stgprod01 --name uploads \
  --permissions rw --expiry 2026-09-20T00:00Z --auth-mode login --as-user

# ---- PowerShell ----
Update-AzVM -ResourceGroupName Prod-Web -VM $vm -IdentityType SystemAssigned

New-AzRoleAssignment -ObjectId $vm.Identity.PrincipalId `
  -RoleDefinitionName "Storage Blob Data Contributor" `
  -Scope (Get-AzStorageAccount -ResourceGroupName Prod-Data -Name stgprod01).Id

New-AzStorageContainerSASToken -Name uploads -Permission rw `
  -ExpiryTime (Get-Date).AddDays(1) -Context $ctx
  
```

Scenario: "A web app on a VM needs to write log files to Blob storage. No secrets should be stored in code or config." → Enable a **system-assigned managed identity** on the VM, grant it `Storage Blob Data Contributor` scoped to the specific container/storage account (least privilege), and use the Azure SDK's `DefaultAzureCredential` which auto-discovers the managed identity token — zero secrets anywhere.

File shares: SMB/NFS access from VMs

- **Azure Files (SMB):** mount directly from Windows/Linux VMs using the storage account key or, in hybrid AD-joined scenarios, **Azure Files identity-based authentication** (Entra Domain Services, Entra Kerberos, or on-prem AD DS) for per-user NTFS permissions.
- **Azure Files (NFS)** 4.1: network-only auth (no identity layer) — secure via VNet/Private Endpoint, requires Premium (FileStorage) accounts.
- Port 445 (SMB) is blocked by many ISPs/corporate networks outbound — a common real-world VM-can't-mount-share issue; mitigate via Private Endpoint + VPN/ExpressRoute or Azure File Sync.

```
# Mount an Azure Files SMB share on a Linux VM
sudo mkdir -p /mnt/filesshare
sudo mount -t cifs //stgprod01.file.core.windows.net/myshare /mnt/filesshare \
  -o vers=3.0,username=stgprod01,password=<account-key>,serverino

# Windows PowerShell: persist a mount as a drive letter
$connectTestResult = Test-NetConnection -ComputerName stgprod01.file.core.windows.net -Port 445
New-PSDrive -Name Z -PSProvider FileSystem -Root "\\stgprod01.file.core.windows.net\myshare" `
  -Credential (New-Object PScredential "localhost\stgprod01", (ConvertTo-SecureString "<account-key>" -AsPlainText -Force)) -Persist
```

3.5 Storage Security & Network Firewall Model

Storage network firewall model

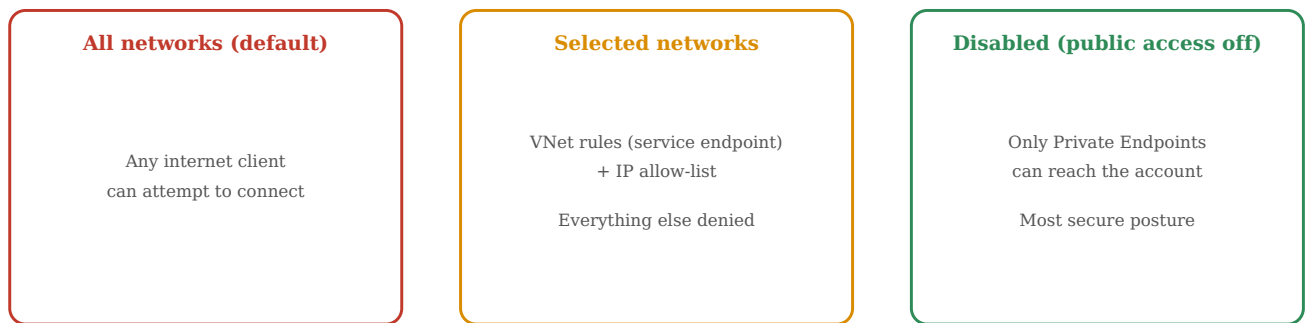


Figure 3.3 — Three network access postures for a storage account, least to most restrictive

By default a new storage account allows traffic from **all networks**. Locking down: Networking blade → set to **Selected networks**, then add either Virtual Network rules (service endpoint) or rely on Private Endpoints, plus optional IP allow-list for on-prem ranges. Setting to "Disabled" (public network access off) forces all traffic through Private Endpoints only.

Encryption

Layer	Mechanism	Key management
At rest (default)	Storage Service Encryption (SSE), AES-256, always on, cannot be disabled	Microsoft-managed keys (default) or Customer-managed keys (CMK) in Key Vault for compliance requiring key control
In transit	HTTPS/TLS enforced via <code>supportsHttpsTrafficOnly</code>	N/A
Client-side (optional, app responsibility)	Encrypt before upload using the Storage Client library's client-side encryption	App-managed keys, often via Key Vault
Infrastructure encryption (double encryption)	A second layer of encryption at the infrastructure level, distinct key from SSE	Set at account creation, cannot be enabled after the fact

Data protection features

- **Soft delete** (blobs, containers, file shares) — retain deleted items for a configurable period (1-365 days) and allow undelete.
- **Versioning** — automatically keeps prior versions of a blob on every overwrite.
- **Point-in-time restore** (blob) — roll an entire container (or account) back to a specific timestamp; requires versioning + change feed + soft delete enabled together.
- **Immutable storage (WORM)** — time-based retention or legal hold policies that prevent modification/deletion even by an account Owner, used for compliance (e.g., SEC 17a-4, financial records).
- **Change feed** — ordered, durable log of every create/update/delete on blobs, consumable for audit/replication pipelines.

Security best-practice checklist

Control	Setting
Minimum TLS version	TLS 1.2 or higher
Shared key access	Disabled where feasible; force Entra ID auth
Public blob access (anonymous)	Disabled at the account level unless a specific container intentionally needs public read
Network access	Selected networks or Disabled + Private Endpoints
Soft delete	Enabled, 7-30+ days depending on RPO needs
Defender for Storage	Enabled for malware scanning / anomalous access alerts
Diagnostic logging	Sent to Log Analytics for audit trail of data-plane operations

4. Compute

4.1 VMs, Availability Sets & Zones

Availability building blocks

Construct	Protects against	SLA (uptime)	Key facts
Single VM (Premium SSD)	N/A	99.9%	No redundancy
Availability Set	Rack/host hardware failure within one datacenter	99.95%	Fault Domains (power/network, up to 3) × Update Domains (up to 20, patching groups); all VMs must be in the set at creation time; cannot add an existing VM to a set after creation
Availability Zone	Entire datacenter failure	99.99% (VM SLA when using Zones)	Physically separate datacenters with independent power/cooling/network within a region; pick zone 1/2/3 explicitly per VM
Virtual Machine Scale Set (VMSS) across zones	Both host and datacenter failure + enables autoscale	99.99% (zonal, 2+ VMs)	Identical VM instances behind a Load Balancer, scale in/out on metrics/schedule

Fault Domains & Update Domains explained

Availability Set (2 Fault Domains x 3 Update Domains example)

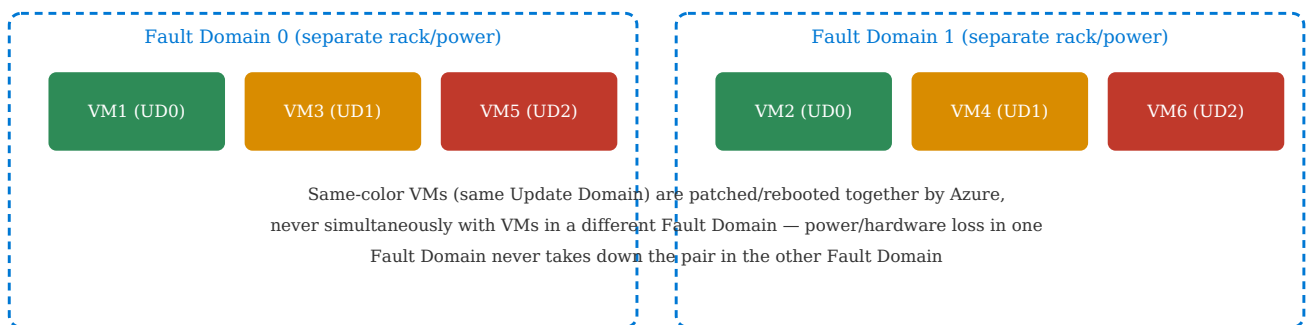


Figure 4.1 — Fault Domains isolate hardware failure; Update Domains isolate planned maintenance

Sizing families cheat table

Series	Optimized for	Typical use
B-series	Burstable, low baseline CPU with credits	Dev/test, low-traffic web servers
D-series	General purpose, balanced CPU/memory	Most application servers
E-series	Memory-optimized	In-memory databases, caching layers
F-series	Compute-optimized (high CPU:memory ratio)	Batch processing, gaming servers, CPU-bound workloads
L-series	Storage-optimized, high disk throughput	Big data, NoSQL databases
N-series	GPU-enabled	ML training/inference, visualization
M-series	Massive memory	SAP HANA, very large in-memory workloads

VM lifecycle states (know these for exam wording)

State	Billed for compute?	Billed for disk?	Notes
Running	Yes	Yes	
Stopped (via OS, not Azure)	Yes	Yes	VM still allocated to hardware — common billing mistake
Stopped (Deallocated)	No	Yes	Only <code>az vm deallocate</code> / Portal "Stop" releases the underlying hardware allocation; public IP (Basic SKU, dynamic) may be released

```
# ---- CLI ----
az vm create -g Prod-Web -n WebVM01 --image Ubuntu2204 --size Standard_D2s_v5 \
  --availability-set webAvSet --admin-username azureuser --generate-ssh-keys

az vm deallocate -g Prod-Web -n WebVM01
az vm start -g Prod-Web -n WebVM01

# ---- PowerShell ----
New-AzVM -ResourceGroupName Prod-Web -Name WebVM01 -Image UbuntuLTS `
  -Size Standard_D2s_v5 -AvailabilitySetName webAvSet

Stop-AzVM -ResourceGroupName Prod-Web -Name WebVM01 -Force
Start-AzVM -ResourceGroupName Prod-Web -Name WebVM01
```

4.2 Virtual Machine Scale Sets & Autoscale

Region: East US

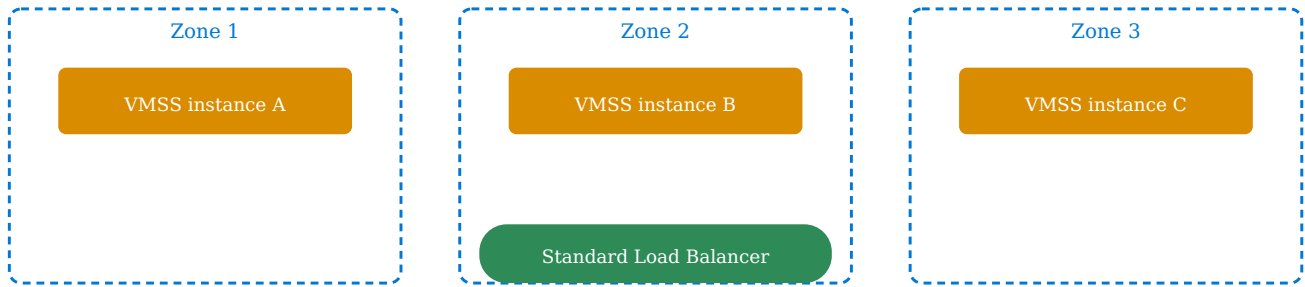


Figure 4.2 — Zone-redundant VMSS: identical instances spread across 3 Availability Zones behind one Load Balancer

Orchestration modes

Mode	Description	Use when
Uniform	Identical VM instances from a single model; scales fast, simpler management	Stateless, homogeneous workloads (web front ends)
Flexible	Mix of VM sizes, can mix Spot + Standard, individual instance management, higher instance count limits (up to 1,000)	Heterogeneous workloads, need per-instance customization, higher availability guarantees closer to individually placed VMs

Autoscale triggers

Trigger type	Example
Metric-based	Scale out when average CPU > 75% for 10 minutes
Schedule-based	Scale to 10 instances every weekday at 8 AM, scale back to 2 at 8 PM
Custom metric	Scale based on Application Insights request queue length or a custom Log Analytics query

```
# ---- CLI ----
az vmss create -g Prod-Web -n webVmss --image Ubuntu2204 \
  --instance-count 3 --zones 1 2 3 --vm-sku Standard_D2s_v5 \
  --upgrade-policy-mode Automatic

az vmss scale -g Prod-Web -n webVmss --new-capacity 6

az monitor autoscale create -g Prod-Web --resource webVmss --resource-type Microsoft.Compute/virtualMachineScaleSets \
  --min-count 2 --max-count 10 --count 3

az monitor autoscale rule create -g Prod-Web --autoscale-name webVmssAutoscale \
  --condition "Percentage CPU > 75 avg 10m" --scale out 2

# ---- PowerShell ----
New-AzVmss -ResourceGroupName Prod-Web -VMSSetName webVmss `
  -Location eastus -InstanceCount 3 -VmSize Standard_D2s_v5

Update-AzVmss -ResourceGroupName Prod-Web -VMSSetName webVmss -SkuCapacity 6
```

Upgrade policies

Mode	Behavior
Manual	Admin must manually trigger updates to existing instances after model changes
Automatic	Azure updates all instances immediately when the model changes (can cause brief simultaneous downtime)
Rolling	Batched update across instances with health checks and pauses between batches — minimizes downtime

4.3 Disks, Snapshots, Backup vs. Replication

Managed disk tiers

Tier	Type	Use case
Ultra Disk	SSD, sub-ms latency, adjustable IOPS/throughput independent of size	SAP HANA, top-tier SQL, latency-critical workloads
Premium SSD v2	SSD, independently scalable performance	Production workloads needing flexible perf tuning without resizing disk
Premium SSD	SSD, performance tied to disk size	Production VMs, most databases
Standard SSD	SSD, lower consistent performance	Web servers, dev/test, light workloads
Standard HDD	Spinning disk	Backup targets, infrequently accessed data

Tip: Disks can be resized (mostly) online for Premium SSD v2/Ultra; older SKUs (Standard/Premium SSD) generally require the VM to be **deallocated** first to change size tier.

Disk types by role

Disk role	Notes
OS disk	Boot volume, max 4 TiB (Gen2), one per VM
Temp disk	Local, ephemeral, NOT persisted — data lost on deallocate/redeploy; used for pagefile/swap
Data disk	Additional persistent storage, up to 32 TiB each, multiple per VM (count depends on VM size)
Shared disk	Attachable to multiple VMs simultaneously (SCSI PR) for clustering scenarios (e.g., Windows Server Failover Clustering)

Snapshots

- **Full snapshot:** complete point-in-time copy of a disk, billed for the full size.
- **Incremental snapshot:** only stores changed blocks since the last snapshot, significantly cheaper, and supports cross-region copy for DR staging.
- Snapshots are standalone resources, independent of the source disk's lifecycle — deleting the original disk does not delete its snapshots.

```
# ---- CLI: incremental snapshot ----
az snapshot create -g Prod-Web -n webvm-osdisk-snap01 \
  --source webvm-osdisk --incremental true

# ---- PowerShell ----
$disk = Get-AzDisk -ResourceGroupName Prod-Web -DiskName webvm-osdisk
$snapConfig = New-AzSnapshotConfig -SourceUri $disk.Id -Location eastus -CreateOption Copy -Incremental
New-AzSnapshot -ResourceGroupName Prod-Web -SnapshotName webvm-osdisk-snap01 -Snapshot $snapConfig
```

Backup vs. Replication vs. Snapshot — know the difference

Mechanism	Protects against	Restore granularity	Where stored
Managed Disk Snapshot	Point-in-time capture before risky change	Whole disk	Same region, standalone resource
GRS/geo-replication	Regional outage (availability)	Whole account, eventual consistency	Paired region
Azure Backup	Accidental deletion, corruption, ransomware, retention compliance	File/VM/DB-level, multiple recovery points over time	Recovery Services Vault (isolated, immutable options)
Azure Site Recovery (ASR)	Full regional/datacenter disaster (DR), not accidental deletion	Whole VM failover to secondary region	Recovery Services Vault

4.4 ARM Templates & Bicep

Deployment options compared

Method	Language	Idempotent?	Notes
ARM (JSON) Templates	Declarative JSON	Yes	Native Azure format; verbose
Bicep	Declarative DSL, transpiles to ARM JSON	Yes	Cleaner syntax, Microsoft-recommended default going forward
Terraform	HCL	Yes	Multi-cloud, third-party but widely used with Azure Provider
Azure CLI / PowerShell scripts	Imperative	No (unless idempotency coded manually)	Good for ad hoc/one-off tasks and automation glue

ARM template anatomy

```
{
  "$schema": "https://schema.management.azure.com/schemas/2019-04-01/deploymentTemplate.json#",
  "contentVersion": "1.0.0.0",
  "parameters": {
    "vmSize": {
      "type": "string",
      "defaultValue": "Standard_D2s_v5"
    }
  },
  "variables": {
    "nicName": "[concat(parameters('vmName'), '-nic')]"
  },
  "resources": [
    /* one object per Azure resource */
  ],
  "outputs": {
    "vmId": {
      "type": "string",
      "value": "[resourceId('Microsoft.Compute/virtualMachines', parameters('vmName'))]"
    }
  }
}
```

The equivalent in Bicep (same result, less noise)

```
param vmSize string = 'Standard_D2s_v5'
param vmName string

resource nic 'Microsoft.Network/networkInterfaces@2023-05-01' = {
  name: '${vmName}-nic'
  location: resourceGroup().location
  properties: { /* ipConfigurations, etc. */ }
}

resource vm 'Microsoft.Compute/virtualMachines@2023-09-01' = {
  name: vmName
  location: resourceGroup().location
  properties: {
    hardwareProfile: { vmSize: vmSize }
    networkProfile: { networkInterfaces: [ { id: nic.id } ] }
  }
}

output vmId string = vm.id
```

Deployment modes

Mode	Behavior
Incremental (default)	Adds/updates resources defined in the template; leaves existing resources in the RG that aren't in the template untouched
Complete	Deletes any resource in the target RG that is <u>not</u> defined in the template — use with caution

Gotcha: Complete mode has deleted production resources when admins forgot a resource existed outside the template. Always run `what-if` first.

```
# ---- CLI ----
az deployment group what-if -g Prod-Web --template-file main.bicep --parameters main.bicepparam
az deployment group create -g Prod-Web --template-file main.bicep --parameters main.bicepparam

# ---- PowerShell ----
New-AzResourceGroupDeployment -ResourceGroupName Prod-Web -TemplateFile main.bicep -WhatIf
New-AzResourceGroupDeployment -ResourceGroupName Prod-Web -TemplateFile main.bicep
```

Resource move & ownership rules

- Moving a resource between Resource Groups or Subscriptions requires **Contributor (minimum) on both source and destination**, and the resource type must support move (most do; a few like VNet Gateways, Log Analytics-linked resources have restrictions).

- Moving does not change the resource's region.
- Resource IDs change on move (subscription/RG segment updates) — scripts/automation with hardcoded resource IDs will break.

4.5 Compute Ownership & Accessibility Rules

Action	Minimum RBAC needed	Notes
Start/Stop/Restart a VM	Virtual Machine Contributor (or Contributor)	Does not require network-level access
RDP/SSH into a VM (network layer)	N/A at RBAC level — governed by NSG rules + Bastion/VPN reachability	Separate from Azure RBAC entirely; this is network + OS-level auth
Sign in to a VM via Entra ID (Login extension)	Virtual Machine Administrator/User Login (data-plane RBAC)	Distinct from Owner/Contributor; requires the VM to have the Entra ID login extension installed
Attach/detach a data disk	Virtual Machine Contributor + Disk access on the disk resource	Disks can be owned/scoped separately from the VM they're attached to
Resize a VM	Virtual Machine Contributor	May require deallocation depending on target size/current hardware cluster
Move a VM to a different Availability Set	Not supported in-place	Must delete and recreate (keeping disks) — the exam tests this as a hard limitation

Tip: Boot diagnostics, serial console access, and VM extensions each have their own minor permission nuances (e.g., serial console access requires Contributor-level rights plus specific extension permissions) — if an exam question specifically calls out "cannot access serial console," check whether the account also needs Virtual Machine Contributor at the VM (not just the RG) and that no Policy blocks the boot diagnostics storage account.

5. Networking

5.1 VNets, Subnets, NSGs, UDRs

Ownership & association map

Relationship	Cardinality	Notes
Subscription → VNet	1 : many	A VNet lives in exactly one region and one subscription
VNet → Subnet	1 : many	Subnets are non-overlapping CIDR ranges carved from the VNet address space; cannot span VNets
Subnet → NIC/Resource	1 : many	A NIC (and therefore a VM) belongs to exactly one subnet
Subnet ↔ NSG	many : many (but typically 1 NSG per subnet)	An NSG can be associated with multiple subnets/NICs; a subnet can have at most 1 NSG, a NIC at most 1 NSG
Subnet ↔ Route Table (UDR)	many : 1	At most one route table per subnet; routes merge with system routes by longest-prefix-match
VNet ↔ VNet (Peering)	many : many	Peering is NOT transitive — A-B and B-C peered does NOT let A talk to C

Address space, subnetting & reserved addresses

Gotcha (frequently tested): Azure reserves **5 IP addresses** in every subnet: network address (.0), default gateway (.1), two reserved for Azure DNS mapping (.2, .3), and broadcast (.255, last address). A /24 subnet (256 addresses) yields only 251 usable.

CIDR	Total addresses	Usable (Azure)
/29	8	3
/28	16	11
/27	32	27
/26	64	59
/24	256	251

Special-purpose subnets (naming is mandatory, exam-relevant)

Subnet	Required exact name	Purpose
Azure Bastion	AzureBastionSubnet	Browser-based RDP/SSH without a public IP on the target VM; minimum /26
VPN/ExpressRoute Gateway	GatewaySubnet	Hosts the Gateway's own NICs; minimum /27 recommended (/29 technically allowed but not future-proof)
Azure Firewall	AzureFirewallSubnet	Dedicated subnet for the firewall's data plane; minimum /26
Delegated subnet (e.g., for App Service VNet Integration, SQL MI)	Any name, but must be "delegated" to the specific service	Grants a PaaS service the ability to inject NICs into that subnet

NSG evaluation logic

- Rules have Priority (100-4096, lower number = higher priority, first match wins), Source, Destination, Port, Protocol, Action (Allow/Deny).
- Default rules (cannot be deleted, only overridden by higher-priority custom rules): Allow VNet-to-VNet inbound/outbound, Allow Azure Load Balancer inbound, Deny all other inbound; Allow Internet outbound.
- NSG can attach at **subnet level** and/or **NIC level** — if both exist, traffic must be allowed by both for inbound, and by both for outbound (most restrictive wins).

Inbound traffic to VM:



Both checkpoints must ALLOW — a single Deny at either layer blocks the packet

Figure 5.1 — Dual NSG evaluation: subnet-level AND NIC-level must both permit inbound traffic

Application Security Groups (ASGs)

ASGs let you group VM NICs logically (e.g., "WebTier", "DbTier") and write NSG rules against the group name instead of hardcoded IPs — rules stay valid as VMs scale in/out. Example: an NSG rule "Allow port 1433 from ASG-WebTier to ASG-DbTier" self-updates as VMs join/leave either group.

User-Defined Routes (UDR) & effective routing

Route source	Priority when overlapping prefix length
User-Defined Route (custom Route Table)	Highest
BGP routes (VPN/ExpressRoute Gateway)	Middle
System routes (default, e.g., VNet-local, Internet)	Lowest

Longest prefix match always wins regardless of source; ties broken by the table above. Common pattern: UDR with 0.0.0.0/0 next hop = **Virtual Appliance** (firewall IP) to force all egress through inspection ("hairpinning").

```
# ---- CLI: create VNet, subnet, NSG, associate ----
az network vnet create -g Prod-Web -n vnet1 --address-prefix 10.0.0.0/16 \
  --subnet-name snet-web --subnet-prefix 10.0.1.0/24
az network nsg create -g Prod-Web -n nsg-web
az network nsg rule create -g Prod-Web --nsg-name nsg-web -n Allow-HTTPS \
  --priority 100 --direction Inbound --access Allow --protocol Tcp --destination-port-ranges 443
az network vnet subnet update -g Prod-Web --vnet-name vnet1 -n snet-web --network-security-group nsg-web

# ---- PowerShell equivalent ----
$subnet = New-AzVirtualNetworkSubnetConfig -Name snet-web -AddressPrefix 10.0.1.0/24
$vnet = New-AzVirtualNetwork -ResourceGroupName Prod-Web -Name vnet1 -AddressPrefix 10.0.0.0/16 -Location eastus -Subnet $subnet
$rule = New-AzNetworkSecurityRuleConfig -Name Allow-HTTPS -Priority 100 -Direction Inbound -Access Allow -Protocol Tcp -
DestinationPortRange 443
$nsg = New-AzNetworkSecurityGroup -ResourceGroupName Prod-Web -Name nsg-web -Location eastus -SecurityRules $rule
Set-AzVirtualNetworkSubnetConfig -VirtualNetwork $vnet -Name snet-web -AddressPrefix 10.0.1.0/24 -NetworkSecurityGroup $nsg
$vnet | Set-AzVirtualNetwork
```

5.2 VNet Peering & Cross-Subscription/Tenant Visibility

VNet Peering vs. VPN Gateway (site-to-site) vs. VNet-to-VNet Gateway

Method	Bandwidth/Latency	Transitive?	Encryption	Cost model
VNet Peering	Backbone speed, low latency	No	Not encrypted by default (Microsoft backbone, private though)	Pay for data transferred, no gateway cost
VPN Gateway (VNet-to-VNet)	Limited by gateway SKU (up to ~10 Gbps)	Depends on routing config	IPsec encrypted	Gateway hourly cost + data
Global VNet Peering	Backbone speed, cross-region	No	Private backbone	Higher cross-region data cost

Peering across subscription and tenant boundaries

Scenario	Supported?	Requirements
Peering two VNets in the same subscription	Yes	Network Contributor on both VNets
Peering VNets across subscriptions, same tenant	Yes	Network Contributor (or equivalent) on both VNets; each subscription owner creates their own peering link (both sides must configure)
Peering VNets across different Entra tenants	Yes	Requires the resource ID of the remote VNet and appropriate RBAC in both tenants (guest access or Lighthouse-style delegation may be needed to create the peering from a single identity); otherwise each tenant's own admin creates their half

Peering settings that control visibility/accessibility

Setting	Effect
Allow virtual network access	Master switch — must be enabled on both sides for any traffic to flow
Allow forwarded traffic	Permits traffic that <i>originated outside</i> the peered VNet (e.g., via an NVA) to pass through — needed for hub-spoke transitive routing via UDR
Allow gateway transit	Lets a spoke use the hub's VPN/ExpressRoute Gateway instead of deploying its own
Use remote gateways	The consuming side's equivalent toggle to actually use the gateway shared via transit

Gotcha: "Allow forwarded traffic" must be enabled on *both* peering links along the path for hub-spoke transitive routing (via NVA/Firewall) to work — a very common exam trip-up when a spoke-to-spoke-via-hub scenario "isn't working" despite correct UDRs.

5.3 Hybrid Connectivity: VPN Gateway & ExpressRoute

Method	Transport	Typical bandwidth	SLA	Use when
Site-to-Site VPN	IPsec over public internet	Up to ~10 Gbps (gateway SKU dependent)	99.9-99.95%	Quick setup, moderate bandwidth, cost-sensitive
Point-to-Site VPN	IPsec/SSTP/OpenVPN from a single device	Per-client limited	N/A	Remote/individual admin access, no on-prem VPN device
ExpressRoute	Private circuit via connectivity provider, not over internet	50 Mbps – 100 Gbps	99.95% (single circuit), 99.99% (dual)	Mission-critical, high-bandwidth, predictable latency, compliance requiring no internet transit
ExpressRoute + VPN failover	Both	—	Highest	ExpressRoute primary with VPN as backup path

VPN Gateway SKUs (know the relative ordering)

SKU tier	Aggregate throughput (approx.)	Max S2S tunnels
Basic (legacy)	100 Mbps	10
VpnGw1	650 Mbps	30
VpnGw2	1 Gbps	30
VpnGw3	1.25 Gbps	30
VpnGw4/5	up to 10 Gbps	100

ExpressRoute peering types

Peering type	Purpose
Private Peering	Connects on-prem network to Azure VNets (the main IaaS/PaaS-in-VNet path)
Microsoft Peering	Connects to Microsoft 365 and Azure PaaS public endpoints without traversing the internet

Connectivity architecture

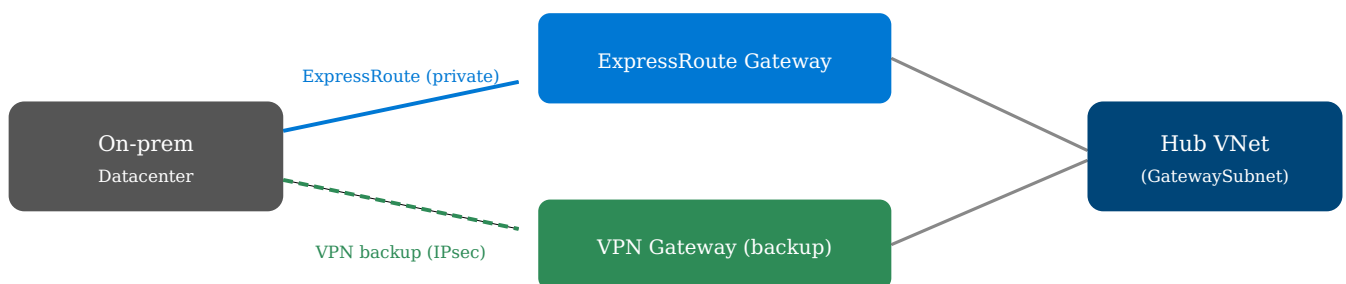


Figure 5.2 — ExpressRoute as primary path with VPN Gateway as automatic failover

```
# ---- CLI: Site-to-Site VPN Gateway creation (abridged) ----
az network public-ip create -g Hub-RG -n vpngw-pip --sku Standard
az network vnet-gateway create -g Hub-RG -n vpngw1 --public-ip-address vpngw-pip \
  --vnet hubVnet --gateway-type Vpn --vpn-type RouteBased --sku VpnGw1

az network local-gateway create -g Hub-RG -n onprem-gw --gateway-ip-address <on-prem-public-ip> \
  --local-address-prefixes 192.168.0.0/16

az network vpn-connection create -g Hub-RG -n onprem-to-azure \
  --vnet-gateway1 vpngw1 --local-gateway2 onprem-gw --shared-key "<preshaed-key>"
```

5.4 Private Link, Service Endpoints & DNS

Private Link vs. Service Endpoint (deep dive)

Aspect	Service Endpoint	Private Endpoint (Private Link)
What it does	Optimizes routing + restricts PaaS firewall to a subnet's identity	Injects a private NIC/IP for the PaaS resource directly into your subnet
Target still has a public IP?	Yes	Can be fully disabled (private-only)
Cross-VNet/on-prem access	No (subnet-local only)	Yes — reachable via VPN/ExpressRoute/peering since it's a real private IP
DNS	No change needed	Requires Private DNS Zone (<code>privatelink.*</code>) linked to the VNet, or on-prem DNS conditional forwarding
Granularity	Whole service (e.g., all storage in a region)	Specific resource instance (one storage account, one SQL server)

Private DNS resolution flow for a Private Endpoint



Figure 5.3 — Public FQDN CNAMEs to the privatelink zone, which resolves to the endpoint's private IP

Gotcha: Creating a Private Endpoint does NOT automatically create/link the Private DNS Zone unless you check that option during setup (Portal) or explicitly include it in your Bicep/ARM. Forgetting this is the #1 real-world "Private Endpoint doesn't work" support case — clients still resolve the public IP.

Azure DNS

Zone type	Resolves for	Notes
Public DNS Zone	Internet-facing domain names	Standard authoritative DNS hosting
Private DNS Zone	Internal VNet resolution	Link to one or more VNets; supports auto-registration of VM records
<code>privatelink.*</code> zones	Private Endpoint FQDNs	One well-known zone name per service (e.g., <code>privatelink.database.windows.net</code>)

Cross-subscription/tenant DNS visibility

Scenario	Resolution behavior
Private DNS Zone linked to multiple VNets in the same subscription	All linked VNets resolve the same records
Private DNS Zone linked to a VNet in another subscription	Fully supported — linking is a subscription-crossing operation (requires Private DNS Zone Contributor on the zone + Network Contributor on the VNet)
On-prem clients needing to resolve <code>privatelink.*</code> names	Requires conditional forwarding from on-prem DNS to an Azure DNS Private Resolver (or a VM running DNS forwarding) — on-prem DNS servers cannot query Azure Private DNS Zones directly

```

# ---- CLI: Private Endpoint + DNS zone linkage ----
az network private-endpoint create -g Prod-Data -n pe-storage \
  --vnet-name vnet1 --subnet snet-data \
  --private-connection-resource-id $(az storage account show -g Prod-Data -n stgprod01 --query id -o tsv) \
  --group-id blob --connection-name stgprod01-connection

az network private-dns zone create -g Prod-Data -n privatelink.blob.core.windows.net
az network private-dns link vnet create -g Prod-Data -n dns-link \
  --zone-name privatelink.blob.core.windows.net --virtual-network vnet1 --registration-enabled false
az network private-endpoint dns-zone-group create -g Prod-Data --endpoint-name pe-storage \
  --name default --private-dns-zone privatelink.blob.core.windows.net --zone-name blob
  
```

5.5 Hub-Spoke Architecture & Secure Access Patterns

Hub-and-Spoke reference architecture

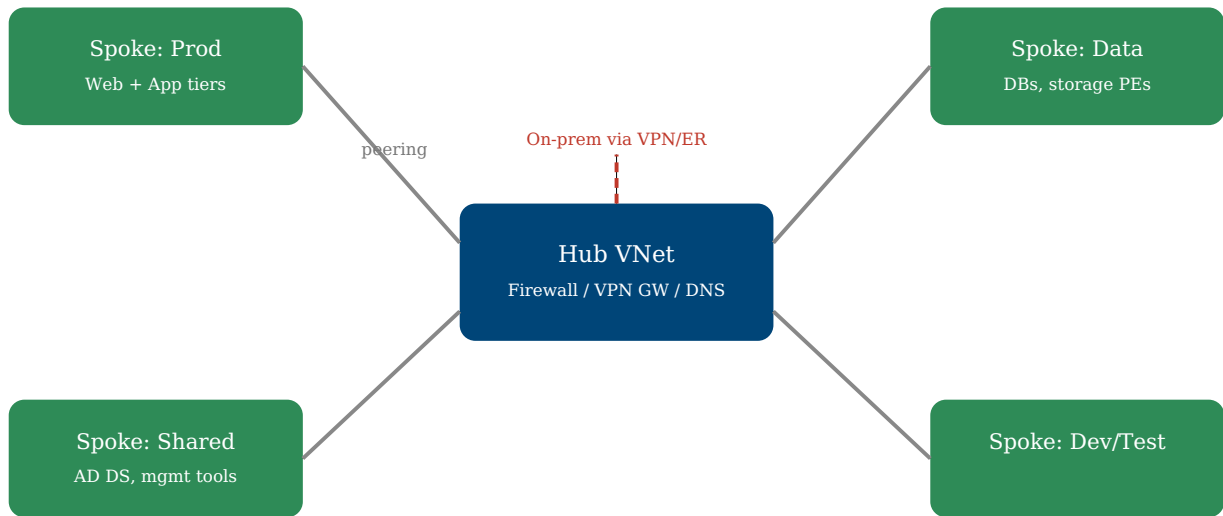


Figure 5.4 — Hub-spoke topology: spokes peer only to the hub; spoke-to-spoke traffic routes through the hub firewall via UDR (peering itself is non-transitive)

Tip: To let spoke-to-spoke traffic flow through a hub firewall, you must combine peering with a **UDR in each spoke** pointing default/cross-spoke traffic to the hub firewall's private IP as next hop — peering alone never provides transitive routing. Also enable "Allow forwarded traffic" on both peering links (see §5.2).

Secure access pattern: internet-facing app with private backend



Each hop's NSG only allows the specific previous hop — defense in depth, no direct internet-to-data-tier path exists

Figure 5.5 — Layered network security: each tier's NSG only trusts the tier immediately upstream

Network Watcher toolset (troubleshooting these patterns)

Tool	Purpose
IP Flow Verify	Tests whether a specific packet would be allowed/denied by NSGs
Effective Security Rules	Shows the merged, effective NSG rules on a NIC (subnet + NIC combined)
Next Hop	Shows the actual routing decision for traffic from a VM to a destination
Connection Troubleshoot	End-to-end connectivity test with latency/hop data
NSG Flow Logs	Records allowed/denied flows over time for auditing/analysis
Packet Capture	Remote packet capture on a VM without installing tools manually
Topology	Visual map of a VNet's resources and their relationships

5.6 Managed Identities & Secure Access

System-assigned vs. User-assigned

Type	Lifecycle	Shareable?	Use when
System-assigned	Created & destroyed with the resource (VM, App Service, etc.)	No, 1:1 with the resource	Single resource needing its own identity; simplest cleanup
User-assigned	Independent Azure resource, exists on its own	Yes, attach to many resources	Multiple VMs/functions needing the same permission set; identity must outlive a specific resource

How a managed identity actually gets a token (IMDS flow)



3. token returned to app → presented to target resource (Storage, Key Vault, SQL...) → 4. resource checks RBAC and honors/denies

Figure 5.6 — No secrets in code: the platform brokers the token via the instance metadata service

Common managed-identity access patterns

Source	Target	Role example
VM / VMSS	Key Vault	Key Vault Secrets User (or access policy: Get/List)
App Service / Function	Storage / SQL	Storage Blob Data Contributor, SQL AAD auth as a contained DB user mapped to the identity
Automation Account (Runbook)	Any resource it manages	Assign per task, e.g. Virtual Machine Contributor to start/stop VMs on a schedule
Azure Policy remediation	Target resources	Contributor or specific role matching the Modify/DINE effect
ADF / Logic App	Storage, Databricks, Key Vault	Scoped data-plane roles per connector

Exam tip: Managed identities cannot be used from outside Azure (they're bound to the Azure control plane's metadata service) — for on-prem or multi-cloud workloads, use a Service Principal with a certificate/secret, or Workload Identity Federation where supported.

```

# ---- CLI: user-assigned identity, attach to multiple VMs ----
az identity create -g Prod-Web -n shared-app-identity
IDID=$(az identity show -g Prod-Web -n shared-app-identity --query id -o tsv)
az vm identity assign -g Prod-Web -n WebVM01 --identities $IDID
az vm identity assign -g Prod-Web -n WebVM02 --identities $IDID

# ---- PowerShell ----
$identity = New-AzUserAssignedIdentity -ResourceGroupName Prod-Web -Name shared-app-identity
Update-AzVM -ResourceGroupName Prod-Web -VM $vm1 -IdentityType UserAssigned -IdentityId $identity.Id
Update-AzVM -ResourceGroupName Prod-Web -VM $vm2 -IdentityType UserAssigned -IdentityId $identity.Id
  
```

6. Monitoring & Backup

6.1 Azure Monitor, Log Analytics & Alerts

Azure Monitor data model



Figure 6.1 — Three data types feed a shared Log Analytics workspace, which powers alerting/visualization/export

Data type	Retention (default)	Source examples
Platform Metrics	93 days, near real-time (1 min granularity typical)	CPU %, disk IOPS, network in/out — auto-collected, no agent needed
Resource/Diagnostic Logs	Configurable, sent to Log Analytics/Storage/Event Hub	NSG flow logs, Storage diagnostic logs, App Service logs
Activity Log	90 days (free, queryable); export for longer	Who did what control-plane operation, when (create/delete/update)

Log Analytics Workspace ownership & visibility

Access mode	Controls
Workspace-context	User has access to all data in the workspace (via Log Analytics Reader/Contributor RBAC at the workspace)
Resource-context	User only sees data for resources they have Reader access to, even without workspace-level RBAC — useful for multi-tenant/multi-team workspaces
Table-level RBAC	Fine-grained: grant read access to specific tables only (e.g., Security team sees SecurityEvent, app team sees AppTraces)

Alerts

- **Metric alerts:** near-real-time, threshold-based (e.g., CPU > 80%).
- **Log alerts:** run a KQL query on a schedule, alert on result count/threshold.
- **Activity Log alerts:** fire on specific control-plane events (e.g., "any VM deleted", "role assignment created").
- **Action Groups:** reusable notification/action bundle (email, SMS, webhook, Logic App, Automation runbook, ITSM) attached to any alert rule.

```

# ---- CLI: metric alert ----
az monitor metrics alert create -g Prod-Web -n high-cpu-alert \
  --scopes $(az vm show -g Prod-Web -n WebVM01 --query id -o tsv) \
  --condition "avg Percentage CPU > 80" --window-size 10m --evaluation-frequency 5m

# ---- CLI: Activity Log alert on role assignment creation ----
az monitor activity-log alert create -g Prod-Web -n role-assignment-watch \
  --condition category=Administrative and operationName=Microsoft.Authorization/roleAssignments/write \
  --action-group myActionGroup

# ---- PowerShell equivalent ----
Add-AzMetricAlertRuleV2 -ResourceGroupName Prod-Web -Name high-cpu-alert `
  -TargetResourceId $vmId -Condition $condition -WindowSize 00:10:00 -Frequency 00:05:00
  
```

```

# Sample KQL: VMs with high CPU in the last hour
Perf
| where ObjectName == "Processor" and CounterName == "% Processor Time"
| where TimeGenerated > ago(1h)
| summarize AvgCPU = avg(CounterValue) by Computer
  
```

```
| where AvgCPU > 80  
| order by AvgCPU desc
```

6.2 Recovery Services Vaults & Backup Policies

Recovery Services Vault — architecture



Figure 6.2 — One vault can hold multiple backup policies protecting many item types

Redundancy option (vault-level)	Notes
LRS	Cheapest, single datacenter — not recommended for production
ZRS	Resilient to zone failure, data stays in-region
GRS (default)	Cross-region resilience; can enable Cross Region Restore to restore into the secondary region

Gotcha: Storage redundancy for a Recovery Services Vault can only be changed before the first backup is taken — plan this at vault creation time.

Vault ownership & RBAC roles

Role	Can do
Backup Contributor	Create/modify policies, enable/disable protection, trigger backup jobs — cannot delete backup data
Backup Operator	Trigger on-demand backups/restores but cannot create policies or change protection configuration
Backup Reader	View-only across backup items, jobs, and policies
Owner/Contributor on the vault resource	Full management-plane control including delete — combine with soft delete/MUA to prevent accidental or malicious data loss

VM backup workflow

1. Create/select a **Recovery Services Vault** in the same region as the VM.
2. Create a **Backup Policy**: schedule (daily/weekly), retention (daily/weekly/monthly/yearly — GFS grandfather-father-son), and optionally enable instant restore snapshot retention (1-30 days, kept locally on the VM's disk for fast restores).
3. Enable backup on the VM — installs the **VM backup extension** (Windows: VSS-based app-consistent snapshot; Linux: pre/post scripts for app consistency, else file-consistent/crash-consistent).
4. Restore options: **Create new VM**, **Replace existing disks**, or **Restore disks only** (then attach manually) — also supports **file-level recovery** without a full VM restore.

```

# ---- CLI ----
az backup vault create -g Prod-Data -n rsv-prod --location eastus

az backup policy create -g Prod-Data -v rsv-prod --name DailyPolicy \
  --backup-management-type AzureIaaSVM --policy dailyPolicy.json

az backup protection enable-for-vm -g Prod-Data -v rsv-prod \
  --vm WebVM01 --policy-name DailyPolicy

az backup restore restore-disks -g Prod-Data -v rsv-prod \
  --container-name IaasVMContainer;iaasvmcontainerv2;Prod-Web;WebVM01 \
  --item-name WebVM01 --rp-name <recovery-point-id> --storage-account restore01

# ---- PowerShell ----
$vault = New-AzRecoveryServicesVault -Name rsv-prod -ResourceGroupName Prod-Data -Location eastus
Set-AzRecoveryServicesVaultContext -Vault $vault
$policy = Get-AzRecoveryServicesBackupProtectionPolicy -Name DailyPolicy
  
```

```
Enable-AzRecoveryServicesBackupProtection -Policy $policy -Name WebVM01 -ResourceGroupName Prod-Web
```

Immutability & ransomware protection

- **Soft delete** (default, 14 days): deleted backup data is retained and recoverable even if an attacker deletes the vault's backup items.
- **Multi-user authorization (MUA)**: critical operations (disable soft delete, reduce retention, delete backup data) require a second approver via Azure Resource Guard.
- **Immutable vaults**: lock retention settings so even an Owner cannot reduce them for a defined period.

Other backup surfaces to know

Workload	Mechanism
Azure Files	Native Azure Backup integration (snapshot-based, no agent)
SQL/SAP HANA on Azure VM	Azure Backup workload extension, log-shipping for point-in-time restore
Azure SQL Database (PaaS)	Automated backups built-in (7-35 days PITR); long-term retention (LTR) up to 10 years separately configured
On-prem servers/files	MARS agent (Microsoft Azure Recovery Services agent) backs up directly to a vault
Managed Disks (ad hoc)	Snapshot (full or incremental) — incremental snapshots only store changed blocks since the last snapshot, cheaper

6.3 Site Recovery vs. Backup vs. Replication

Aspect	Azure Backup	Azure Site Recovery	Storage Geo-Replication
Purpose	Point-in-time recovery of data/VMs	Whole-workload disaster recovery/failover	Availability of the storage service across regions
Granularity	File, VM, database	Whole VM(s), orchestrated multi-tier failover	Whole storage account
RPO (data loss window)	As frequent as the backup schedule (typically hours)	Near-continuous (seconds to low minutes) via continuous replication	Minutes (asynchronous), not customer-controlled
Failover is...	N/A (it's a restore, not a failover)	Customer-triggered, orchestrated via Recovery Plans	Customer-triggered (or Microsoft in true regional loss)
Protects against accidental deletion	Yes (this is its primary purpose)	Not directly (it replicates deletes too, unless delayed)	No
Protects against regional disaster	Only if vault uses GRS and target region is viable	Yes (this is its primary purpose)	Yes (availability of the storage layer only)

Azure Site Recovery (ASR) at a glance

ASR continuously replicates VM disk writes to a target region via a **Mobility service** agent (or agentless for some scenarios), maintaining crash-consistent + app-consistent recovery points. A **Recovery Plan** orchestrates failover order (e.g., start DB tier before web tier) and can run scripts/Automation runbooks mid-failover. Test failovers run in an isolated network without impacting production or the replication stream.

ASR concept	Detail
Recovery Point type	Crash-consistent (frequent, like a power-loss snapshot) vs. App-consistent (includes in-memory/transaction state, less frequent)
Test Failover	Non-disruptive validation in an isolated network, doesn't interrupt ongoing replication
Planned Failover	Zero data loss, orderly shutdown of source before failover — used for planned maintenance/migration
Unplanned Failover	Used during an actual outage; may lose the most recent, unreplicated writes
Failback	Reverse replication once the primary region recovers, then fail back

```
# ---- CLI: enable ASR replication (abridged, typically done via Portal wizard) ----
az backup vault create -g DR-RG -n asr-vault --location westus2

# ASR configuration is primarily done via Portal or PowerShell due to complexity
# ---- PowerShell: create a protection container mapping ----
New-AzRecoveryServicesAsrProtectionContainerMapping -Name "primary-to-dr" `
-PrimaryProtectionContainer $primaryContainer -RecoveryProtectionContainer $drContainer `
-Policy $replicationPolicy
```

Choosing the right tool — decision guide

Need	Use
"Someone deleted a VM/file last week, restore it"	Azure Backup
"A whole datacenter/region just went down, get production running elsewhere in minutes"	Azure Site Recovery
"Storage must remain readable during a regional outage without any admin action"	RA-GRS/RA-GZRS
"Test a disaster scenario without impacting production"	ASR Test Failover
"Comply with a 7-year retention mandate"	Azure Backup with GFS long-term retention policy, possibly Immutable Vault

7. Practical Scenario Bank

Exam case studies are almost always "given this business requirement, pick the right combination of services." Use these worked scenarios as templates — they mirror the phrasing and reasoning pattern of real exam questions.

1. Restrict outbound internet, allow only approved PaaS. "VMs in Subnet-A must never reach the internet directly, but must reach Storage and SQL PaaS."

Remove/avoid a default outbound NAT/public IP; add a UDR sending `0.0.0.0/0` to a Firewall/NVA (or explicitly block via NSG Deny Internet outbound); add Private Endpoints for Storage/SQL in the same VNet so PaaS traffic never needs the internet.

2. Temporary external access to one container. "Contractors need temporary read access to one blob container for 7 days, no Azure account required."

Generate a **SAS token** scoped to the container, Read-only permission, 7-day expiry. Prefer a **user delegation SAS** (Entra-backed, revocable by removing the granting identity's access) over an account-key SAS (harder to revoke individually).

3. Self-service RG with guardrails. "A department needs full self-service over their own resource group but must not be able to create resources outside 2 approved regions or delete the RG itself."

Assign **Contributor** at the RG scope for the team; apply an **Azure Policy** (Deny effect) restricting `location` to the 2 approved regions, assigned at the RG or a management group above it; apply a `CanNotDelete` **Resource Lock** on the RG.

4. Tenant-wide enforcement, zero manual review. "All new storage accounts across 40 subscriptions must have HTTPS-only traffic enforced, with no exceptions and no manual review needed."

Assign an **Azure Policy** with Deny effect for `supportsHttpsTrafficOnly != true`, scoped at the **Management Group** level covering all 40 subscriptions — enforced at deployment, not just audited.

5. Retiring on-prem file servers. "An existing on-prem file server needs to be phased out; branch offices need low-latency access to the same files."

Migrate data to **Azure Files**, deploy **Azure File Sync** with a Storage Sync Service + cloud endpoint (the Azure file share) and server endpoints at each branch (cache hot files locally, tiering cold files to cloud) — branches keep fast local access while Azure becomes the authoritative copy.

6. Resilience + maintenance control. "A VM must automatically restart if it becomes unresponsive, and must be excluded from planned Azure host maintenance windows when possible."

Use **Boot diagnostics** + appropriate health monitoring; for maintenance control use **Maintenance Configurations** (scheduled patching/host maintenance windows) attached to the VM or a Dedicated Host.

7. Departmental cost visibility. "Finance wants to see the last 3 months of spend broken down by department, and get alerted before a subscription exceeds its monthly budget."

Tag resources with a `department` tag (enforce via Policy), use **Cost Analysis** grouped by tag, and create a **Budget** in Cost Management with alert thresholds (e.g., 80%/100%) tied to an Action Group.

8. Federated subsidiaries under central oversight. "Two subsidiaries (both with existing Azure subscriptions) need centrally managed identity, but each subsidiary's IT team must keep control of their own subscriptions day-to-day."

Keep separate tenants OR consolidate identity into one Entra tenant with the subsidiaries as sub-scopes; place each subsidiary's subscriptions under its own **Management Group**; grant subsidiary IT **Owner** at their management group/subscription, while a central platform team holds **Reader/Policy Contributor** at the root for oversight without daily interference.

9. Automated regional failover. "A regionally-outaged primary datacenter must fail over web VMs to a secondary region within minutes, automatically, with minimal data loss."

Azure Site Recovery with continuous replication to the secondary region; configure a **Recovery Plan** with automated failover groups and scripted pre/post actions; pair with **Traffic Manager** or **Front Door** for automatic DNS/endpoint failover of user traffic.

10. Just-in-time production access. "Developers should be able to deploy resources into a sandbox subscription freely, but production subscriptions must require a change ticket reviewed by a second person."

Sandbox: assign **Contributor** broadly. Production: use **Privileged Identity Management (PIM)** for **just-in-time, time-bound, approval-required** elevation to Contributor/Owner, with justification and optional MFA re-auth — removes standing access entirely.

11. Lift-and-shift with minimal downtime. "300 on-prem VMs need to move to Azure with the least possible downtime and minimal re-architecture."

Use **Azure Migrate** for discovery/assessment (sizing, dependency mapping), then agentless or agent-based replication for the

actual migration — this uses the same replication engine as ASR, allowing a test migration before final cutover with a short final sync window.

12. Multi-region active-active web app. "A public web app must remain available even if an entire Azure region fails, with automatic traffic redirection."

Deploy VMSS/App Service in 2+ regions behind **Azure Front Door** (global HTTP(S) load balancing with health probes and automatic failover) or **Traffic Manager** (DNS-based) depending on latency/feature needs; ensure data tier uses geo-replicated storage/SQL with appropriate consistency guarantees.

13. Locking down a Key Vault. "Only the app's managed identity and 2 named admins should be able to read secrets; everyone else, including subscription Owners, should be blocked."

Switch the Key Vault to the **RBAC permission model** (not legacy access policies) and grant Key Vault Secrets User only to the app's managed identity and the 2 admins — being Owner on the vault resource does not itself grant secret read (data-plane separation, same principle as storage).

14. Auditing who deleted a resource. "A production VM was deleted overnight; you must determine who did it and prevent recurrence."

Query the **Activity Log** filtered to Microsoft.Compute/virtualMachines/delete for the timeframe to identify the caller identity; going forward, apply a CanNotDelete lock on critical resource groups and create an **Activity Log Alert** that notifies on any future delete operations in that scope.

15. Segregating dev/test costs and access. "Developers need broad rights to experiment, but their spend and blast radius must be fully isolated from production."

Separate **subscription** for dev/test (hard billing + quota isolation, not just a resource group) with its own budget/alerts; assign Contributor broadly within that subscription; apply a Policy initiative limiting VM SKUs/regions to control cost; keep it under a distinct "Non-Production" management group with looser policies than "Production."

16. Securing a database migration. "An on-prem SQL Server must move to Azure with encryption in transit, private connectivity, and no public exposure at any point."

Use **Azure Database Migration Service** over a Site-to-Site VPN or ExpressRoute connection; deploy target as Azure SQL Managed Instance (in a delegated subnet, inherently private) or SQL on a VM with a Private Endpoint; disable public network access on the target from the outset.

17. Automated tag governance. "All resources must have a CostCenter tag; resources deployed without one should get a default value automatically rather than being blocked."

Use an Azure Policy with the Modify effect (not Deny) and a system-assigned managed identity with Contributor at the remediation scope, so missing tags are appended automatically rather than blocking legitimate deployments.

18. Cross-region data residency compliance. "Data must never leave the EU region, even for backups or replication."

Use **LRS or ZRS only** (never GRS, which replicates to a paired region that may be outside the EU) for storage; for Azure Backup, select a Recovery Services Vault in an EU region with LRS/ZRS redundancy and disable cross-region restore; apply a Policy denying resource creation outside approved EU regions at the management group level.

19. Emergency break-glass access. "If Conditional Access or PIM approval workflows fail and lock out all admins, there must be a way back in."

Maintain 1-2 **break-glass accounts**: cloud-only (not synced from on-prem AD), excluded from all Conditional Access policies, with Global Administrator as a **standing (not PIM-eligible)** assignment, strong unique credentials stored offline/physically secured, and monitored with alerts on any sign-in.

20. Autoscaling a queue-driven backend. "A backend processes messages from a queue; capacity should scale with queue depth, not CPU."

Deploy workers as a **VMSS (Flexible orchestration)** or Azure Functions with a custom autoscale rule/trigger based on **Storage Queue length** (a custom metric via Log Analytics or native Functions queue trigger scaling) rather than the default CPU-based rule.

21. Least-privilege automation account. "An Automation Account runbook needs to stop/start VMs nightly across 5 resource groups, nothing else."

Enable a **system-assigned managed identity** on the Automation Account; create a **custom role** with only Microsoft.Compute/virtualMachines/start/action, .../deallocate/action, and .../read; assign it at each of the 5 resource groups (not the subscription) to keep blast radius minimal.

22. Merger with conflicting IP address spaces. "After acquiring Fabrikam, both companies' VNets use overlapping 10.0.0.0/16 address space and now need to connect."

VNet Peering requires non-overlapping address spaces, so peering directly is impossible without re-addressing. Options: re-IP one side's VNets (disruptive but cleanest long-term), or insert a **NAT-capable NVA/Azure Firewall** between the networks to translate

addresses, or use separate non-overlapping subnets for any newly shared services while keeping the rest isolated until a re-IP project completes.

8. Master Cheat Sheet

Scope hierarchy (top to bottom)

Tenant → Management Group → Subscription → Resource Group → Resource — RBAC, Policy, and Locks all inherit downward only.

RBAC quick reference

Role	Grants access?	Manages resources?
Owner	✓	✓
Contributor	✗	✓
Reader	✗	✗
User Access Admin	✓	✗

Locks

Lock	Blocks
CanNotDelete	Delete only
ReadOnly	Modify + Delete

Policy effects (strength order)

Deny > DeployIfNotExists > Modify > Append > Audit > AuditIfNotExists > Disabled

Storage redundancy

SKU	Copies/Scope
LRS	3, 1 DC
ZRS	3, 3 zones
GRS	6, 2 regions
RA-GRS	6, 2 regions, readable
GZRS	6, ZRS+geo

Blob access tiers

Tier	Min days
Hot	0
Cool	30
Cold	90
Archive	180

Availability SLAs

Config	SLA
Single VM (Premium disk)	99.9%
Availability Set	99.95%
Availability Zone(s)	99.99%

Subnet reserved IPs

Azure always reserves 5 per subnet: .0 (network), .1 (gateway), .2 & .3 (Azure DNS), .255 (broadcast, last address). /24 = 251 usable.

VM access to storage — pick the right method

Need	Use
App code needs no secrets	Managed Identity + RBAC data role
Temporary delegated access, external party	SAS token (prefer user-delegation SAS)
Network-level isolation, cross-VNet/on-prem	Private Endpoint + Private DNS zone
Simple subnet-scoped firewall allow	Service Endpoint
Legacy/quick full access (avoid in production)	Account key

Networking essentials

Item	Key fact
NSG evaluation	Lowest priority number wins; first match stops evaluation; subnet AND NIC NSGs both must allow
Route priority	UDR > BGP > System route (ties by longest prefix match first)
VNet Peering	Not transitive; no encryption by default (private backbone); no bandwidth limit like a gateway
Private Endpoint vs Service Endpoint	Private Endpoint = private IP in your subnet, cross-network reachable; Service Endpoint = optimized route + firewall rule, subnet-local only
ExpressRoute	Private circuit, not over internet, highest SLA/bandwidth
Special subnet names	AzureBastionSubnet, GatewaySubnet, AzureFirewallSubnet

Backup vs. DR vs. Replication

Need	Use
------	-----

Restore an accidentally deleted VM/file/blob from last week	Azure Backup (Recovery Services Vault)
Fail over an entire region during a disaster in minutes	Azure Site Recovery
Protect against regional storage outage (availability, not deletion)	GRS/RA-GRS/GZRS
Quick point-in-time rollback of a single disk before a risky change	Managed Disk Snapshot (incremental)

Identity acquisition/merge cheat card

Goal	Tool
Manage another tenant's subscription without moving it	Azure Lighthouse
Auto-provision partner/subsidiary users as guests	Cross-tenant synchronization (Entra ID)
Control trust between two tenants (MFA, device compliance)	Cross-tenant access settings
Fully merge org into your tenant	Subscription transfer + Entra ID user migration + RBAC re-assignment
Just-in-time elevation with approval	Privileged Identity Management (PIM)
Conditional sign-in restrictions (device, location, risk)	Conditional Access

Governance ownership summary (quick recall)

Scope	Owned by	Cannot directly contain
Tenant	Global Administrator	Resources (only identities/directory objects)
Management Group	RBAC Owner at that MG (root MG owner elevated via Entra)	Resources directly — only subscriptions
Subscription	Account Admin / Owner	N/A — the billing boundary itself
Resource Group	Owner/Contributor at the RG	Other Resource Groups (no nesting)
Resource	Owner/Contributor at the resource or inherited	N/A — the leaf node

Essential CLI + PowerShell command reference

```
### ---- Azure CLI ---- ###
az login
az account set --subscription "<sub-id>"
az group create -n rg-demo -l eastus
az vm create -g rg-demo -n vm1 --image Ubuntu2204 --admin-username azureuser --generate-ssh-keys
az vm deallocate -g rg-demo -n vm1
az network vnet create -g rg-demo -n vnet1 --address-prefix 10.0.0.0/16 --subnet-name snet1 --subnet-prefix 10.0.0.0/24
az network nsg create -g rg-demo -n nsg1
az network vnet peering create -g rg-demo -n hub-to-spoke --vnet-name hubVnet --remote-vnet spokeVnetId --allow-vnet-access
az storage account create -g rg-demo -n stgdemo01 --sku Standard_GRS --kind StorageV2
az role assignment create --assignee <principal> --role Contributor --scope <resourceId>
az policy assignment create --name enforce-tag --policy <policyDefId> --scope <mgScopeId>
az lock create --name NoDelete --resource-group rg-demo --lock-type CanNotDelete
az backup protection enable-for-vm -g rg-demo -v rsv1 --vm vm1 --policy-name DefaultPolicy
az monitor metrics alert create -g rg-demo -n high-cpu --scopes <vmId> --condition "avg Percentage CPU > 80"
az vmss create -g rg-demo -n vmss1 --image Ubuntu2204 --instance-count 3 --zones 1 2 3
az deployment group create -g rg-demo --template-file main.bicep

### ---- PowerShell (Az module) ---- ###
Connect-AzAccount
Set-AzContext -Subscription "<sub-id>"
New-AzResourceGroup -Name rg-demo -Location eastus
New-AzVM -ResourceGroupName rg-demo -Name vm1 -Image UbuntuLTS
Stop-AzVM -ResourceGroupName rg-demo -Name vm1 -Force
New-AzVirtualNetwork -ResourceGroupName rg-demo -Name vnet1 -AddressPrefix 10.0.0.0/16 -Location eastus
New-AzNetworkSecurityGroup -ResourceGroupName rg-demo -Name nsg1 -Location eastus
Add-AzVirtualNetworkPeering -Name hub-to-spoke -VirtualNetwork $hubVnet -RemoteVirtualNetworkId $spokeVnetId
New-AzStorageAccount -ResourceGroupName rg-demo -Name stgdemo01 -SkuName Standard_GRS -Kind StorageV2 -Location eastus
New-AzRoleAssignment -ObjectId <principal> -RoleDefinitionName Contributor -Scope <resourceId>
New-AzPolicyAssignment -Name enforce-tag -PolicyDefinition $def -Scope <mgScopeId>
New-AzResourceLock -LockName NoDelete -LockLevel CanNotDelete -ResourceGroupName rg-demo
Enable-AzRecoveryServicesBackupProtection -Policy $policy -Name vm1 -ResourceGroupName rg-demo
Add-AzMetricAlertRuleV2 -ResourceGroupName rg-demo -Name high-cpu -TargetResourceId $vmId -Condition $condition
New-AzVmss -ResourceGroupName rg-demo -VMScaleSetName vmss1 -InstanceCount 3
New-AzResourceGroupDeployment -ResourceGroupName rg-demo -TemplateFile main.bicep
```

Top 20 exam gotchas, consolidated

#	Gotcha
1	Management Groups cannot hold resources directly, only subscriptions
2	A subscription belongs to exactly one management group and one tenant at a time
3	Owner/Contributor on a storage account ≠ blob data access (management vs. data plane)
4	VNet Peering is never transitive; needs UDR + "allow forwarded traffic" for hub-spoke routing
5	Both subnet AND NIC NSGs must allow traffic — most restrictive wins
6	Stopping a VM from the guest OS still bills compute; only Deallocate stops it
7	Resource Locks override RBAC entirely, even for Owners
8	GRS/geo-replication is availability, not backup — doesn't protect against accidental deletion
9	Geo-failover is manual/customer-triggered, not automatic or instant
10	Azure reserves 5 IPs per subnet (.0, .1, .2, .3, .255)
11	Private Endpoint DNS requires an explicit Private DNS Zone link — not automatic
12	Complete ARM deployment mode deletes resources not in the template — always run what-if first
13	Vault storage redundancy can only be set before the first backup
14	Moving a subscription to a new tenant orphans all existing RBAC assignments
15	Availability Set membership must be set at VM creation — cannot add an existing VM later
16	ReadOnly locks can silently break Autoscale, Key Vault purge, and disk resize
17	Temp disks are ephemeral and lost on deallocate/redeploy — never store persistent data there
18	Key Vault legacy access policies vs. RBAC permission model are mutually exclusive per vault
19	A ReadOnly-locked RG's resources can still be read/listed, but writes silently fail even for automation
20	SAS tokens derived from an account key are invalidated when that key is rotated/regenerated

"On exam day: identify the scope first, the ownership boundary second, and the least-privilege/least-blast-radius option third. Azure almost always rewards the narrowest solution that still meets every stated requirement."

9. Hybrid Identity Deep Dive

Most enterprise AZ-104 scenarios assume an existing on-prem Active Directory that must coexist with Entra ID. This section covers the synchronization and authentication options in depth — a recurring source of case-study questions about acquisitions, mergers, and hybrid environments.

Synchronization tools compared

Tool	Runs where	Best for	Key limits
Microsoft Entra Connect Sync	On-prem server (full install)	Complex sync topologies, custom attribute filtering, multiple on-prem forests	Requires a dedicated (often HA-paired) Windows Server; more operational overhead
Microsoft Entra Cloud Sync	Lightweight agent, cloud-managed provisioning logic	Simple topologies, multiple forests without complex filtering, faster feature rollout	Fewer customization options than Connect Sync; newer, still catching up on feature parity for edge cases
Cross-tenant synchronization	Cloud-to-cloud, Entra tenant to Entra tenant	Multi-tenant organizations, acquisitions (see §2.7)	Only works tenant-to-tenant, not on-prem AD to tenant

Authentication methods

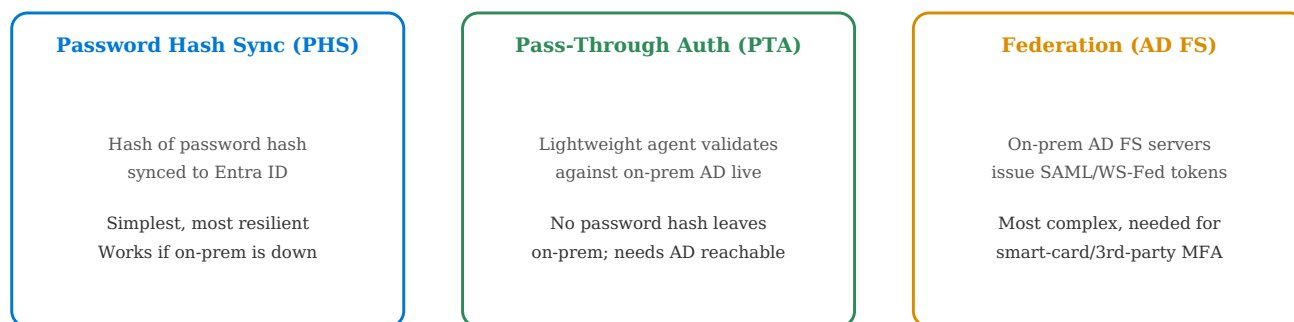


Figure 9.1 — Three authentication methods, simplest/most resilient to most complex

Method	Password ever leaves on-prem?	Works if on-prem network is down?	Microsoft recommendation
Password Hash Sync (PHS)	No (only a hash-of-hash)	Yes — authentication happens entirely in the cloud	Default recommendation for most organizations; also enables Leaked Credentials detection in Entra ID Protection
Pass-Through Authentication (PTA)	No (validated live, never transmitted or stored in the cloud)	No — requires on-prem AD reachable via the PTA agent	Use when password policies/logon-hour restrictions must be enforced by on-prem AD in real time
Federation (AD FS)	No (token-based)	No — requires AD FS farm availability	Only when a specific requirement (smart card auth, third-party MFA, certain compliance mandates) can't be met by PHS/PTA

Exam gotcha: PHS and PTA can both be enabled together with **Seamless SSO** for a smooth on-prem-like sign-in experience without deploying AD FS at all — a very common "right-sized" answer when a case study says "minimize infrastructure" but still wants password-based hybrid sign-in.

Object sync scope & filtering

- By default, Entra Connect/Cloud Sync syncs all users, groups, and contacts from configured OUs — use **OU-based filtering** or **attribute-based filtering** to exclude service accounts or specific business units.
- Writeback** features (password writeback, group writeback, device writeback) let changes made in the cloud flow back to on-prem AD — e.g., a cloud-initiated self-service password reset updates the on-prem account too.
- Staging mode:** a second Connect server can be installed in staging mode as a warm standby — it syncs data but does not export changes, ready to be promoted if the primary fails.

Device identity states

State	Description	Typical scenario
-------	-------------	------------------

Entra ID joined	Cloud-only device identity, no on-prem AD account	Cloud-native organizations, new hardware with no on-prem dependency
Entra ID registered	Personal/BYOD device registered for access, not fully managed	BYOD scenarios needing Conditional Access checks without full management
Hybrid Entra ID joined	Device joined to on-prem AD AND registered in Entra ID	Traditional domain-joined enterprise devices transitioning to hybrid

Scenario: "Fabrikam's on-prem AD must continue to be the source of truth for passwords immediately after acquisition, with minimal new infrastructure, and cloud password resets should flow back on-prem." → Deploy **Entra Connect Sync** (or Cloud Sync) with **Password Hash Sync** and **password writeback** enabled — avoids standing up an AD FS farm while still keeping on-prem AD authoritative for day-to-day changes, and self-service password resets in the cloud sync back down.

10. App Service & Containers

App Service Plan tiers

Tier	Use case	Key features
Free / Shared	Dev/test only	Shared compute, no custom domain SSL, low quotas
Basic	Small production/dev-test	Dedicated compute, custom domains, manual scale only
Standard	Production workloads	Autoscale, deployment slots (5), daily backups
Premium (v2/v3)	Higher-scale production	More deployment slots (20), VNet integration, faster autoscale, higher memory/CPU
Isolated	High-compliance, network-isolated	Runs in an App Service Environment (ASE) — dedicated, single-tenant deployment inside your own VNet

Ownership model: App Service Plan vs. App

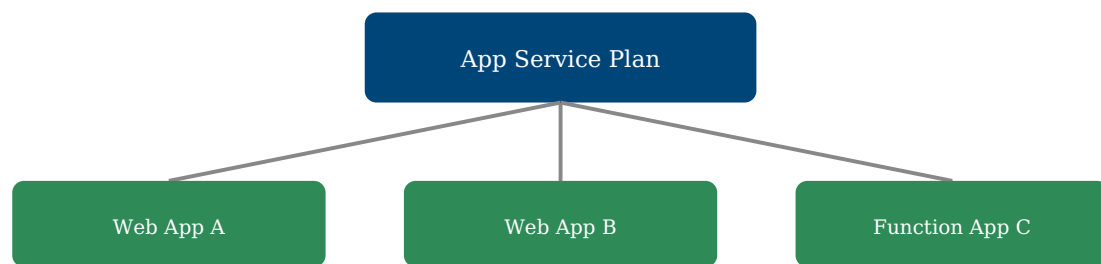


Figure 10.1 — A single App Service Plan (the compute/billing unit) can host multiple apps that share its capacity

The Plan is the unit of compute and billing (a VM-like allocation); individual apps deployed to it share that capacity. Scaling the plan (up or out) affects every app hosted on it simultaneously — a frequent source of "why did App B slow down when App A got busy" scenario questions.

Deployment slots

- A **deployment slot** is a live, fully functional copy of an app with its own hostname, useful for staging before production cutover.
- Swap** operation exchanges the slot's content and configuration with production, with near-zero downtime (warms up the target slot first).
- Some settings are **slot-specific** (marked "deployment slot setting") and do NOT swap — e.g., connection strings pointing to different databases per environment; other settings swap along with the app content.

```

# ---- CLI: create a slot and swap ----
az webapp deployment slot create -g Prod-Web -n mywebapp --slot staging
az webapp deployment slot swap -g Prod-Web -n mywebapp --slot staging --target-slot production

# ---- PowerShell ----
New-AzWebAppSlot -ResourceGroupName Prod-Web -Name mywebapp -Slot staging
Switch-AzWebAppSlot -ResourceGroupName Prod-Web -Name mywebapp -SourceSlotName staging -DestinationSlotName production
  
```

VNet integration & access control

Feature	Purpose
Regional VNet Integration	Outbound-only: app can reach resources inside a VNet (e.g., a private database) via a delegated subnet
Private Endpoint (inbound)	Makes the app itself reachable only from inside a VNet — disables the public inbound endpoint when combined with access restrictions
Access Restrictions	IP allow/deny list applied to the app's public endpoint, evaluated before requests reach the app
Managed Identity	System/user-assigned, same model as VMs (§5.6) — used to reach Key Vault, Storage, SQL without secrets

Containers on Azure — choosing the right compute

Service	Orchestration	Best for	Ops overhead
Azure Container	None (single)	Burst workloads, simple batch jobs, CI/CD	Lowest — no cluster to manage

Instances (ACI)	container/container group)	build agents	
App Service (Web App for Containers)	None (App Service Plan handles scaling)	Web apps packaged as a container, wanting PaaS features (slots, easy auth, autoscale)	Low
Azure Kubernetes Service (AKS)	Full Kubernetes	Complex microservices, need full orchestration control, multi-container coordination	Highest — you manage node pools, upgrades, networking (though control plane is managed free)
Azure Container Apps	Managed Kubernetes under the hood (KEDA-based), abstracted away	Microservices/event-driven apps wanting Kubernetes power without Kubernetes complexity	Low-medium

Azure Container Registry (ACR)

- Private Docker registry for storing container images consumed by ACI, AKS, App Service, or on-prem Docker.
- SKUs: Basic, Standard, Premium (Premium adds geo-replication, higher throughput, Private Endpoint support, content trust).
- **AKS-to-ACR integration:** attach ACR to an AKS cluster (`az aks update --attach-acr`) to grant the cluster's managed identity pull access without storing registry credentials as Kubernetes secrets.

AKS core concepts (exam-relevant basics)

Concept	What it is
Node pool	A group of VMs (nodes) of the same size backing the cluster; system node pool (required) runs core K8s components, user node pools run workloads
Cluster autoscaler	Automatically adds/removes nodes based on pending pod scheduling pressure
Azure CNI vs. Kubenet	Azure CNI gives pods routable VNet IPs (more IP consumption, direct VNet integration); Kubenet uses NAT'd pod IPs (simpler IP planning, less direct integration)
Managed identity for the cluster	AKS control plane uses a managed identity to manage underlying Azure resources (Load Balancers, disks) on your behalf

```
# ---- CLI: ACI and AKS quickstarts ----
az container create -g Prod-Web -n aci-demo --image mcr.microsoft.com/azuredocs/aci-helloworld \
  --cpu 1 --memory 1 --ports 80 --ip-address Public

az aks create -g Prod-Web -n aks-demo --node-count 3 --enable-cluster-autoscaler \
  --min-count 1 --max-count 5 --network-plugin azure --attach-acr myacr

# ---- PowerShell ----
New-AzContainerGroup -ResourceGroupName Prod-Web -Name aci-demo -Image mcr.microsoft.com/azuredocs/aci-helloworld -OsType Linux
New-AzAksCluster -ResourceGroupName Prod-Web -Name aks-demo -NodeCount 3 -EnableClusterAutoscaler -NodeMinCount 1 -NodeMaxCount 5
```

11. Load Balancing & Application Delivery

The four load-balancing/delivery services and how they differ

Service	OSI Layer	Scope	Protocol awareness	Key use case
Azure Load Balancer	Layer 4 (TCP/UDP)	Regional	None — blind to HTTP content	High-performance, low-latency load balancing of any TCP/UDP traffic (VMs, VMSS)
Application Gateway	Layer 7 (HTTP/HTTPS)	Regional (zonal-redundant)	Full — URL path-based routing, cookie-based affinity, WAF	Web app front door needing path/host-based routing, SSL offload, WAF
Traffic Manager	DNS-based (not a proxy)	Global	None — routes DNS queries, not traffic itself	Global DNS-level routing/failover across regions, any protocol
Azure Front Door	Layer 7 (HTTP/HTTPS), global anycast	Global	Full — URL routing, caching (CDN), WAF	Global web app acceleration with a single anycast IP, health-probe failover, edge caching

Decision diagram

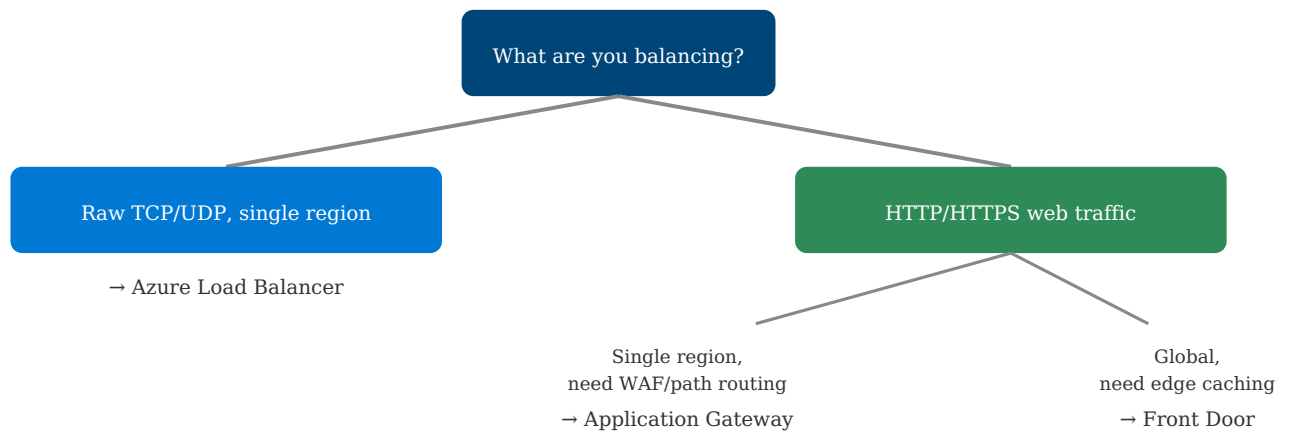


Figure 11.1 — Choosing between the four load-balancing/delivery services

Tip: Traffic Manager is pure DNS — it tells the client which IP to connect to, then gets out of the way. Front Door and Application Gateway are actual reverse proxies that traffic flows *through*, so they can do things DNS-based routing cannot: WAF inspection, path-based routing, SSL termination, response caching. A common exam distinction: "needs to inspect/modify HTTP traffic" always rules out Traffic Manager.

Azure Load Balancer: Basic vs. Standard SKU

Feature	Basic	Standard
SLA	None	99.99%
Availability Zone support	No	Yes (zone-redundant or zonal frontend)
Backend pool size	Up to 300 instances	Up to 1,000 instances
NSG requirement	Optional	Required — Standard LB is secure-by-default (denies unless explicitly allowed by NSG)
Outbound rules	Implicit	Explicit, more control over SNAT port allocation

Load Balancer components

Component	Purpose
Frontend IP configuration	The public or private IP clients connect to
Backend pool	The VMs/VMSS instances receiving traffic
Health probe	Determines which backend instances are healthy enough to receive traffic (TCP, HTTP, or HTTPS probe)
Load balancing rule	Maps a frontend IP:port to a backend pool:port, with a chosen distribution mode
Inbound NAT rule	Maps a specific frontend port to a specific single backend instance (e.g., RDP to VM3 via port 3390)

```
# ---- CLI: Standard Load Balancer with health probe ----
az network lb create -g Prod-Web -n lb-web --sku Standard --public-ip-address lb-pip \
  --frontend-ip-name feConfig --backend-pool-name beWeb

az network lb probe create -g Prod-Web --lb-name lb-web -n http-probe --protocol Http --port 80 --path /healthz

az network lb rule create -g Prod-Web --lb-name lb-web -n http-rule --protocol Tcp \
  --frontend-port 80 --backend-port 80 --frontend-ip-name feConfig --backend-pool-name beWeb --probe-name http-probe

# ---- PowerShell ----
$pip = New-AzPublicIpAddress -ResourceGroupName Prod-Web -Name lb-pip -Sku Standard -AllocationMethod Static -Location eastus
$fe = New-AzLoadBalancerFrontendIpConfig -Name feConfig -PublicIpAddress $pip
$be = New-AzLoadBalancerBackendAddressPoolConfig -Name beWeb
$probe = New-AzLoadBalancerProbeConfig -Name http-probe -Protocol Http -Port 80 -RequestPath /healthz -IntervalInSeconds 15 -
ProbeCount 2
New-AzLoadBalancer -ResourceGroupName Prod-Web -Name lb-web -Sku Standard -Location eastus `
  -FrontendIpConfiguration $fe -BackendAddressPool $be -Probe $probe
```

Application Gateway routing patterns

Pattern	Example
Path-based routing	/images/* → image-service pool, /api/* → api-service pool
Multi-site (host header) routing	shop.contoso.com and blog.contoso.com served from the same Gateway, different backend pools
SSL offload/termination	Gateway decrypts HTTPS, forwards plain HTTP to backend (reduces backend CPU load)
End-to-end SSL	Gateway re-encrypts before forwarding to backend, for compliance requiring encryption throughout

WAF (Web Application Firewall) can run on either Application Gateway (WAF SKU, regional) or Front Door (global) — both use OWASP Core Rule Set-based managed rules plus custom rules, protecting against SQL injection, XSS, and other common web exploits.

12. Azure Firewall, WAF & Network Security

Azure Firewall vs. NSG vs. third-party NVA

Aspect	NSG	Azure Firewall	Third-party NVA
OSI Layer	L3/L4 (IP/port)	L3-L7 (FQDN filtering, threat intelligence)	Varies by vendor, often L3-L7
Stateful?	Yes	Yes	Usually yes
Centralized logging/management	Per-NSG, decentralized	Centralized, single pane via Firewall Policy	Vendor-dependent
FQDN-based outbound filtering	No	Yes (e.g., allow only *.windowsupdate.com)	Yes (vendor-dependent)
Threat intelligence feed	No	Yes (Microsoft threat intel, alert or deny mode)	Vendor-dependent
Deployment model	Free, attached to subnet/NIC	Managed PaaS service in a dedicated subnet (AzureFirewallSubnet)	Self-managed VM(s), you patch/scale it

Azure Firewall SKU tiers

Tier	Adds
Basic	Entry-level L3-L7 filtering for small/medium workloads
Standard	Full L3-L7 filtering, threat intelligence, DNS proxy
Premium	TLS inspection, IDPS (intrusion detection/prevention), URL filtering (vs. FQDN-only)

Defense-in-depth layering (how these fit together)

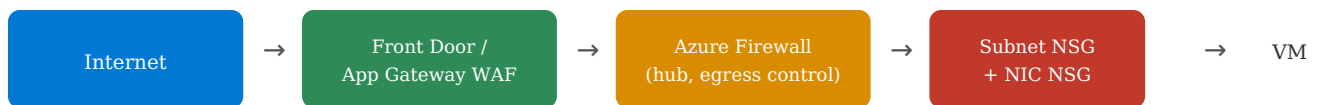


Figure 12.1 — Layered network defense: each layer independently filters before traffic reaches the workload

Azure DDoS Protection

Tier	Coverage	Cost model
Basic (always on)	Automatic, platform-wide, no configuration	Free, included with every VNet
Network Protection (formerly Standard)	Tuned mitigation policies per VNet, attack analytics/alerting, cost protection (credits for scale-out costs during an attack), rapid response support	Paid, per protected VNet resource

Just-in-time (JIT) VM access

A Microsoft Defender for Cloud feature that keeps management ports (RDP 3389, SSH 22) closed by default via NSG, opening them only for a requested time window and source IP when an admin explicitly requests access — reduces the exposure window for brute-force attacks compared to a standing NSG allow rule.

Azure Bastion

A fully managed PaaS bastion host providing browser-based RDP/SSH to VMs **without any public IP** on the target VM and without exposing RDP/SSH ports to the internet at all. Deployed into the dedicated `AzureBastionSubnet` (\$5.1), it is a regional service shared by all VMs in the peered/local VNets.

Aspect	Bastion	Traditional jump box VM
Public IP on target VM needed?	No	Sometimes (or via a separate jump box)

Patching/maintenance	Fully managed by Microsoft	Customer-managed
Client requirement	Just a browser	RDP/SSH client software

13. Additional Storage Tools & Migration

Data movement tools compared

Tool	Interface	Best for	Notes
AzCopy	Command-line	Scripted, high-performance bulk copy/sync (blob, file, and between storage accounts)	Supports parallel transfer, resumable jobs, and Entra ID or SAS auth
Azure Storage Explorer	Desktop GUI	Ad hoc browsing, small transfers, managing ACLs/metadata visually	Cross-platform (Windows/Mac/Linux), connects via Entra ID, SAS, or keys
Azure Data Factory / Synapse Pipelines	Managed ETL/orchestration	Scheduled, large-scale, transformation-in-flight data movement	Connects 90+ sources/sinks, not just Azure Storage
Azure Import/Export service	Physical disk shipping	Very large one-time datasets where network transfer would take too long	Ship your own encrypted disks to a Microsoft datacenter
Azure Data Box (family)	Microsoft-supplied physical appliance	Large-scale offline migration (Data Box: up to ~1 PB via a ruggedized appliance; Data Box Disk/Heavy for smaller/larger variants)	Microsoft ships you the device, you fill it, ship it back

```
# ---- AzCopy: sync a local folder to Blob storage ----
azcopy sync 'C:\localdata' 'https://stgprod01.blob.core.windows.net/uploads?<SAS-token>' --recursive

# ---- AzCopy: copy between two storage accounts, server-side (no local hop) ----
azcopy copy 'https://stgsrc.blob.core.windows.net/data?<SAS>' \
  'https://stgdest.blob.core.windows.net/data?<SAS>' --recursive
```

Static website hosting

Blob storage can serve static content (HTML/CSS/JS) directly via a `$web` system container, with a public endpoint and optional custom domain + CDN/Front Door in front for caching and HTTPS on the custom domain. No compute required — ideal for SPAs and documentation sites.

Object Replication (Blob)

Asynchronously copies block blobs between a source and destination storage account (can be cross-region, cross-subscription) at the *object* level, distinct from account-level GRS replication. Useful for minimizing read latency for users near the destination region, or maintaining a separate copy for compliance without full-account geo-redundancy cost.

Storage migration decision guide

Scenario	Recommended tool
10 GB of files, good internet connection	AzCopy or Storage Explorer
500 TB dataset, limited bandwidth, weeks of transfer time otherwise	Azure Data Box
Ongoing, scheduled ETL from an on-prem SQL Server into Blob/Data Lake	Azure Data Factory with a Self-Hosted Integration Runtime
Migrating VMs and their attached disks as part of a lift-and-shift	Azure Migrate (see scenario §7.11)
One-time export of an entire storage account for offline archive	Azure Import/Export service (export job)

Storage account limits worth memorizing

Limit	Value
Max storage account capacity	~5 PiB (varies by account type/region)
Max storage accounts per subscription (default)	250 (raisable via support request)
Max blob size (block blob)	~190.7 TiB
Max ingress/egress per storage account (default)	Region/account-type dependent, typically tens of Gbps
Max File Share size (standard)	5 TiB (100 TiB for large file shares, premium tiers)

14. Dedicated Hosts, Spot VMs & Cost Optimization

Compute purchasing/deployment options

Option	What it is	Discount vs. Pay-as-you-go	Eviction risk	Use when
Pay-as-you-go	Standard on-demand billing	0% (baseline)	None	Unpredictable/variable workloads
Reserved Instances (1 or 3-yr)	Commit to a specific VM size/region for a term	Up to ~72%	None	Steady-state, predictable production workloads
Savings Plan (1 or 3-yr)	Commit to an hourly \$ spend across eligible compute, flexible across VM families/regions	Up to ~65%	None	Predictable spend but variable VM sizes/families/regions
Spot VMs	Bid for unused Azure capacity at steep discount	Up to ~90%	High — can be evicted with ~30 sec notice when Azure needs the capacity back	Fault-tolerant, interruptible batch/CI workloads, stateless VMSS nodes
Azure Dedicated Host	Physical server dedicated entirely to you (no other tenants)	N/A (priced by host, not discount-based)	None (you control placement)	Compliance requiring physical isolation, license mobility (Bring-Your-Own-License scenarios), control over maintenance timing

Dedicated Host architecture

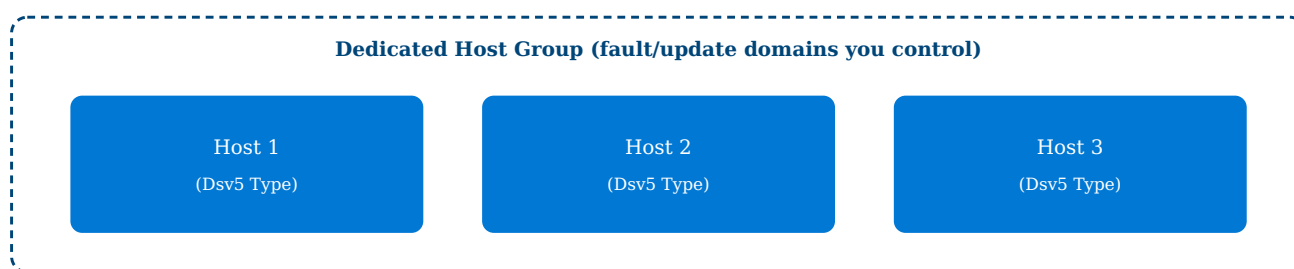


Figure 14.1 — A Dedicated Host Group lets you explicitly control fault/update domain placement across physically isolated hosts

Tip: Dedicated Hosts are billed for the *host* regardless of how many/few VMs you place on it — cost-efficient only when you need the isolation/compliance/licensing benefit, not as a general cost-saving mechanism (Reserved Instances or Savings Plans are for that).

Spot VM configuration essentials

Setting	Options
Eviction policy	Deallocate (default, disk preserved, can restart when capacity returns) or Delete (VM and disks removed entirely)
Max price	Set a price cap (or "-1" to pay up to the on-demand rate and only be evicted for capacity, not price)
Eviction notice	~30 seconds via Scheduled Events (IMDS) — applications should watch this to gracefully checkpoint/shutdown

```
# ---- CLI: Spot VM ----
az vm create -g Prod-Batch -n spotVM01 --image Ubuntu2204 \
  --priority Spot --eviction-policy Deallocate --max-price -1 --size Standard_D2s_v5

# ---- CLI: Dedicated Host ----
az vm host group create -g Prod-Web -n hostGroup1 --platform-fault-domain-count 2
az vm host create -g Prod-Web -n host1 --host-group hostGroup1 --sku Dsv5-Type1

# ---- PowerShell: Reserved Instance purchase (via portal typically, CLI shown for reference) ----
New-AzReservationOrder -SkuName "Standard_D2s_v5" -Term "P1Y" -BillingScope $subscriptionId
```

Cost optimization checklist (Advisor recommendations, condensed)

- Rightsize or shut down underutilized VMs (Advisor flags VMs with consistently low CPU/network use).
- Delete unattached managed disks (common orphaned-cost source after VM deletion).

- Use Auto-shutdown schedules for dev/test VMs outside business hours.
- Move infrequently accessed blobs to Cool/Cold/Archive tiers via Lifecycle Management.
- Buy Reservations/Savings Plans for steady-state production compute once usage patterns are established (typically after 1-3 months of stable usage data).
- Consider Spot VMs for CI/CD build agents, batch rendering, and other interruption-tolerant workloads.

15. Service Health & Governance Deep Dive

Service Health vs. Resource Health vs. Azure Status

Tool	Scope	Answers
Azure Status page (status.azure.com)	Global, public, all customers	"Is Azure having a widespread outage right now?"
Service Health (in-portal)	Personalized to your subscriptions' regions/services	"Is an active Azure issue affecting <i>my</i> resources specifically?" — includes Planned Maintenance and Health Advisories
Resource Health	Per-resource	"Is this specific VM/resource healthy right now, and if not, why (platform issue vs. something the customer did)?"

Tip: Configure a **Service Health alert** (Action Group-backed) so your team is notified automatically of Azure-caused incidents/maintenance affecting your specific subscriptions/regions/services — this is the "proactive" answer whenever a scenario says "we found out about an Azure outage too late."

Governance deep dive: Blueprints, Template Specs & Deployment Stacks

Tool	Status	Bundles	Notes
Azure Blueprints	Deprecated (retirement announced; migrate to Template Specs)	RBAC + Policy + ARM templates + RG structure, versioned as one artifact	Assignments could "lock" artifacts from deletion; still appears in legacy exam content
Template Specs	Current	A single ARM/Bicep template packaged as a versioned, shareable Azure resource	Simpler than Blueprints; distributed via RBAC like any resource, consumed via standard deployment commands
Deployment Stacks	Current	A managed collection of resources deployed together, with lifecycle tracking and "deny settings" to protect managed resources from out-of-band changes	Closest modern replacement for what Blueprints' "lock assignments" did — recommended pairing with Template Specs for full landing-zone-as-code

Management Group design patterns

Pattern	Structure	Use when
Environment-based	Root → Production / Non-Production → individual subscriptions	Simple orgs where the main governance difference is env-based policy strictness
Business-unit-based	Root → BU-Sales / BU-Engineering / BU-Finance → each with their own Prod/Non-Prod split	Larger orgs with distinct budget owners and differing compliance needs per BU
Landing zone (Cloud Adoption Framework)	Root → Platform (Identity/Management/Connectivity) + Landing Zones (Corp/Online) + Sandbox + Decommissioned	Enterprise-scale, following Microsoft's published reference architecture

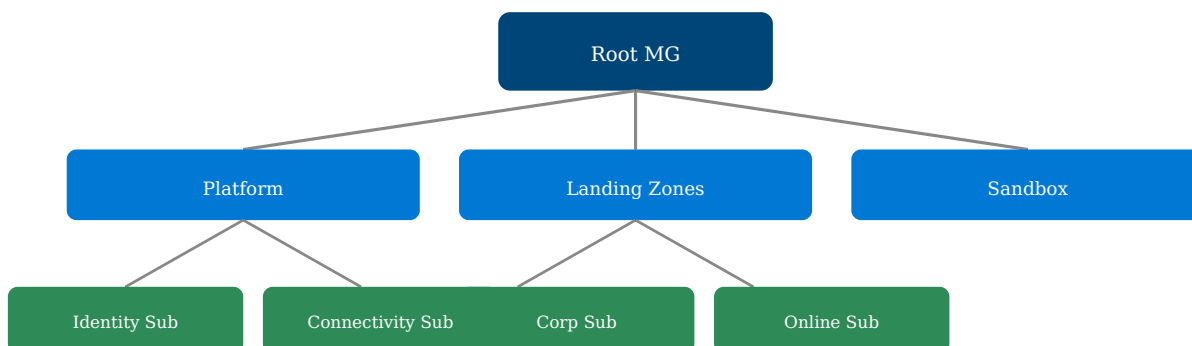


Figure 15.1 — Cloud Adoption Framework landing zone pattern, simplified

Azure Advisor

A free, continuous recommendation engine spanning five pillars: **Cost**, **Security** (surfaces Defender for Cloud findings), **Reliability** (resiliency gaps like missing zone redundancy), **Operational Excellence**, and **Performance**. Recommendations can be tracked,

postponed, or dismissed, and are aggregated at Management Group scope for portfolio-wide visibility.

16. Exam Practice Questions (25 with Explanations)

These questions mirror AZ-104 style and difficulty. Attempt each before reading the explanation.

Q1. Management Group limits

Q: You need to organize 200 subscriptions for policy inheritance. What is the maximum depth of management groups beneath the root management group?

- A) 3 B) 6 C) 10 D) Unlimited

Answer: B. Up to 6 levels of management groups are allowed beneath the always-present root management group.

Q2. Data-plane access

Q: A user has the Owner role on a storage account. Which statement is true regarding their ability to read blob contents via their Entra ID identity?

- A) They can always read blob contents because Owner includes all permissions
B) They cannot read blob contents without an explicit data-plane role like Storage Blob Data Reader
C) They can read blobs only if MFA is enabled
D) Owner role is automatically mapped to Storage Blob Data Owner

Answer: B. Management-plane roles (Owner/Contributor) do not grant data-plane access; a separate `Storage Blob Data * role` or key/SAS is required.

Q3. VNet Peering transitivity

Q: VNet A is peered with VNet B. VNet B is peered with VNet C. Can a VM in VNet A communicate with a VM in VNet C using only these peerings?

- A) Yes, automatically B) No, peering is not transitive C) Only if both use the same NSG D) Only within the same region

Answer: B. VNet Peering is never transitive. A-to-C connectivity requires either a direct A-C peering or routing through an NVA/Firewall in B with appropriate UDRs and "allow forwarded traffic" enabled.

Q4. VM billing states

Q: A VM is stopped from within the Windows guest OS (Start menu → Shut down), not from the Azure Portal. Is the VM still billed for compute?

- A) No, stopping the OS always deallocates the VM
B) Yes, it remains billed until explicitly deallocated via Azure
C) Only billed for the first hour
D) Billing depends on the VM size

Answer: B. Only an Azure-level Deallocate (Portal "Stop" or `az vm deallocate`) releases the hardware allocation and stops compute billing.

Q5. Resource Locks and RBAC

Q: A resource group has a CanNotDelete lock applied. A user with the Owner role at the subscription attempts to delete a resource in that group. What happens?

- A) The delete succeeds because Owner overrides locks
B) The delete fails; the lock must be removed first, regardless of RBAC role
C) The delete succeeds but generates a warning
D) The delete succeeds only with MFA

Answer: B. Resource Locks apply regardless of RBAC role — even a subscription Owner must remove the lock before deleting.

Q6. Storage redundancy for read access during regional outage

Q: Which storage redundancy option allows you to read data from the secondary region while the primary region is unavailable?

- A) LRS B) ZRS C) GRS D) RA-GRS

Answer: D. Standard GRS replicates to a secondary region but does not allow reads there; RA-GRS (Read-Access GRS) explicitly enables

reading the secondary.

Q7. NSG evaluation with dual attachment

Q: A subnet has NSG-1 attached, and a NIC within that subnet has NSG-2 attached. NSG-1 allows inbound port 443; NSG-2 denies it. Can inbound HTTPS traffic reach the VM?

- A) Yes, subnet NSG takes priority
- B) Yes, NIC NSG takes priority so it's actually allowed
- C) No, both NSGs must allow the traffic; a deny at either layer blocks it
- D) It depends on the rule priority number only

Answer: C. When both subnet-level and NIC-level NSGs are present, traffic must be permitted by both. A deny at either layer blocks the packet.

Q8. Custom role components

Q: Which property of a custom role definition controls which scopes the role can be assigned to?

- A) DataActions
- B) NotActions
- C) AssignableScopes
- D) Actions

Answer: C. `AssignableScopes` defines where the custom role definition can be used when creating a role assignment.

Q9. ARM deployment mode risk

Q: Your team deploys a template in Complete mode to a resource group that also contains manually created resources not in the template. What happens to those manual resources?

- A) They are left untouched
- B) They are deleted
- C) They are moved to a new resource group
- D) The deployment fails

Answer: B. Complete mode deletes any resource in the target resource group not defined in the template — always run `what-if` first to catch this.

Q10. Availability Set limitation

Q: You have an existing standalone VM and want to add it to an Availability Set for higher SLA. What must you do?

- A) Simply edit the VM's Availability Set property in the Portal
- B) Delete and recreate the VM inside the Availability Set (disks can be reused)
- C) It's not possible to ever move a VM to an Availability Set
- D) Use `Move-AzResource`

Answer: B. Availability Set membership must be set at VM creation time; an existing VM must be deleted and recreated (keeping its disks) inside the set.

Q11. Private Endpoint DNS

Q: After creating a Private Endpoint for a storage account, clients in the VNet still resolve the storage account's public IP. What is the most likely cause?

- A) The Private Endpoint was created in the wrong region
- B) The Private DNS Zone was not linked/configured
- C) The storage account SKU doesn't support Private Endpoints
- D) NSG is blocking DNS traffic

Answer: B. Creating a Private Endpoint does not automatically create or link the required Private DNS Zone unless explicitly configured — the #1 real-world Private Endpoint support issue.

Q12. Cross-tenant subscription transfer impact

Q: After transferring a subscription to a new Entra tenant, existing role assignments referencing the old tenant's users fail to resolve. Why?

- A) RBAC assignments are region-locked
- B) The old-tenant object IDs no longer exist in the new tenant's directory
- C) Role assignments transfer automatically but need 24 hours to propagate
- D) This never happens; assignments always transfer correctly

Answer: B. RBAC assignments reference identity object IDs scoped to a specific tenant; after a tenant transfer, those object IDs don't exist in the destination tenant and must be re-created.

Q13. Policy effect selection

Q: You want non-compliant resources to be automatically remediated by deploying a missing monitoring agent, without blocking the original deployment. Which policy effect fits best?

- A) Deny B) Audit C) DeployIfNotExists D) Disabled

Answer: C. DeployIfNotExists triggers a remediation deployment for a related missing resource, without blocking the original request the way Deny would.

Q14. Backup vault redundancy timing

Q: When can you change a Recovery Services Vault's storage redundancy setting (e.g., from GRS to LRS)?

- A) At any time B) Only before the first backup is taken C) Only during a support-assisted migration D) Only from PowerShell, never the Portal

Answer: B. Vault redundancy can only be changed before the first backup is taken in that vault.

Q15. Temp disk data persistence

Q: An application stores session cache data on a VM's temp (D:) disk. What happens to that data if the VM is redeployed to different host hardware?

- A) It is preserved automatically B) It is lost — temp disks are not persistent C) It is backed up automatically to the OS disk D) It depends on the VM size only

Answer: B. Temp disks are local, ephemeral storage and are not preserved across deallocation/redeploy operations — never store persistent data there.

Q16. Choosing Service Endpoint vs Private Endpoint

Q: A company needs an on-premises server (connected via ExpressRoute) to securely reach an Azure SQL Database without traversing the public internet. Which is required?

- A) Service Endpoint alone is sufficient
B) Private Endpoint, since Service Endpoints only work for traffic originating inside the VNet
C) Either works identically
D) Neither; use a Public IP with an NSG rule

Answer: B. Service Endpoints only apply to traffic originating from within the configured subnet; they cannot be reached from on-premises. A Private Endpoint provides a real private IP reachable via ExpressRoute/VPN.

Q17. Azure Backup vs. Site Recovery

Q: Which service is designed specifically to fail over an entire application's VMs to a secondary Azure region during a regional disaster?

- A) Azure Backup B) Azure Site Recovery C) Storage GRS D) Azure Monitor

Answer: B. Azure Site Recovery is purpose-built for orchestrated, whole-workload regional disaster recovery with Recovery Plans; Azure Backup restores point-in-time data but isn't a failover orchestration tool.

Q18. PIM activation

Q: A user has an "eligible" assignment for the Contributor role via PIM on a subscription. What must they do before performing Contributor actions?

- A) Nothing, eligible assignments are always active
B) Explicitly activate the role, potentially requiring MFA/approval/justification
C) Wait 24 hours for automatic activation
D) Contact Microsoft Support

Answer: B. An eligible PIM assignment requires explicit, time-bound activation (which may require MFA, justification, and/or approval) before the permissions take effect.

Q19. Application Gateway vs Load Balancer

Q: You need to route `/api/*` requests to one backend pool and `/images/*` to another, all under one public hostname. Which service provides this natively?

- A) Azure Load Balancer (Standard SKU)
- B) Application Gateway (path-based routing)
- C) Traffic Manager
- D) Network Security Group

Answer: B. Application Gateway operates at Layer 7 and supports URL path-based routing; Load Balancer is Layer 4 and has no concept of URL paths.

Q20. Storage account key rotation impact

Q: You rotate (regenerate) a storage account's primary access key. What happens to SAS tokens that were generated using that key?

- A) Nothing, they remain valid until their stated expiry
- B) They are all immediately invalidated
- C) Only read-permission SAS tokens are invalidated
- D) They become read-only automatically

Answer: B. Any SAS token derived from an account key is invalidated when that specific key is regenerated — a useful emergency revocation technique, but it also breaks all other SAS tokens tied to that key.

Q21. Subnet special names

Q: You are deploying Azure Bastion into an existing VNet. What must the subnet be named, and what is the minimum size?

- A) BastionSubnet, /24
- B) AzureBastionSubnet, /26 minimum
- C) GatewaySubnet, /27
- D) Any name is fine, /28 minimum

Answer: B. Azure Bastion requires the exact subnet name `AzureBastionSubnet` with a minimum /26 address space.

Q22. Autoscale trigger types

Q: A retail company wants their VMSS to scale to a fixed higher capacity every day at 8 AM regardless of current load, ahead of anticipated traffic. Which autoscale mechanism fits?

- A) Metric-based rule only
- B) Schedule-based profile
- C) Manual scaling only
- D) Azure Advisor recommendation

Answer: B. A schedule-based autoscale profile lets you define fixed capacity changes at specific times, independent of live metrics.

Q23. Key Vault access models

Q: A Key Vault is configured with the RBAC permission model. A user with Owner on the vault resource, but no data-plane role assignment, tries to read a secret. What happens?

- A) Succeeds, Owner includes all data access
- B) Fails; a data-plane role like Key Vault Secrets User is required
- C) Succeeds only if the secret has no expiry date
- D) Fails only if soft-delete is enabled

Answer: B. Same management-plane vs. data-plane separation as storage: Key Vault's RBAC model requires an explicit data-plane role even for resource Owners.

Q24. Cost Management budget alerts

Q: You want stakeholders notified by email when a subscription's forecasted spend is projected to exceed its monthly budget, even before it actually happens. What feature supports this?

- A) Cost Analysis only
- B) Budgets with a forecasted-cost alert condition
- C) Azure Advisor
- D) Resource Locks

Answer: B. Budgets support alert conditions based on both actual and forecasted spend, notifying stakeholders proactively before the threshold is actually reached.

Q25. Hub-spoke transitive routing requirement

Q: In a hub-spoke topology with an NVA firewall in the hub, spoke-to-spoke traffic is not flowing even though UDRs point to the firewall's private IP. What setting is most likely missing?

- A) NSG priority number too high
- B) "Allow forwarded traffic" not enabled on both peering links
- C) The firewall's public IP is missing
- D) DNS is not configured

Answer: B. Peering links must have "Allow forwarded traffic" enabled on both sides of the path for traffic that didn't originate in the peered VNet (i.e., traffic being forwarded by the NVA) to pass through.

17. Glossary of Key Terms

Term	Definition
ARM (Azure Resource Manager)	The deployment and management layer that processes all requests to create/update/delete Azure resources, regardless of the tool used (Portal, CLI, PowerShell, SDK)
ASG	Application Security Group — logical grouping of NICs used inside NSG rules instead of hardcoded IPs
ASR	Azure Site Recovery — disaster recovery orchestration service
Availability Set	Logical grouping ensuring VMs are spread across fault and update domains within one datacenter
Availability Zone	A physically separate datacenter with independent power/cooling/network within an Azure region
Blueprint	Deprecated packaging of RBAC + Policy + ARM templates as one versioned artifact; superseded by Template Specs + Deployment Stacks
CIDR	Classless Inter-Domain Routing — notation for IP address ranges (e.g., 10.0.0.0/24)
Conditional Access	Entra ID feature enforcing if-then sign-in rules based on user, device, location, and risk signals
Custom Role	A user-defined RBAC role combining specific Actions/DataActions not covered by a built-in role
Data plane	Operations on the actual content/data of a resource (e.g., reading a blob), as opposed to the management plane
Deny Assignment	A rare RBAC construct that explicitly blocks an action regardless of granted roles, typically system-generated by Blueprints/managed apps
Entra Connect / Cloud Sync	Tools that synchronize on-prem Active Directory objects to Microsoft Entra ID
ExpressRoute	Private, dedicated network circuit connecting on-premises infrastructure to Azure without traversing the public internet
Fault Domain	A group of hardware (rack/power/network) that can fail independently of other fault domains within an Availability Set
Federation (AD FS)	Hybrid authentication method where on-prem AD FS servers issue security tokens for cloud sign-in
GRS / RA-GRS / ZRS / GZRS	Storage redundancy options; see §3.2 for full comparison
Hierarchical Namespace (HNS)	Feature turning Blob storage into Azure Data Lake Storage Gen2 with real directory semantics
Hub-spoke	Network topology pattern where a central "hub" VNet hosts shared services (firewall, gateway) and "spoke" VNets peer to it
Initiative	A grouped set of Azure Policy definitions assigned and tracked together
Management Group	A container above subscriptions used to apply RBAC/Policy across many subscriptions at once
Managed Identity	An automatically managed Entra ID identity for an Azure resource, used for secretless authentication to other services
NSG	Network Security Group — stateful, rule-based traffic filter applied at subnet or NIC level
PIM	Privileged Identity Management — enables just-in-time, time-bound, approval-gated role activation
Private Endpoint / Private Link	A private IP injected into your VNet that represents a specific PaaS resource instance, removing the need for a public endpoint
RBAC	Role-Based Access Control — Azure's authorization system based on (principal + role + scope)
Resource Group	A logical container and lifecycle/deployment boundary for resources within a subscription
Resource Lock	A setting (CanNotDelete or ReadOnly) that overrides RBAC to prevent accidental changes
SAS (Shared Access Signature)	A signed URL granting scoped, time-limited access to storage resources without sharing account keys
Service Endpoint	A subnet-level setting that optimizes routing and firewall-restricts a PaaS service to specific subnets, without a private IP
Service Principal	The identity representation of an application/service within a specific Entra ID tenant, used for non-interactive authentication
Tenant	The top-level Microsoft Entra ID directory containing users, groups, and devices for an organization
UDR (User-Defined Route)	A custom routing rule in a Route Table that overrides Azure's default system routes for a subnet
VMSS	Virtual Machine Scale Set — a group of identical, load-balanced VM instances that can autoscale
VNet Peering	Direct, non-transitive network connectivity between two Virtual Networks over the Azure backbone
WAF	Web Application Firewall — L7 protection against common web exploits (SQLi, XSS), available on App Gateway and Front Door
Update Domain	A logical grouping determining which VMs in an Availability Set are rebooted together during planned maintenance

"Fluency with this vocabulary is itself half the exam — many AZ-104 questions are really just testing whether you know which term describes which mechanism."

18. Extended Case Studies (Exam-Format)

Real AZ-104 case studies present a company background, existing environment, requirements, and then ask several questions against that shared context. Practice reading the whole scenario before answering — here are two full-length examples.

Case Study 1: Contoso Ltd. Retail Expansion

Background

Contoso Ltd. is a retail company migrating its e-commerce platform to Azure. The company has one Entra ID tenant (`contoso.onmicrosoft.com`) and two Azure subscriptions: `Contoso-Prod` and `Contoso-NonProd` .

Existing environment

- A hub VNet (`hub-vnet` , `10.0.0.0/16`) in `Contoso-Prod` contains a VPN Gateway connected to the on-prem datacenter and an Azure Firewall.
- A spoke VNet (`web-vnet` , `10.1.0.0/16`) is peered to the hub and hosts a VMSS running the web tier behind a Standard Load Balancer.
- A separate spoke VNet (`data-vnet` , `10.2.0.0/16`) hosts an Azure SQL Managed Instance in a delegated subnet.
- A storage account `contosoimages` (StorageV2, GRS) stores product images, currently with public network access enabled from all networks.
- The engineering team has Contributor access at the `Contoso-NonProd` subscription; only two senior admins have Owner on `Contoso-Prod` .

Requirements

1. Product images must only be accessible from within the Azure network (hub and spokes) or via ExpressRoute/VPN from on-prem — never over the public internet.
2. Developers must be able to deploy to `Contoso-NonProd` freely but require approval and MFA for any elevated access to `Contoso-Prod` .
3. The web tier VMSS must remain available even if an entire Availability Zone fails.
4. All resources across both subscriptions must have a `CostCenter` tag; missing tags should be filled automatically rather than blocking deployment.
5. A recent incident showed a critical VM was deleted by mistake; this must not be possible for the SQL Managed Instance's resource group going forward.

Case Study 1 Questions

Q1.1 Which single change satisfies Requirement 1 with the least additional infrastructure?

- A) Enable a Service Endpoint for Storage on both spoke subnets
- B) Deploy a Private Endpoint for `contosoimages` into `data-vnet` , link the appropriate Private DNS Zone to all three VNets, and set the storage account's public network access to Disabled
- C) Add an NSG rule denying all internet traffic on the web subnet
- D) Move the storage account into the hub VNet's resource group

Answer: B. Only a Private Endpoint provides a real private IP reachable from all peered VNets and on-prem via the hub's gateway; setting public access to Disabled closes the last gap. Storage accounts aren't "in" a VNet (option D is a category error), and Service Endpoints alone don't help on-prem access.

Q1.2 How should Requirement 2 be implemented for `Contoso-Prod` ?

- A) Grant all developers Contributor directly on `Contoso-Prod`
- B) Configure Contributor as a PIM-eligible assignment on `Contoso-Prod` , requiring MFA and approval on activation
- C) Create a new subscription just for approvals
- D) Use a Resource Lock on the subscription

Answer: B. This is the textbook PIM use case: standing access in Non-Prod, just-in-time approval-gated elevation in Prod.

Q1.3 What satisfies Requirement 3 for the web tier?

- A) Move all VMSS instances into a single Availability Zone for consistency
- B) Configure the VMSS as zone-redundant across zones 1, 2, and 3 behind the existing Standard Load Balancer
- C) Switch to Basic Load Balancer
- D) Add more VMSS instances within the same zone

Answer: B. Zone-redundant VMSS with a Standard (not Basic) Load Balancer is required for zone-failure resilience; Basic LB doesn't support Availability Zones.

Q1.4 Which policy effect satisfies Requirement 4?

A) Deny B) Audit C) Modify D) Disabled

Answer: C. Modify auto-appends the missing tag rather than blocking the deployment the way Deny would.

Q1.5 What addresses Requirement 5?

- A) Remove Owner role from all users on that resource group
- B) Apply a CanNotDelete lock on the resource group containing the SQL Managed Instance
- C) Enable soft delete on the SQL Managed Instance
- D) Move the resource group to a different subscription

Answer: B. A CanNotDelete lock blocks deletion regardless of RBAC role, directly addressing the "must not be possible" requirement.

Case Study 2: Fabrikam Health Systems Acquisition

Background

Fabrikam Health Systems has just acquired a smaller competitor, Litware Diagnostics. Litware operates its own Entra ID tenant (`litware.onmicrosoft.com`) with a single Azure subscription containing all of its clinical scheduling application resources.

Existing environment

- Fabrikam's tenant follows a management group structure: Root → `MG-Fabrikam-Prod` and `MG-Fabrikam-NonProd`, each with an Azure Policy initiative enforcing an approved-regions list and mandatory diagnostic logging.
- Litware's subscription (`Litware-Clinical`) has its own VNet, a storage account holding patient scheduling exports, and a Recovery Services Vault backing up its application VMs nightly.
- Fabrikam's compliance team requires that patient data storage accounts never use GRS with replication outside the country's borders, per regulatory requirements.
- Litware's three IT staff need to keep managing their own environment day-to-day for at least 12 months post-acquisition, per the acquisition agreement.
- Fabrikam's security team wants immediate read-only visibility into Litware's resources for a compliance audit, without waiting for a full tenant migration.

Requirements

1. Fabrikam's security team must get Reader access across all of Litware's resources within days, with minimal disruption to Litware's existing operations.
2. Litware's IT staff must retain full administrative control of their own subscription for the agreed 12-month period.
3. Any new patient-data storage accounts in Litware's subscription must comply with Fabrikam's data-residency requirement immediately.
4. Eventually (post 12 months), Litware's subscription should be fully absorbed into Fabrikam's tenant and management group structure.
5. Litware's nightly VM backups must continue uninterrupted throughout any transition.

Case Study 2 Questions

Q2.1 What best satisfies Requirement 1 with minimal disruption?

- A) Immediately transfer the Litware subscription into Fabrikam's tenant
- B) Deploy an Azure Lighthouse delegation granting Fabrikam's security team a Reader role across Litware's subscription, without moving the subscription
- C) Ask Litware IT to manually export a report of all resources
- D) Create guest accounts for every Fabrikam security team member in Litware's tenant with Owner access

Answer: B. Lighthouse is designed exactly for this: cross-tenant delegated access without moving billing/subscription ownership, minimizing disruption, and scoped precisely (Reader only, per Requirement 1's "read-only visibility").

Q2.2 How does the chosen approach in Q2.1 satisfy Requirement 2 simultaneously?

- A) It doesn't; a separate step is needed
- B) Because Lighthouse delegation does not remove or alter Litware's own existing RBAC assignments — Litware IT keeps their full administrative rights untouched
- C) Litware IT must be granted new roles to compensate
- D) Litware's subscription becomes read-only for Litware IT during delegation

Answer: B. Lighthouse purely *adds* delegated access for the managing tenant; it does not touch or reduce the existing tenant's own role assignments.

Q2.3 What satisfies Requirement 3?

- A) Manually review each new storage account after creation
- B) Assign Fabrikam's existing data-residency Policy initiative (or an equivalent Deny-effect policy restricting storage redundancy/regions) directly to the Litware subscription (or an appropriate scope covering it)
- C) Wait until the full tenant migration to apply any policy
- D) Ask Litware to self-certify compliance

Answer: B. Policy can be assigned to Litware's subscription immediately (even while it remains in its own tenant/management group structure, since Policy assignment targets a scope Fabrikam now has delegated Owner/Policy Contributor rights to via Lighthouse, or directly if governance access was included in the delegation) — enforcing the residency rule going forward without waiting for full migration.

Q2.4 What is the correct sequence of steps for Requirement 4 after the 12-month period?

- A) Delete Litware's subscription and recreate all resources in Fabrikam's tenant
- B) Transfer subscription ownership/directory association to Fabrikam's tenant, re-create RBAC assignments using Fabrikam-tenant identities, then re-parent the subscription under the appropriate Fabrikam management group
- C) Simply add the subscription's resource IDs to Fabrikam's Policy scope
- D) Ask Litware users to individually request guest access to Fabrikam's tenant

Answer: B. This is the standard full-consolidation sequence: subscription-to-tenant transfer, RBAC re-creation (since old-tenant object IDs won't resolve), then management group re-parenting for governance inheritance.

Q2.5 What should be verified regarding Requirement 5 during the eventual tenant transfer?

- A) Nothing; backups are subscription-scoped and automatically continue
- B) Recovery Services Vault RBAC assignments and any automation identities tied to the old tenant must be re-validated/re-created, since backup *schedules* continue but *management access* to the vault follows the same RBAC re-assignment rules as any other resource
- C) Backups must be manually re-uploaded after transfer
- D) The vault must be deleted and recreated post-transfer

Answer: B. The underlying backup policy/schedule keeps running (it's configuration on the vault resource, unaffected by tenant identity), but anyone who needs to *manage* the vault (trigger restores, change policy) needs RBAC re-established with new-tenant identities, exactly like every other resource in the subscription.

19. Global Infrastructure, Regions & Quotas

Azure's physical hierarchy

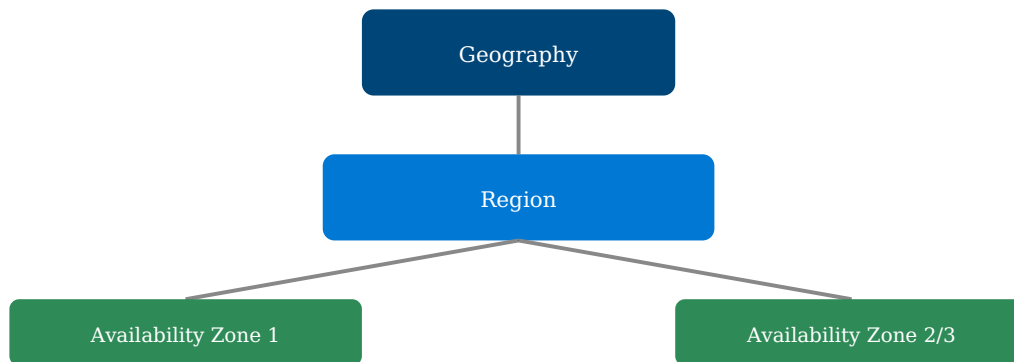


Figure 19.1 — Geography (e.g., "United States") contains Regions (e.g., "East US"), which may contain 3+ Availability Zones

Concept	Definition
Geography	A discrete market with one or more regions, respecting data residency/compliance boundaries (e.g., "United States," "Europe")
Region	A set of datacenters within a geography, deployed within a defined latency perimeter, connected by a dedicated low-latency network
Availability Zone	A physically separate location within a region with independent power, cooling, and networking (not every region has zones)
Region Pair	Every region is paired with another region ≥ 300 miles away in the same geography (where possible), used for GRS replication and staged platform updates
Sovereign/Government cloud	Physically and logically isolated Azure instances (e.g., Azure Government, Azure China) for specific regulatory requirements

Why region pairs matter for the exam

- GRS/RA-GRS/GZRS/RA-GZRS storage replication always targets the **paired region**, never an arbitrary region — you cannot choose an unpaired target.
- Microsoft prioritizes recovering one region of a pair before the other during a broad service issue, and schedules planned maintenance updates sequentially across paired regions (never both simultaneously) to reduce dual-region risk.
- Some services (e.g., certain Backup/Site Recovery configurations) specifically document their supported target as "the paired region" — recognizing the term is often enough to answer a question.

Resource naming & tagging constraints (quick reference)

Object	Constraint
Storage account name	3-24 characters, lowercase letters and numbers only, globally unique
Resource Group name	Up to 90 characters, unique within the subscription
VM name (Windows)	Up to 15 characters
VM name (Linux)	Up to 64 characters
Tag key/value	512 chars for key, 256 for value; max 50 tags per resource
VNet/Subnet name	Up to 80 characters, unique within the VNet (for subnets) or resource group (for VNets)

Common default quotas (regional vCPU and resource-count limits)

Resource	Default limit	Raisable via support request?
vCPUs per subscription per region (by VM family)	Varies (often 10-20 for new subscriptions)	Yes
Resource Groups per subscription	~980	No (hard limit)
Storage accounts per subscription	250	Yes, up to 500
VNets per subscription (per region)	1,000	Yes
NSGs per subscription (per region)	5,000	Yes

Role assignments per subscription	4,000	Yes (to a documented ceiling)
Management groups per directory	10,000	Yes
Tags per resource	50	No (hard limit)

Tip: The exam won't expect you to memorize exact numbers for every limit, but it does expect you to know the *category* of each limit (subscription-level vs. tenant-level vs. hard vs. raisable) — e.g., knowing that resource group count is a subscription-level, largely fixed limit while vCPU quota is a raisable, per-region, per-VM-family limit.

```
# Check current quota usage for a region
az vm list-usage --location eastus -o table

# PowerShell equivalent
Get-AzVMUsage -Location eastus
```

20. Advanced Monitoring: VM Insights & Dependency Mapping

Azure Monitor Agent (AMA) vs. legacy agents

Agent	Status	Notes
Azure Monitor Agent (AMA)	Current, recommended	Unified agent replacing Log Analytics (MMA/OMS) agent and Diagnostics extension; uses Data Collection Rules (DCRs) to define exactly what's collected per-machine or per-group
Log Analytics Agent (MMA/OMS)	Deprecated (retired)	Legacy agent, being phased out in favor of AMA — may still appear in exam legacy questions
Diagnostics Extension (WAD/LAD)	Legacy	Platform-specific (Windows/Linux) diagnostics, largely superseded by AMA for new deployments

Data Collection Rules (DCR)

A DCR defines *what* data to collect (specific performance counters, event logs, syslog facilities), *from which* machines (via DCR associations), and *where* to send it (which Log Analytics workspace). This decouples "what to monitor" from "which agent is installed," letting you apply different collection profiles to different VM groups from a shared agent deployment.

VM Insights



Figure 20.1 — VM Insights aggregates performance/health data and, with the Dependency Agent, visualizes process-level connections

VM Insights provides pre-built performance charts (CPU, memory, disk, network) and health monitoring without needing to author custom KQL from scratch. Adding the **Dependency Agent** alongside the monitoring agent enables the **Map** feature, which auto-discovers processes and their network connections — visually showing which VMs talk to which other VMs/external endpoints, extremely useful for planning NSG rules or understanding an undocumented legacy application's architecture before migration.

Application Insights (brief, cross-reference)

Concept	Purpose
Application Insights	Application Performance Monitoring (APM) — request rates, response times, failure rates, dependency calls, exceptions, at the code level
Instrumentation	Via SDK (code-level) or codeless agent-based auto-instrumentation for supported languages/App Service
Live Metrics	Sub-second latency view of a running application's health, useful during deployments
Availability Tests	Synthetic transactions (URL ping tests, multi-step tests) run from multiple global locations to verify uptime

Tip: VM Insights monitors infrastructure health (is the VM/OS okay); Application Insights monitors application health (is the code/request pipeline okay). A scenario asking about "why are requests to our API slow" points to Application Insights; "why is this VM's disk queue length high" points to VM Insights/Azure Monitor metrics.

Diagnostic settings routing

Destination	Use when
Log Analytics Workspace	Need to query/correlate with KQL, build alerts/workbooks
Storage Account	Long-term/cheap archival for compliance, infrequent access
Event Hub	Stream to a third-party SIEM or real-time processing pipeline

Partner solution (e.g., Sentinel)	Security-focused correlation and incident response
-----------------------------------	--

A single diagnostic setting can route to multiple destinations simultaneously (e.g., Log Analytics for operational queries AND Storage for compliance archive AND Event Hub for a partner SIEM, all from one configuration).

21. Additional Practice Questions (26-45)

Q26. App Service scaling

Q: Two web apps run on the same App Service Plan. App A experiences a traffic spike and the plan autoscales out. What happens to App B?

- A) Nothing, apps scale independently even on a shared plan
- B) App B also gets the additional instances, since scaling applies to the whole plan
- C) App B is automatically moved to a new plan
- D) App B is throttled to compensate

Answer: B. Scaling (in or out, up or down) applies to the App Service Plan as a whole; every app hosted on that plan shares the resulting capacity.

Q27. Container orchestration choice

Q: A team needs to run a single, short-lived container for a nightly batch job with no need for orchestration. What is the simplest fit?

- A) AKS
- B) Azure Container Instances (ACI)
- C) Azure Container Apps
- D) App Service for Containers

Answer: B. ACI is the lowest-overhead option for simple, non-orchestrated container workloads like a scheduled batch job.

Q28. Hybrid authentication resilience

Q: Which hybrid identity authentication method continues to allow sign-in even if the on-premises network is completely unreachable?

- A) Pass-Through Authentication
- B) AD FS Federation
- C) Password Hash Sync
- D) None of these

Answer: C. Password Hash Sync validates entirely in the cloud using the synced hash-of-hash, so it keeps working even if on-prem is down; PTA and Federation both require on-prem components to be reachable.

Q29. Load Balancer SKU and Availability Zones

Q: You need your Load Balancer's frontend to survive an Availability Zone failure. Which SKU is required?

- A) Basic
- B) Standard
- C) Either works identically
- D) Gateway SKU

Answer: B. Only the Standard SKU supports Availability Zones; Basic Load Balancer has no zone-redundancy capability and no SLA.

Q30. Azure Firewall FQDN filtering

Q: You need to allow outbound traffic only to *.windowsupdate.com and deny everything else, evaluated at the application layer. Which service provides this natively?

- A) NSG
- B) Azure Firewall
- C) Load Balancer
- D) Traffic Manager

Answer: B. NSGs filter on IP/port only and have no FQDN awareness; Azure Firewall's application rules support FQDN/wildcard-based filtering.

Q31. Data Box use case

Q: A company needs to migrate 400 TB of archival data to Azure but has only a 50 Mbps internet connection. What is the most practical approach?

- A) AzCopy over the existing connection
- B) Azure Data Box
- C) Increase the ExpressRoute circuit temporarily
- D) Storage Explorer with parallel threads

Answer: B. At 50 Mbps, 400 TB would take many months to transfer online; Azure Data Box (physical appliance shipping) is the practical choice for large one-time offline migrations.

Q32. Dedicated Host billing

Q: A Dedicated Host has capacity for 8 VMs but only 3 are currently deployed on it. How is it billed?

- A) Only for the 3 active VMs
- B) For the entire host, regardless of how many VMs are placed on it
- C) Free until fully utilized
- D) Per VM-hour like standard VMs

Answer: B. Dedicated Hosts are billed as a whole physical unit; VM count placed on it doesn't change the cost, which is why it's a compliance/isolation tool rather than a cost-saving one.

Q33. Spot VM eviction

Q: A Spot VM's eviction policy is set to Delete. What happens when Azure reclaims the capacity?

- A) The VM is deallocated but its disks are preserved
- B) The VM and its disks are removed entirely
- C) The VM is migrated to Pay-as-you-go automatically
- D) Nothing, Delete policy prevents eviction

Answer: B. With the Delete eviction policy, both the VM and its associated disks are removed upon eviction — contrast with Deallocate, which preserves the disk for a later restart.

Q34. Resource Health vs Service Health

Q: A specific VM is reporting "Unavailable." You need to know if this is a broader Azure platform issue or something specific to that VM. Which tool answers this directly?

- A) Azure Status page
- B) Resource Health for that VM
- C) Cost Management
- D) Azure Advisor

Answer: B. Resource Health shows the specific resource's health and explicitly differentiates platform-caused issues from user/configuration-caused issues.

Q35. Template Specs vs Blueprints

Q: Your organization wants to package a versioned, shareable ARM/Bicep template as a distributable Azure resource, using the currently recommended (non-deprecated) approach. What should you use?

- A) Azure Blueprints
- B) Template Specs
- C) Azure Policy Initiative
- D) Resource Graph

Answer: B. Blueprints are deprecated; Template Specs is the current recommended mechanism for packaging and sharing versioned templates as an RBAC-governed Azure resource.

Q36. Landing zone management group

Q: Following the Cloud Adoption Framework landing zone pattern, where would a subscription hosting Entra Domain Services and core identity infrastructure typically be placed?

- A) Under Sandbox
- B) Under Landing Zones → Online
- C) Under Platform → Identity
- D) Under Decommissioned

Answer: C. The Platform management group's Identity sub-group is specifically for core identity infrastructure subscriptions in the CAF landing zone pattern.

Q37. VM Insights Dependency Agent

Q: You need to visually map which processes on a VM are communicating with which external endpoints, without any manual documentation. What should you deploy?

- A) Azure Monitor Agent alone
- B) The Dependency Agent alongside the monitoring agent (enables the Map feature)
- C) Network Watcher Packet Capture only
- D) Azure Advisor

Answer: B. The Dependency Agent, paired with the monitoring agent, powers VM Insights' Map feature for automatic process-level dependency visualization.

Q38. Diagnostic settings fan-out

Q: You need resource diagnostic logs sent simultaneously to a Log Analytics workspace for querying AND to a Storage account for 7-year compliance retention. How many diagnostic settings are required?

- A) Two separate settings, one per destination
- B) One setting can route to both destinations simultaneously
- C) This is not possible
- D) Requires a Logic App to duplicate the data

Answer: B. A single diagnostic setting can fan out to multiple destinations (Log Analytics, Storage, Event Hub) at once.

Q39. Region pair replication target

Q: A storage account uses GRS. Can the administrator choose an arbitrary secondary region for replication?

A) Yes, any region worldwide B) No, GRS always replicates to the predefined paired region C) Only within the same continent, any region D) Only if using Premium storage

Answer: B. GRS/RA-GRS/GZRS all replicate specifically to the Azure-defined paired region — this is not a configurable choice.

Q40. Object Replication vs. GRS

Q: A company wants only specific containers replicated to a different Azure region (not the entire account) with an async but object-level copy mechanism, for lower read latency to users near the destination. What feature fits?

A) GRS B) Object Replication C) RA-GRS D) LRS

Answer: B. Object Replication operates at the individual blob/container level between specific storage accounts (potentially cross-region), distinct from whole-account GRS replication.

Q41. AKS network plugin choice

Q: An AKS cluster needs pods to have directly routable IP addresses within the VNet, integrating tightly with existing VNet-based network policies. Which CNI plugin should be selected?

A) Kubenet B) Azure CNI C) Neither supports this D) Flannel

Answer: B. Azure CNI assigns pods routable VNet IPs directly, unlike Kubenet which NATs pod traffic through node IPs.

Q42. ACR and AKS credential-free pulls

Q: How can an AKS cluster pull images from Azure Container Registry without storing registry credentials as Kubernetes secrets?

A) It's not possible; secrets are always required
B) Attach the ACR to the AKS cluster (`--attach-acr`), which grants the cluster's managed identity pull access
C) Make the ACR registry fully public
D) Use a SAS token stored in a ConfigMap

Answer: B. Attaching ACR to AKS grants the cluster's managed identity the AcrPull role, eliminating the need for stored image-pull secrets.

Q43. Federation use case

Q: A company requires smart-card-based authentication for cloud sign-in, which is not supported by Password Hash Sync or Pass-Through Authentication alone. What hybrid identity method should be used?

A) Password Hash Sync B) Cloud Sync only C) Federation via AD FS D) Cross-tenant synchronization

Answer: C. Federation (AD FS) is required for scenarios needing capabilities like smart-card authentication that PHS/PTA alone cannot provide.

Q44. Deployment Stack protection

Q: Which modern feature provides "deny settings" to protect resources deployed as part of a managed collection from out-of-band manual changes, similar to what Blueprint assignment locks once did?

A) Template Specs alone B) Deployment Stacks C) Resource Locks alone D) Azure Policy Initiatives

Answer: B. Deployment Stacks introduce deny settings specifically to protect their managed resources from unauthorized out-of-band changes, the modern equivalent of Blueprint's artifact locking.

Q45. Budget forecast alerts

Q: Finance wants to be alerted when spend is *projected* to exceed budget by month-end, not only once it actually happens. Which Cost Management feature supports this?

A) Cost Analysis export only B) A Budget alert condition based on forecasted cost C) Azure Advisor cost recommendations D) Reservation utilization reports

Answer: B. Budgets support both actual-cost and forecasted-cost alert conditions, enabling proactive notification before the threshold is actually crossed.

22. Study Plan & Exam-Day Checklist

Suggested 6-week study plan

Week	Focus	Activities
1	Identity & Governance (\$2)	Read fully; hands-on: create a management group tree, assign RBAC/Policy at each level, test inheritance in a free-tier subscription
2	Storage (\$3, \$13)	Read fully; hands-on: create a storage account, test each of the 5 VM-access paths, configure lifecycle management
3	Compute (\$4, \$10, \$14)	Read fully; hands-on: build a VMSS with autoscale, deploy a simple ARM/Bicep template, try a Spot VM
4	Networking (\$5, \$11, \$12)	Read fully; hands-on: build a hub-spoke topology with peering, NSGs, and a UDR; test Private Endpoint DNS resolution
5	Monitoring & Backup (\$6, \$20) + Hybrid Identity (\$9)	Read fully; hands-on: configure a backup policy and restore a test VM; write a few KQL queries
6	Review + Practice	Re-read the Cheat Sheet (\$8) and Scenario Bank (\$7) daily; complete all 45 practice questions (\$16, \$21) and both case studies (\$18) under timed conditions

Final-week review checklist

- ✓ Can you draw the scope hierarchy from memory, including inheritance direction?
- ✓ Can you explain management vs. data plane for both Storage and Key Vault without looking it up?
- ✓ Can you list all 5 storage redundancy tiers and when each is appropriate?
- ✓ Can you explain why VNet Peering is non-transitive and how to work around it?
- ✓ Can you distinguish Azure Backup, Site Recovery, and geo-replication in one sentence each?
- ✓ Do you know the exact required subnet names (AzureBastionSubnet, GatewaySubnet, AzureFirewallSubnet)?
- ✓ Can you explain PIM eligible vs. active assignments and when each is appropriate?
- ✓ Have you completed all practice questions and reviewed every wrong answer's explanation?

Exam-day logistics

Item	Recommendation
Arrival (test center)	15-30 minutes early for check-in and ID verification
Online proctored exam	Test your webcam/mic and run the system check tool the day before; clear your desk per proctoring rules
ID requirements	Two forms of ID, at least one government-issued with photo, matching your registration name exactly
Scratch material	Test centers typically provide a whiteboard/marker or digital notepad — useful for sketching scope hierarchies or network diagrams from memory
Time management	With ~120-150 minutes for ~40-60 questions, budget roughly 2-3 minutes per question, more for case studies; mark-and-review rather than stalling

After the exam

- A preliminary pass/fail result is typically shown immediately at the test center; the official score report follows within a few business days via the certification portal.
- If you pass, AZ-104 contributes toward the **Microsoft Certified: Azure Administrator Associate** certification and is also a prerequisite/complementary credential for several expert-level paths (e.g., AZ-305 Solutions Architect).
- If you don't pass, review your score report's skill-area breakdown to target weak domains, respect the retake waiting period (24 hours after attempt 1, 14 days after attempt 2), and revisit this guide's Scenario Bank and Practice Questions for the flagged domains specifically.

"Good luck — you've now covered every domain in depth, seen the ownership and visibility model at every scope, worked through 45 practice questions and 2 full case studies, and have a command reference for both CLI and PowerShell. That's a genuinely complete preparation for AZ-104."

23. Full-Length Mock Exam (40 Questions)

Simulate real exam conditions: 75 minutes, no notes, answer all 40 before checking the answer key at the end of this section. Questions are mixed across all domains, matching the real exam's distribution.

#	Question
1	Which scope, if any, can contain Azure resources directly? (A) Tenant (B) Management Group (C) Resource Group (D) None of the above can contain resources
2	A user is Reader at the subscription and Contributor at one specific resource group. What is their effective access to a resource in a sibling resource group? (A) Contributor (B) Reader (C) Owner (D) No access
3	Which Azure Policy effect blocks a non-compliant deployment outright? (A) Audit (B) Append (C) Deny (D) Disabled
4	What is required, in addition to a role assignment, to allow a subscription Owner to read blob data via their Entra ID identity? (A) Nothing, Owner is sufficient (B) A Storage Blob Data role (C) MFA (D) A Resource Lock
5	How many IP addresses does Azure reserve in every subnet? (A) 2 (B) 3 (C) 5 (D) 10
6	Which storage redundancy option provides the highest durability while allowing reads from the secondary region? (A) LRS (B) ZRS (C) GRS (D) RA-GZRS
7	A VM is stopped from within the guest OS. Is compute still billed? (A) No (B) Yes (C) Only for 1 hour (D) Depends on VM size
8	What must be true for VNet A to reach VNet C when A-B and B-C are peered? (A) Nothing extra, peering is transitive (B) A UDR via an NVA in B plus "allow forwarded traffic" on both links (C) They must be in the same region (D) It is never possible
9	Which Azure Firewall capability is NOT available on the Basic tier? (A) FQDN filtering (B) TLS inspection (C) Stateful filtering (D) Network rules
10	Which service is purpose-built for orchestrated regional disaster recovery of entire VM workloads? (A) Azure Backup (B) Azure Site Recovery (C) GRS (D) Azure Monitor
11	What subnet name is mandatory for Azure Bastion? (A) BastionSubnet (B) AzureBastionSubnet (C) GatewaySubnet (D) Any name works
12	Which PIM concept requires explicit activation before permissions take effect? (A) Active assignment (B) Eligible assignment (C) Conditional Access (D) Custom role
13	Which load-balancing service is DNS-based rather than a traffic proxy? (A) Application Gateway (B) Azure Load Balancer (C) Traffic Manager (D) Azure Firewall
14	What happens to SAS tokens derived from a storage account key when that key is regenerated? (A) Nothing (B) They are all invalidated (C) Only write-permission tokens are invalidated (D) They convert to user-delegation SAS
15	Which Availability construct protects against an entire datacenter failing? (A) Fault Domain (B) Update Domain (C) Availability Zone (D) Availability Set
16	What is the minimum requirement to move a resource between resource groups? (A) Reader on both (B) Contributor (minimum) on both source and destination (C) Owner only on the destination (D) No permission needed if in the same subscription
17	Which ARM deployment mode can delete resources not defined in the template? (A) Incremental (B) Complete (C) Validate (D) What-if
18	Which hybrid identity method keeps working if the on-prem network becomes unreachable? (A) Federation (B) Pass-Through Authentication (C) Password Hash Sync (D) None of these
19	What does a Resource Lock of type ReadOnly prevent, that CanNotDelete does not? (A) Deletion (B) Both modification and deletion (C) Nothing extra (D) Read access
20	Which container service has no orchestration layer and is best for a single short-lived job? (A) AKS (B) ACI (C) Azure Container Apps (D) App Service for Containers
21	What must be configured for a Private Endpoint's FQDN to resolve correctly inside a VNet? (A) A Service Endpoint (B) A Private DNS Zone linked to the VNet (C) A public DNS A record (D) Nothing, it's automatic
22	Which Azure Backup feature retains deleted backup data recoverable even if an attacker deletes the backup items? (A) GRS (B) Soft delete (C) Snapshot (D) Lifecycle management
23	What is the primary distinction between Service Endpoints and Private Endpoints? (A) Service Endpoints work from on-prem, Private Endpoints don't (B) Private Endpoints assign a private IP reachable cross-network; Service Endpoints only optimize routing for subnet-local traffic (C) They are functionally identical (D) Private Endpoints are free, Service Endpoints are not
24	In a hub-spoke topology, what additionally is required (beyond peering) for spoke-to-spoke traffic via a hub NVA? (A) Nothing, peering alone suffices (B) A UDR pointing to the NVA plus "allow forwarded traffic" on both peering links (C) A VPN Gateway (D) A Public IP on the NVA
25	Which management/billing hierarchy applies specifically to Enterprise Agreement subscriptions? (A) Tenant → Management Group → Subscription (B) Enrollment → Department → Account → Subscription (C) Region → Zone → Resource (D) Root → Landing Zone → Sandbox
26	What is the effect of the Azure Policy "Modify" action? (A) Blocks deployment (B) Adds/updates properties or tags on resources automatically (C) Only reports non-compliance (D) Disables the resource
27	Which VM disk type is NOT persistent across deallocation? (A) OS disk (B) Data disk (C) Temp disk (D) Shared disk
28	What tool lets you test whether a specific packet would be allowed or denied by current NSG rules, without generating real traffic? (A)

	Connection Troubleshoot (B) IP Flow Verify (C) Packet Capture (D) Topology
29	Which of these is NOT one of the five Azure Advisor recommendation pillars? (A) Cost (B) Security (C) Reliability (D) Licensing
30	What must happen to RBAC assignments after a subscription is transferred to a new Entra tenant? (A) Nothing, they transfer automatically (B) They must be re-created using destination-tenant identities (C) They are converted to Policy assignments (D) They remain valid for 90 days then expire
31	Which redundancy tier is appropriate when data must never leave a specific country/region, even during failover? (A) GRS (B) RA-GRS (C) LRS or ZRS (D) GZRS
32	What Azure feature provides browser-based RDP/SSH without exposing a public IP on the target VM? (A) VPN Gateway (B) Azure Bastion (C) ExpressRoute (D) Network Watcher
33	Which VMSS orchestration mode supports mixing VM sizes and Spot + Standard instances? (A) Uniform (B) Flexible (C) Classic (D) Manual
34	What does Azure Lighthouse allow an MSP/parent tenant to do? (A) Migrate a subscription instantly (B) Manage a customer's subscription with delegated RBAC, without moving it into their own tenant (C) Bypass all RBAC checks (D) Merge two tenants automatically
35	Which KQL table would you query to investigate who deleted an Azure resource? (A) Perf (B) AzureActivity (Activity Log) (C) Heartbeat (D) AppTraces
36	What is required for AKS to pull images from ACR without storing credentials as Kubernetes secrets? (A) A SAS token in a ConfigMap (B) Attaching the ACR to the cluster, granting the managed identity AcrPull (C) Making the registry public (D) It's not supported
37	Which cost-optimization option offers the deepest discount but carries eviction risk? (A) Reserved Instances (B) Savings Plans (C) Spot VMs (D) Dedicated Hosts
38	What must be true for a Management Group Policy assignment to affect a specific subscription? (A) The subscription must be a direct or indirect child of that management group (B) Nothing, all policies apply tenant-wide regardless of hierarchy (C) The subscription must have an Owner login within 24 hours (D) The policy must be manually re-applied to each subscription
39	Which service combines Layer 7 global routing, edge caching, and WAF in one offering? (A) Traffic Manager (B) Azure Load Balancer (C) Azure Front Door (D) Network Watcher
40	What is the correct sequence for retiring an on-prem file server with Azure Files while preserving branch-office performance? (A) Just move files to Blob storage (B) Deploy Azure Files + Azure File Sync with server endpoints at each branch (C) Use Azure Migrate (D) Use Data Box only

Mock Exam Answer Key

#	Answer
1	C — only Resource Groups (and below) hold resources
2	B — Reader only, inherited from subscription; the RG Contributor grant doesn't apply to a sibling RG
3	C — Deny
4	B — a Storage Blob Data role (data plane)
5	C — 5 (.0, .1, .2, .3, .255)
6	D — RA-GZRS (zone + geo, secondary readable)
7	B — Yes, still billed until Azure-level Deallocate
8	B — UDR via NVA + allow forwarded traffic on both links
9	B — TLS inspection is Premium-tier only
10	B — Azure Site Recovery
11	B — AzureBastionSubnet
12	B — Eligible assignment
13	C — Traffic Manager (DNS-based)
14	B — all are invalidated
15	C — Availability Zone
16	B — Contributor minimum on both source and destination
17	B — Complete mode
18	C — Password Hash Sync
19	B — ReadOnly blocks both modification and deletion; CanNotDelete only blocks deletion
20	B — Azure Container Instances

#	Answer
21	B — a linked Private DNS Zone
22	B — Soft delete
23	B — Private Endpoint = private IP, cross-network; Service Endpoint = subnet-local routing optimization
24	B — UDR + "allow forwarded traffic" on both peering links
25	B — Enrollment → Department → Account → Subscription
26	B — adds/updates properties or tags automatically
27	C — Temp disk
28	B — IP Flow Verify
29	D — Licensing is not one of the five pillars (Cost, Security, Reliability, Operational Excellence, Performance)
30	B — must be re-created with destination-tenant identities
31	C — LRS or ZRS (never GRS, which crosses to a paired region)
32	B — Azure Bastion
33	B — Flexible orchestration mode
34	B — manage without moving the subscription (Lighthouse)
35	B — AzureActivity (Activity Log)
36	B — attach ACR to the cluster (managed identity gets AcrPull)
37	C — Spot VMs (up to ~90% off, but evictable)
38	A — the subscription must be a direct or indirect child of that management group
39	C — Azure Front Door
40	B — Azure Files + Azure File Sync with server endpoints per branch

Scoring guide

Score	Interpretation
36-40 correct (90%+)	Exam-ready; do a final pass over the Cheat Sheet (§8) and any missed questions' topics
30-35 correct (75-89%)	Close; revisit the specific domains behind your misses using the section cross-references above
Below 30 correct (<75%)	Return to the full domain sections for your weakest areas before attempting another mock exam

24. Azure Service Quick Reference (60+ Services)

A one-line memory-jogger for every service mentioned in this guide, organized by category, for rapid pre-exam review.

Identity & Governance

Service	One-line purpose
Microsoft Entra ID	Cloud identity directory: users, groups, apps, devices
Entra Connect / Cloud Sync	Synchronizes on-prem AD objects to Entra ID
Conditional Access	If-then sign-in policies based on user/device/location/risk
Privileged Identity Management	Just-in-time, approval-gated privileged role activation
Azure RBAC	Who can do what, at what scope
Azure Policy	Enforces/audits resource configuration compliance
Azure Blueprints (deprecated)	Legacy bundling of RBAC+Policy+templates
Template Specs	Versioned, shareable ARM/Bicep templates as a resource
Deployment Stacks	Managed resource collections with deny-setting protection
Resource Locks	CanNotDelete/ReadOnly protection overriding RBAC
Management Groups	Container above subscriptions for RBAC/Policy at scale
Cost Management + Billing	Budgets, cost analysis, reservations, savings plans
Azure Lighthouse	Cross-tenant delegated resource management
Cross-tenant synchronization	Auto-provisions B2B guests between Entra tenants

Storage

Service	One-line purpose
Storage Account (Blob/File/Queue/Table)	Multi-service managed storage namespace
Azure Data Lake Storage Gen2	Hierarchical-namespace Blob storage for analytics
Azure Files	Managed SMB/NFS file shares
Azure File Sync	Syncs on-prem file servers with an Azure Files cloud endpoint
AzCopy	CLI tool for high-performance bulk data transfer
Azure Storage Explorer	GUI tool for browsing/managing storage
Azure Import/Export	Ship physical disks to/from a Microsoft datacenter
Azure Data Box	Microsoft-supplied appliance for large offline data migration
Object Replication	Async, object-level replication between storage accounts

Compute

Service	One-line purpose
Virtual Machines	IaaS compute instances
Availability Sets	Fault/Update Domain grouping within one datacenter
Availability Zones	Physically separate datacenters within a region
Virtual Machine Scale Sets	Autoscaling groups of identical VM instances
Azure Dedicated Host	Physical server dedicated to a single customer
Spot VMs	Steeply discounted, evictable compute capacity
Managed Disks	Block storage volumes for VMs (OS/Data/Temp)
Azure Migrate	Discovery, assessment, and migration tool for lift-and-shift
ARM Templates / Bicep	Declarative infrastructure-as-code for Azure
App Service	PaaS web app hosting with deployment slots, autoscale
Azure Container Instances (ACI)	Single/simple container groups, no orchestration

Azure Kubernetes Service (AKS)	Managed Kubernetes container orchestration
Azure Container Apps	Serverless containers with managed Kubernetes underneath
Azure Container Registry (ACR)	Private Docker image registry

Networking

Service	One-line purpose
Virtual Network (VNet)	Private, isolated network in Azure
Network Security Group (NSG)	Stateful L3/L4 traffic filter
Application Security Group (ASG)	Logical NIC grouping for NSG rules
User-Defined Route (UDR)	Custom routing overriding system routes
VNet Peering	Direct, non-transitive VNet-to-VNet connectivity
VPN Gateway	IPsec site-to-site/point-to-site hybrid connectivity
ExpressRoute	Private circuit to Azure, bypassing the internet
Azure Load Balancer	L4 TCP/UDP load balancing, regional
Application Gateway	L7 HTTP(S) load balancing with WAF, regional
Traffic Manager	DNS-based global traffic routing
Azure Front Door	Global L7 routing, edge caching, WAF
Azure Firewall	Managed, centralized L3-L7 network firewall
Azure DDoS Protection	Volumetric attack mitigation for VNets
Azure Bastion	Browser-based RDP/SSH without public IPs
Private Link / Private Endpoint	Private IP access to specific PaaS resource instances
Service Endpoint	Subnet-scoped optimized routing + firewall rule to PaaS
Azure DNS	Public and private authoritative DNS hosting
Azure DNS Private Resolver	Enables on-prem to Azure Private DNS Zone resolution
Network Watcher	Diagnostics: flow verify, effective rules, packet capture

Monitoring & Backup

Service	One-line purpose
Azure Monitor	Umbrella platform for metrics, logs, alerts
Log Analytics Workspace	KQL-queryable store for logs/metrics
Azure Monitor Agent (AMA)	Current unified data collection agent
Data Collection Rules	Defines what/from-where/to-where for AMA collection
VM Insights	Pre-built VM performance/health monitoring + dependency Map
Application Insights	APM: request/response/exception tracking for apps
Alerts & Action Groups	Threshold/condition-based notifications and automated actions
Service Health	Personalized Azure incident/maintenance notifications
Resource Health	Per-resource health status and platform-vs-customer cause
Azure Advisor	Cost/Security/Reliability/Performance/Ops recommendations
Recovery Services Vault	Container for Azure Backup and Site Recovery
Azure Backup	Point-in-time backup/restore for VMs, files, DBs
Azure Site Recovery	Orchestrated regional disaster recovery failover

Security & Compliance

Service	One-line purpose
Microsoft Defender for Cloud	Cloud security posture management and workload protection
Key Vault	Managed secrets/keys/certificates storage
Managed Identity	Auto-managed Entra ID identity for secretless resource auth
Just-in-time VM Access	Time-limited, on-demand opening of management ports

25. KQL Query Reference Appendix

Kusto Query Language (KQL) powers Log Analytics queries used throughout Monitor, Sentinel, and Application Insights. AZ-104 expects familiarity with reading/adapting simple queries, not writing complex ones from scratch.

Core syntax building blocks

Operator	Purpose	Example
where	Filter rows	where TimeGenerated > ago(1h)
summarize	Aggregate rows into groups	summarize count() by Computer
project	Select specific columns	project TimeGenerated, Computer, CounterValue
order by	Sort results	order by TimeGenerated desc
join	Combine rows from two tables on a key	join kind=inner (OtherTable) on Computer
render	Visualize results as a chart	render timechart
ago()	Relative time reference	ago(24h) , ago(7d)
let	Define a reusable variable	let threshold = 80;

Frequently useful queries

```
# Who deleted a resource in the last 7 days?
AzureActivity
| where TimeGenerated > ago(7d)
| where OperationNameValue has "delete"
| where ActivityStatusValue == "Success"
| project TimeGenerated, Caller, ResourceGroup, Resource, OperationNameValue
| order by TimeGenerated desc

# VMs with sustained high CPU (candidates for rightsizing)
Perf
| where ObjectName == "Processor" and CounterName == "% Processor Time"
| where TimeGenerated > ago(24h)
| summarize AvgCPU = avg(CounterValue) by Computer
| where AvgCPU > 80
| order by AvgCPU desc

# Denied NSG flows in the last hour (requires NSG Flow Logs sent to this workspace)
AzureNetworkAnalytics_CL
| where TimeGenerated > ago(1h)
| where FlowStatus_s == "D"
| summarize count() by SrcIP_s, DestIP_s, DestPort_d
| order by count_ desc

# Failed sign-ins by user in the last 24 hours (requires Sign-in logs)
SigninLogs
| where TimeGenerated > ago(24h)
| where ResultType != 0
| summarize FailCount = count() by UserPrincipalName, ResultDescription
| order by FailCount desc

# Storage account availability trend (Metrics-based)
AzureMetrics
| where ResourceProvider == "MICROSOFT.STORAGE"
| where MetricName == "Availability"
| summarize AvgAvailability = avg(Average) by bin(TimeGenerated, 1h)
| render timechart

# Backup job failures across all protected items
AddonAzureBackupJobs
| where TimeGenerated > ago(7d)
| where JobStatus == "Failed"
| project TimeGenerated, BackupItemUniqueId, JobOperation, JobFailureCode
```

Common tables cheat-list

Table	Contains
AzureActivity	Control-plane operations (the queryable form of the Activity Log)
Perf	Performance counters from VM agents (CPU, memory, disk, network)
Heartbeat	Agent connectivity/health check-ins, used to detect offline VMs

SignInLogs / AuditLogs	Entra ID sign-in and directory audit events (requires diagnostic setting to the workspace)
AzureNetworkAnalytics_CL	Parsed NSG Flow Log data (requires Traffic Analytics enabled)
AppRequests / AppTraces / AppExceptions	Application Insights telemetry
AddonAzureBackupJobs	Azure Backup job history and status
AzureMetrics	Platform metrics forwarded into Log Analytics via diagnostic settings

Exam tip: You won't be asked to write complex KQL from scratch, but you should recognize what a given query is checking for (e.g., spotting that a query filters `AzureActivity` for delete operations means "this answers who deleted something") and know which table a given monitoring scenario points to.

26. Real-World Troubleshooting Playbooks

These playbooks walk through diagnosing common Azure administration problems step by step — the same reasoning pattern the exam expects, applied to realistic operational incidents.

Playbook 1: "I can't RDP/SSH into my VM"



Figure 26.1 — Connectivity troubleshooting funnel, outside-in

Step	Check	Tool
1	Is the VM actually running (not stopped/deallocated)?	<code>az vm get-instance-view</code> / Portal Overview blade
2	Does the VM have a reachable path — Public IP with correct NAT rule, or Bastion deployed?	Portal Networking blade, or check for <code>AzureBastionSubnet</code>
3	Do both the subnet AND NIC-level NSGs allow the port inbound from your source?	Network Watcher → Effective Security Rules, or IP Flow Verify
4	Is a UDR silently routing the response traffic somewhere unexpected (e.g., through a firewall that then drops it)?	Network Watcher → Next Hop
5	Is the guest OS's firewall (Windows Firewall / iptables) blocking the port, or is the RDP/SSH service not running?	Boot diagnostics / Serial console (works even without network path)

Root cause frequency in practice: NSG misconfiguration (missing rule or wrong priority ordering) accounts for the majority of real-world "can't connect" tickets, followed by guest-OS firewall, then genuinely stopped/deallocated VMs.

Playbook 2: "A user can see a resource but can't modify it"

Step	Check	Tool
1	What is their effective RBAC role at this exact resource and every ancestor scope?	<code>az role assignment list --assignee <user> --all -o table</code>
2	Is a Resource Lock (ReadOnly or CanNotDelete) present anywhere in the scope chain?	Portal Locks blade at resource/RG/subscription level
3	Is an Azure Policy with a Deny effect blocking the specific write operation?	Portal Policy → Compliance, or attempt the change and read the returned error (Policy denials include the policy name)
4	Is this a data-plane operation requiring a separate data role (common for Storage/Key Vault)?	Check whether the resource type has a management/data-plane split (§3.3, Key Vault section)

Gotcha to remember here: A Policy Deny and a missing RBAC permission produce different error messages — a Policy denial explicitly names the policy assignment that blocked it, while an RBAC failure returns a generic "AuthorizationFailed." Reading the actual error text tells you which of the two you're dealing with immediately.

Playbook 3: "Storage costs unexpectedly spiked"

Step	Check	Tool
1	Did access tier distribution shift (more Hot, less Archive than expected)?	Cost Analysis filtered by the storage account, grouped by meter
2	Did redundancy change (e.g., someone switched LRS to GRS)?	Storage account Configuration blade / Activity Log for a "Set Storage Account" event
3	Is there unexpected egress (e.g., a misconfigured job repeatedly downloading the	Storage account Metrics → Egress, broken down by

	same large dataset)?	hour
4	Are soft-deleted blobs/snapshots/versions accumulating and still being billed?	Storage account Data Protection blade, review retention settings
5	Did a Lifecycle Management rule stop working (e.g., after a filter/prefix change) leaving Hot data that should have tiered down?	Review Lifecycle Management rule run history

Playbook 4: "VNet peering shows Connected but resources still can't communicate"

Step	Check
1	Confirm "Allow virtual network access" is enabled on <i>both</i> peering links, not just one side
2	Confirm address spaces don't overlap (peering would have failed to establish if they did, so this is more relevant if peering itself won't connect)
3	Check NSGs on both sides — peering being "Connected" only means the network path exists, not that NSGs allow the specific traffic
4	If traffic needs to traverse a third VNet (hub) between two spokes, verify "Allow forwarded traffic" and a correct UDR exist on both relevant peering links
5	Check for a conflicting UDR that sends traffic somewhere other than the peering connection's default system route

Playbook 5: "Backup jobs are failing"

Step	Check
1	Review the specific failure code in the vault's Backup Jobs blade or via the <code>AddonAzureBackupJobs</code> KQL table (\$25)
2	For VM backups: confirm the VM backup extension is installed and reporting healthy, and that the VM has network access to the required Azure Backup service endpoints (or appropriate Private Endpoint if network-isolated)
3	Check whether a ReadOnly lock on the VM's resource group is blocking the snapshot operation (a classic ReadOnly-lock side effect)
4	For app-consistent backups on Windows, confirm the VSS writer for the relevant application (SQL, Exchange, etc.) is functioning inside the guest
5	Confirm the vault hasn't hit a soft-delete/retention conflict, or that Multi-User Authorization isn't blocking an automated operation awaiting approval

Playbook 6: "Autoscale isn't scaling out despite high load"

Step	Check
1	Confirm the metric threshold and aggregation window actually match current conditions (e.g., "avg over 10 minutes" won't react to a 2-minute spike)
2	Check whether the scale set has already hit its configured maximum instance count
3	Verify a ReadOnly lock isn't blocking the write operation autoscale needs to add instances (autoscale genuinely fails silently under a ReadOnly lock)
4	Check regional vCPU quota — autoscale can be blocked by hitting a subscription's per-region vCPU limit for that VM family
5	Review the Autoscale "Run history" in the Portal, which explicitly logs why a scale action was or wasn't taken at each evaluation

"Every troubleshooting playbook follows the same shape as the scope-and-ownership reasoning from §2.2: work outside-in from network to identity to configuration, and check every layer rather than assuming the first plausible cause is the actual one."

27. Naming Conventions & Enterprise Tagging Standards

Not directly examined, but foundational to every enterprise Azure deployment and frequently the backdrop of scenario questions that mention "following our naming standard." A consistent convention makes ownership, environment, and purpose identifiable from the resource name alone.

Recommended naming pattern

<resource-type-abbreviation>-<workload/app>-<environment>-<region>-<instance>

Example: `vm-webapp-prod-eus-001`, `st-webappprodeus001` (storage accounts strip hyphens due to naming constraints), `rg-webapp-prod-eus`

Common resource-type abbreviations

Resource type	Abbreviation
Resource Group	rg
Virtual Machine	vm
Virtual Network	vnet
Subnet	snet
Network Security Group	nsg
Storage Account	st
Key Vault	kv
Load Balancer	lb
Public IP	pip
App Service Plan	asp
App Service (Web App)	app
Recovery Services Vault	rsv
Log Analytics Workspace	log
Azure Kubernetes Service	aks
Container Registry	acr
Virtual Machine Scale Set	vmss
Application Gateway	agw
Azure Firewall	afw
Management Group	mg

Common environment abbreviations

Environment	Abbreviation
Production	prod
Staging	stg
Development	dev
Test/QA	test / qa
Sandbox	sbx
Shared/Platform	shd

Recommended baseline mandatory tags

Tag key	Example value	Purpose
Environment	Production / Staging / Development	Filter cost, apply differentiated policy strictness
CostCenter	CC-4471	Chargeback/showback to the owning business unit
Owner	jane.doe@contoso.com	Accountable contact for the resource

Application	WebApp-Checkout	Groups resources belonging to the same logical application
DataClassification	Public / Internal / Confidential / Restricted	Drives security/compliance controls and audit scope
CreatedDate	2026-09-01	Lifecycle tracking, cleanup automation eligibility

Tip: Enforce mandatory tags via an Azure Policy initiative with the `Modify` effect for auto-population of low-risk defaults (e.g., `CreatedDate`) and `Deny` for tags that must always be explicitly and correctly set by the deployer (e.g., `CostCenter`, `DataClassification`) — mixing the two effects appropriately balances friction against governance rigor.

Naming constraints quick-reference (recap from §19, consolidated with abbreviations)

Resource	Allowed characters	Length
Storage account	Lowercase letters, numbers only	3-24
Resource Group	Alphanumerics, underscores, hyphens, periods, parentheses	Up to 90
VM (Windows)	Alphanumerics, hyphens (no trailing period/hyphen)	Up to 15
VM (Linux)	Alphanumerics, hyphens, periods, underscores	Up to 64
Key Vault	Alphanumerics, hyphens; must start with a letter	3-24, globally unique

28. Capstone: End-to-End Reference Architecture Walkthrough

This closing chapter ties every domain in this guide into one coherent, governed, production-grade deployment — the kind of environment a working Azure Administrator actually builds and operates. Use it as a mental checklist: for any new workload, can you name the governance, storage, compute, network, and monitoring decision at each layer the way this walkthrough does?

The scenario

Contoso Retail is deploying a new order-processing platform: a public web front end, an internal API tier, a SQL database, and Blob storage for invoice documents. It must be highly available within one region, secured end-to-end, cost-governed, and recoverable from both accidental deletion and a regional outage.

Full architecture diagram

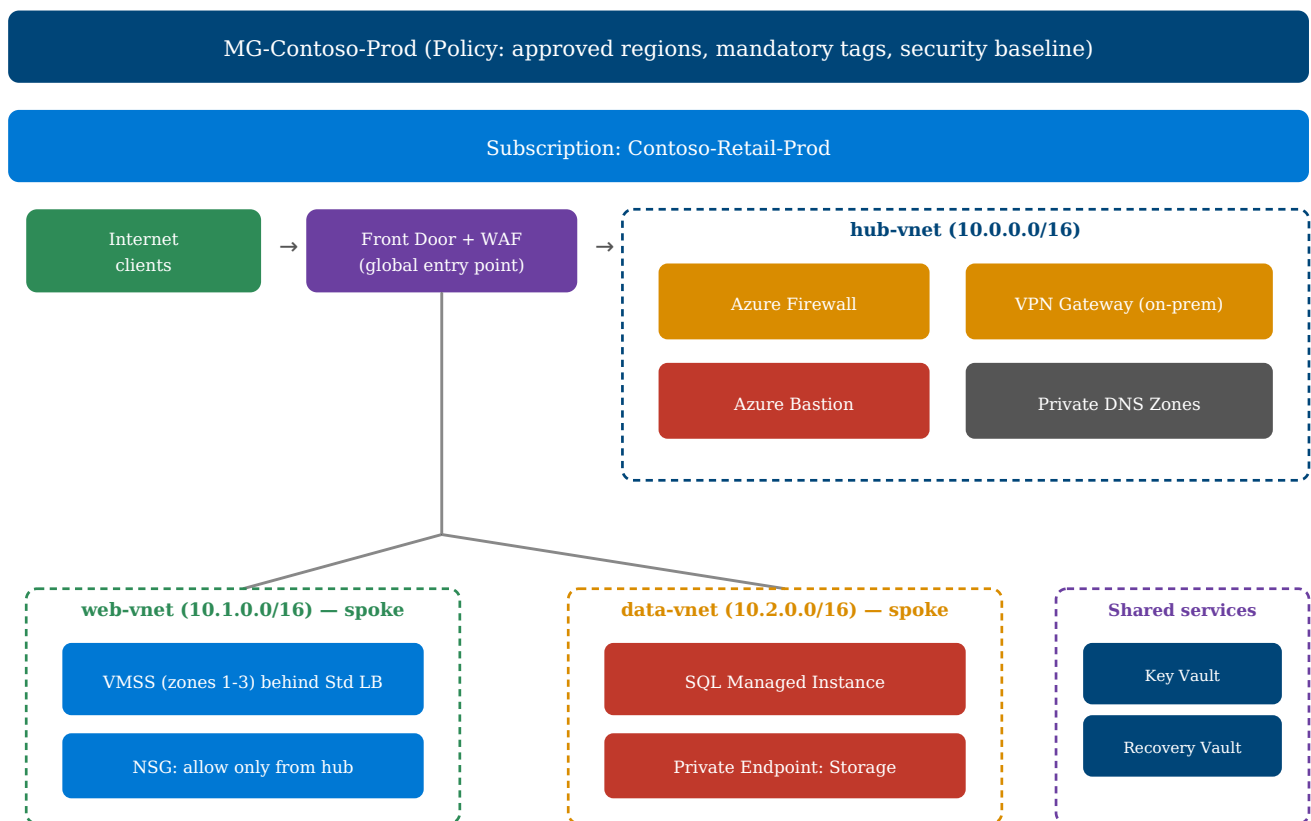


Figure 28.1 — Complete governed hub-spoke deployment: every domain in this guide represented in one architecture

Layer-by-layer decisions and why (cross-referenced to earlier sections)

Layer	Decision	Reasoning
Governance (\$2)	Subscription placed under MG-Contoso-Prod with a Policy initiative enforcing approved regions, mandatory tags (Modify effect), and a security baseline (Deny effect for public storage access)	Applying guardrails at the management group means every future subscription added to Prod inherits them automatically
RBAC (\$2.3)	Platform team: Owner at the subscription. App team: Contributor scoped to <code>rg-webapp-prod</code> only. DBAs: a custom role with SQL-specific actions, scoped to <code>rg-data-prod</code>	Least-privilege, narrowest-blast-radius assignment per team's actual responsibility
Network (\$5)	Hub-spoke topology; Azure Firewall in the hub inspects all egress; spokes reach each other only via hub-routed UDRs; Bastion provides admin access with no public IPs anywhere	Centralized security inspection and zero exposed management ports, consistent with defense-in-depth (\$12)
Compute (\$4)	Web tier: zone-redundant VMSS with autoscale on CPU + a scheduled morning scale-out profile	Survives a full Availability Zone failure; anticipates known traffic patterns proactively
Storage (\$3)	Invoice documents: StorageV2, GRS, Cool tier with Lifecycle Management to Archive after 90 days; accessed only via Private	Cost-optimized tiering, cross-region durability, zero secrets, and no public network exposure

	Endpoint from data-vnet, using the app's managed identity with Storage Blob Data Contributor	
Database	SQL Managed Instance in a delegated subnet inside data-vnet, inherently private with no public endpoint option even needed	Simplifies the private-by-default requirement compared to SQL on a VM needing its own NSG/Private Endpoint setup
Secrets (\$5.6)	Connection strings and API keys in Key Vault; VMSS instances and the app tier use managed identities to retrieve them, granted Key Vault Secrets User only	No credentials in code, config files, or CI/CD pipeline variables
Backup & DR (\$6)	Recovery Services Vault (GRS) backs up VMSS instance disks and the SQL MI nightly with 30-day retention; a documented ASR-based failover runbook exists for the rare full-region-loss scenario	Covers both "someone deleted something" (Backup) and "the region is gone" (ASR readiness) per §6.3's decision guide
Monitoring (\$20)	Azure Monitor Agent + VM Insights on every VMSS instance, Application Insights instrumented in the API tier, Action Groups routing critical alerts to the on-call team's phone and a ticketing webhook	Full visibility from infrastructure to application code, with actionable alerting rather than passive dashboards
Cost governance (\$2.5, \$14)	A Budget with 80%/100% forecasted alerts on the subscription; Reserved Instances purchased for the baseline VMSS capacity once usage stabilized; Spot VMs considered only for a separate, unrelated batch-reporting workload	Predictable production workloads get committed-use discounts; interruption-tolerant workloads get Spot pricing — matching the cost tool to the workload's tolerance for disruption

What this walkthrough demonstrates

No single decision here is exotic — every choice traces directly back to a principle covered earlier in this guide: scope-appropriate governance, least-privilege RBAC, defense-in-depth networking, zone-redundant compute, tiered and privately-accessed storage, managed-identity secrets access, layered backup/DR, and full-stack monitoring. That is precisely the level of synthesis the AZ-104 exam's case studies are testing: not whether you know each fact in isolation, but whether you can assemble them correctly under a real set of business requirements.

"An Azure Administrator's job is rarely to invent something novel — it's to correctly combine well-understood building blocks for the specific requirements in front of them. Master the building blocks in this guide, and the combinations become straightforward."